

Documentation

Project: Parque Industrial Taiwan
Order no: 13018435

Machine:
RoK4

Data sheet..... 3

Manual..... 4

Handling of stainless steel..... 49

Declaration of Incorporation..... 51

Spare parts drawing..... 55

Spare parts list..... 56

Geared motor - Manual..... 57

Motor data sheet screw drive BF40-74W - 51718583..... 145

Geared motor: Spare parts drawing/-list gear..... 146

Geared motor: Spare parts drawing/-list motor..... 148

Huber Technology

Data sheet

Project:	Parque Industrial Taiwan
Order number:	13018435
Machine:	HUBER Pumping Stations Screen ROTAMAT® RoK4
Type:	RoK4
Size:	300
Material:	1.4307 (304L)
Hole diameter:	6 mm
Length:	6500 mm
Geared motor:	BF40-74W
Voltage / Frequency:	400 Y V / 50 Hz
Protection class:	IP65

Operating Instructions

HUBER Pumping Stations Screen ROTAMAT® RoK4



HUBER SE
Industriepark Erasbach A1

92334 Berching

Original operating instructions
Version 01/25

	<table border="1"><tr><th data-bbox="359 376 1447 443">NOTICE</th></tr><tr><td data-bbox="359 443 1447 640"> This manual is part of the plant and must be available for the operators any time. The safety instructions must be observed. In case of selling the plant, the manual must be included. For a better overview this manual may contain photos/graphics of the machine without safety devices. Operation of the machine is only permitted when all prescribed safety devices as e.g. covers or railings are mounted. </td></tr></table>	NOTICE	This manual is part of the plant and must be available for the operators any time. The safety instructions must be observed. In case of selling the plant, the manual must be included. For a better overview this manual may contain photos/graphics of the machine without safety devices. Operation of the machine is only permitted when all prescribed safety devices as e.g. covers or railings are mounted.
NOTICE			
This manual is part of the plant and must be available for the operators any time. The safety instructions must be observed. In case of selling the plant, the manual must be included. For a better overview this manual may contain photos/graphics of the machine without safety devices. Operation of the machine is only permitted when all prescribed safety devices as e.g. covers or railings are mounted.			

Translation

Supplies to EEA countries require translation of the operating instructions into the language of the country of destination.

Should there emerge any inconsistencies in the translation, the original (German) operating instructions shall be used for clarification or the supplier consulted.

Copyright

Circulation, copying or use of this document, or disclosure of its contents, shall be prohibited unless otherwise expressly agreed. Copyright infringement will result in a liability to pay compensation for damages.

All rights reserved.

1	PRODUCT SPECIFICATION	4
1.1	Intended use	4
1.2	EMC advice	4
1.3	Definition of terms for machine components.....	5
1.4	Specification of plant components	6
1.5	Functional description	6
1.6	Functional description of (partly optional) components	7
1.6.1	Press zone	7
1.6.2	Bagging unit (option)	9
1.6.3	Internal heating or trace heating (optional)	9
1.6.4	Level control system	10
2	EC-DECLARATION OF CONFORMITY RESP. OF INCORPORATION.....	12
3	SAFETY	13
3.1	General safety instructions	13
3.1.1	Operator's duty of care.....	13
3.1.2	Definition of safety symbols	14
3.1.3	Qualification and training of personnel.....	14
3.1.4	Safety instructions for maintenance, inspection, installation	15
3.1.5	Residual risk.....	17
3.1.6	Unauthorised rebuilding and production of spare parts	17
3.2	Machine identification.....	17
3.3	Incorporated safety systems	17
3.4	Safety measures	19
3.5	Operator's duty of care.....	19
3.6	Safety tests	20
4	HANDLING AND TRANSPORTING	21
4.1	Dimensions and weight.....	21
4.2	Permitted transport devices and auxiliaries	21
4.3	Machine storage.....	22
4.4	Transport to the installation place	23
5	INSTALLATION.....	24
5.1	Acceptable environmental conditions	24
5.2	Site requirements	25
5.3	General instructions for installation	25
5.4	Assembly and installation.....	26
5.4.1	Mechanical installation	26
5.4.2	Electrical installation.....	26
6	START-UP	28
6.1	Customer-supplied connections.....	28
6.1.1	Electrical connection	28
6.1.2	Connection for wash water	28
6.2	Checks prior to initial start-up	29
7	OPERATION	30
7.1	Control philosophy	30
7.2	Operation possibilities.....	30
7.2.1	Operation by means of adjacent control switch	30
7.2.2	Time-dependent automatic start control (option)	31
7.2.3	Run-on time.....	31
10	SHUTDOWN	44
10.1	Temporary shutdown	44
10.2	Final shutdown / Disposal	44
11	ADDITIONAL INFORMATION	45

1 Product specification

1.1 Intended use

The equipment is intended to separate solids from wastewater, and compact and wash out separated screenings – depending on the equipment version.

For solids separation the machine is applied:

- in wastewater treatment plants (municipal and industrial), especially manholes
- for industrial process and partial flows

The intended use also includes:

- Observance of the start-up, operation and maintenance conditions as set out in these operating instructions
- Taking foreseeable malpractice into account
- Operation by skilled workers only (who are familiar with the correct procedures and know the dangers)

WARNING

The machine is intended exclusively for the above specified use.

Any additional use or rebuilding of the equipment without prior written approval by the manufacturer does not comply with the intended use.

The manufacturer will not assume liability for consequential damage. The operator alone will bear the risk.

Do not start up the machine before there has been ensured that all safety devices are completely mounted and operable, and that the plant into which the machine may be incorporated complies with the rules.

The machine is suitable to be applied in EX protection areas if a separate ATEX Declaration of Conformity has been issued and one of the below additional specifications is made on the identification plate.

It is mandatory to perform the ATEX-related maintenances (see chapter Maintenance).

The operator must check that gas is never discharged into the non Ex-protection area.

Specification on identification plate for Ex protection areas zone 1:

  II 2 G Ex h IIB T3 Gb

Specification on identification plate for Ex protection areas zone 2 inside / no zone outside:

  II 3/- G c IIC T3

1.2 EMC advice

NOTICE

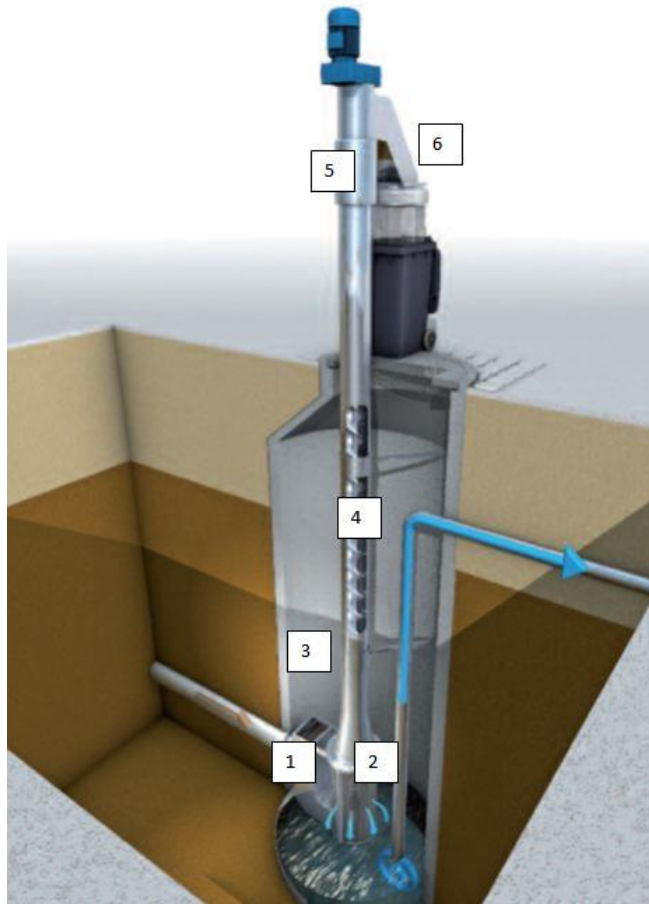
Advice for applying the EC rule EMV 2014/30/EU:

The plant is prepared for operation in industrial areas acc. to DIN EN 61000-6-4 (generic standards for transient emissions in industrial areas). Wiring and control technology have to meet additionally the requirements of DIN EN 61000-6-3 (transient emission in living areas) for operation in living areas, in business and commercial areas and in small companies.

1.3 Definition of terms for machine components

Definition of terms:

Terms for equipment components according to the following drawing:



- | | |
|----------------------|---------------------------|
| 1 Inlet | 4 Vertical screw conveyor |
| 2 Screen basket | 5 Press zone |
| 3 Emergency overflow | 6 Discharge chute |

The stationary part of the screen comprises (from top to bottom):

- Lower vertical support
- Closed screen basket with inlet chamber
- Rising pipe with guide bars
- Horizontal support
- Press zone, press liquor collection chamber, press water return
- Screenings discharge
- Gear motor

These components form the machine body, to which optional equipment can be attached.

The rotating part of the machine consists of the following components:

- Screw with central shaft and flights
- Upper shaft pinion

These components form a rotating unit which is driven by a gear motor with hollow shaft fitted at the upper end of the machine.

1.4 Specification of plant components

Components of the stationary equipment and their functions:

Screen basket

The screen basket is a precisely welded construction. The screening surface, which is retaining the screenings, is a self-supported integral part that is supported in its exactly defined form by the welded front plate at the bottom a flange on top. The screen basket is axially divided and flanged. The inlet chamber is welded to one of the basket halves. It is important not to damage the screening surface during screen installation or operation, respectively immediately repair a damage, if any.

Rising pipe:

The rising pipe is the central and stationary component and therefore has a stable design:

- Internal guide bars prevent that the screenings turn with the screw while the screw transports them out of the screen basket and conveys them upwards.
- The compaction chamber where the screenings are dewatered is positioned at the upper end of the conveyance area prior to the screenings discharge. The rising pipe is perforated in this section and encased by the press water collection chamber.
- A cover in the rising pipe allows access to the compaction chamber. Access to that cover is possible by removing the whole press water collection tank or the inspection cover attached to the press water collection tank.
- The machine supports are single C-profiles. The horizontal supports are clamp mounted to the rising pipe.

Gear motor:

The gear motor is directly flanged to the top of the rising pipe. The torque is transferred onto the shaft pivot by means of an adjusting spring connection. The gear motor has also the function of the axial and upper radial bearing of the screw.

Components of the rotating screen parts and their functions:

Auger:

- The central screw shaft transmits the torque onto the screw flights.
- The shaft pinion is equipped with a fitting key connection.
- The screw flights have a different design and pitch and end in front of the press zone.
- One oppositely directed flight with doctor blade at the upper screw end supports screenings discharge.

1.5 Functional description

The wastewater flowing in through the feed line streams into the inlet chamber, which is a rectangular open top chamber so that it can also serve as an emergency overflow in the event of a power failure. A rectangular opening in the inlet chamber bottom connects the inlet chamber with the screen basket. The screen basket is fed with wastewater through this opening. The level measurement equipment for plant control is mounted within the inlet chamber. The clarification of the wastewater takes place within the screen basket whilst the wastewater runs off by gravity through the perforated plate basket. The solids are retained in the vertically arranged screen basket. Screen basket cleaning in this section is achieved by screw segments being equipped with brushes that transport the retained screenings upwards and clean the screening surface. On its way towards the discharge the screenings are further dewatered and compacted in the press zone section prior to being discharged into a container without any odour nuisance. The press liquor collects in the press liquor collection chamber and flows back through the transparent press liquor return hose into the channel. The filtrate can then be lifted by pump.

Optional additional equipment:

- A frost protection allows outdoor operation down to -25°C. The rising pipe is equipped with an insulation and heating rod or heating cable, depending on the individual machine type and size.
- An automatic press zone washing system ensures fully automatic washing of the press liquor collection chamber and reduces cleaning work.

- A screenings bagging unit reduces odour annoyance in the screen building. The screenings bagging unit is attached directly to the screenings discharge. Plastic bags (360 l bags for 120/240 l containers, 1100 l bags for 770/1100 l containers) or a 70 m long sausage-type bag take in the discharged screenings.

Supply limits:

Mechanical:

- Inlet chamber flange
- Screenings discharge chute
- Lifting eyes on the machine:
- Internal coupling of press zone washing system

Electrical:

- Screws in the terminal box
- Terminal strip inside the control panel

1.6 Functional description of (partly optional) components

1.6.1 Press zone

Press liquor collection chamber:

The screenings are dewatered in the press zone which is located on the upper end of the conveying section prior to the screenings discharge. In this area the rising pipe is perforated and encased by the press water collection chamber.

The press water collected in the press water collection chamber is discharged through the outlet hose into the manhole. Along with the water also sludge or faeces are pressed out while the screenings are compacted, which might lead to clogging in the collection chamber and outlet hose. It may therefore become necessary to dismount the press liquor collection chamber.

WARNING

Switch off and lock the mains isolator prior to starting with dismantling!

Wear protective gloves!

How to proceed with dismantling/mounting the plastic chamber:

- Dismount the outlet hose.
- Loosen the lateral screws with hexagonal recessed hole. Be careful to prevent that the washers cast into the plastic turn with the screws.
- Carefully remove the plastic half-shells.
- Pay attention to press the sealing strip into the provided groove. Make sure the sealing surfaces of the rising pipe are clean.
- Thoroughly lubricate the front sealing surfaces.
- Hand-screw the fixing screws with a maximum torque of 5 Nm. Attention: Do not use any screw locking mean!

Press zone washing system: Automatic / manual

Activation of the press zone washing systems effects cleansing of the press zone respectively press liquor collection chamber.

A spray nozzle is mounted in the bottom front end of the press zone. The machine is supplied with a Geka coupling, a 1" shutoff line to be supplied by the customer.

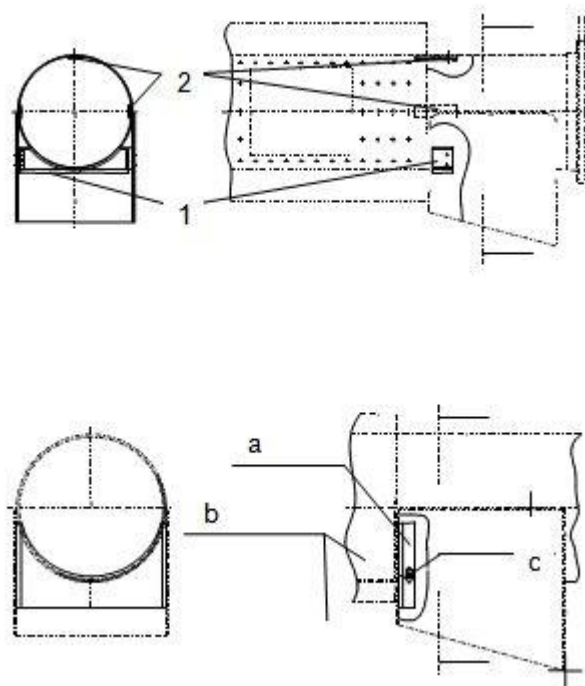
Manual operation of press zone washing system: Open water supply for approx. 30 s once or twice a week. The connection for water is a quick coupling on the press liquor collection chamber.

Automatic operation of press zone washing system (optional): Water supply for press zone washing is automatically activated via a solenoid valve. The standard setting is 30 s twice a day. The washing time can be changed independent of the operational conditions to ensure a sufficient cleaning of the collection chamber.

Water supply: See chapter 6.1.2
 Operational water pressure: min. 5 bar max. 7 bar
 Water quality: below 0.2 mm screened or filtered water

Regulation devices for increased/reduced compaction:

All rising pipe have a 50 mm long detachable extension piece (part 1 in below drawing), which lengthens the press zone at the upper end to increase compaction. This extension piece can be dismantled if the DS content of screenings is too high.



The machine size 700 (355 mm pipe dia.) has a displaceable plate (part a) installed in the discharge which can be displaced by means of fixing screws (part c). Displace the plate downwards if the DS content is too high and upwards if the DS content is too low. Standard setting: 40 mm from the lower discharge end, which reduces the cross section by 8 mm (see b).

	! WARNING
	Switch off and padlock the mains isolator prior to starting to work on the press zone. Attention: Automatic machine start: Never grip into the machine's discharge while the machine is running! Be careful not to get caught!

1.6.2 Bagging unit (option)

The screenings are discharged into a plastic bag fixed at the discharge of the screen. The bagging unit reduces odour annoyance as would be caused by screenings lying in the skip or container.

Design of the bagging unit for individual bags:

The bagging unit comprises a transition piece between the rectangular screenings discharge assembly, a circular fixing device for the bags, and a clamping band to fix the 360 l bags (item no. 706799) or 1100 l plastic bags (item no. 703417).

How to replace the bag: Loosen the clamping band and pull the bag downwards. When attaching the new bag be careful it is pressed tight onto the bagging unit, throughout its full circumference, by the clamping band.

Design of the bagging unit for sausage-type bags:

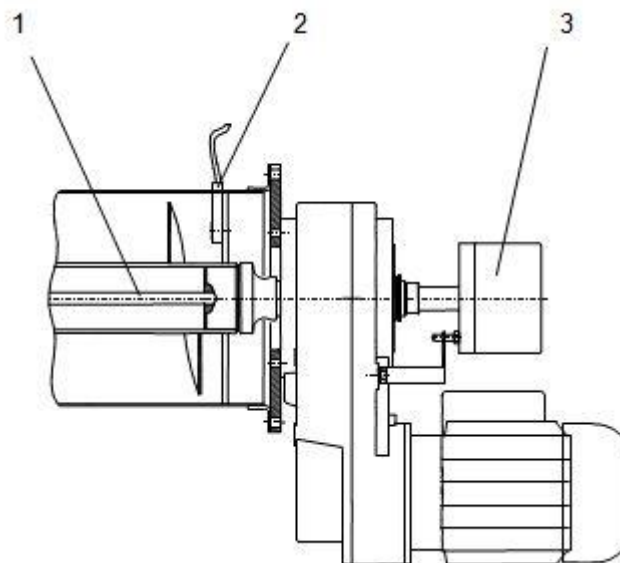
The bagging unit consists of a transition piece to take up the folded 70 m long plastic bag. When you replace the skip, pull the bag (Longo-Pack bag, article no. 706880) approx. 30 cm downwards. Lace the bag and cut it through above the knot. Lace also the end that hangs down so that the hose can again be filled with screenings.

Switch off the machine prior to replacing the bags and wear gloves!

1.6.3 Internal heating or trace heating (optional)

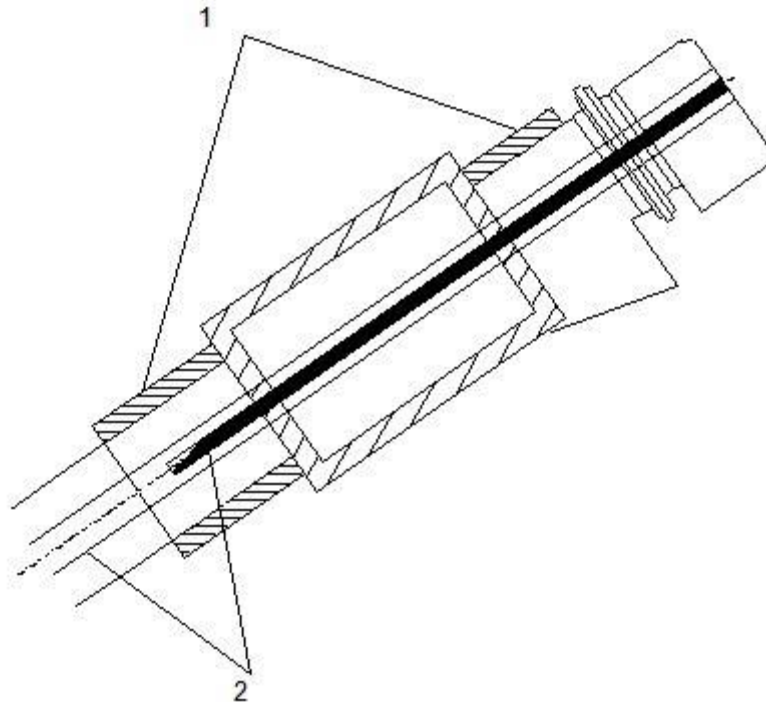
Action principle of internal heating:

A self-regulating heating rod heats the screw shaft effectively down to – 15°C and can be applied in non-explosive areas.



- 1 Heating rod
- 2 Temperature probe
- 3 Slip ring

We offer as an option an additional insulation with covering shells which is effective down to – 25° C.



- 1 Additional insulation
- 2 Heating rod in central screw shaft

WARNING

The heating rod must be connected to a mains voltage of 230 V (+20 V). See also the PTB test report attached in the appendix.

Action principle of heating cable:

A self-regulating heating cable is applied in Ex-protection areas which is wrapped around the rising pipe (trace heating). The rising pipe is additionally covered with 60 mm insulation material and a stainless steel plate. Control of the heating is via a temperature probe in the rising pipe, the connected wattage is approx. 33 W per meter of rising pipe heating.

1.6.4 Level control system

Since the level control system is part of the electrical switchboard and control panel, it is only part of HUBER SE supply if the machine is ordered complete with the electrical control panel. The following description of the standard HUBER level control system is therefore provided to be thoroughly considered if the control panel is customer-supplied.

Water level control by air injection via a submerged pipe; miniature compressor for air supply (voltage: 230V) integrated in the control panel.

Action principle:

- The pressure air generated by the compressor is connected at the pressure switch.
- Prior to the pressure switch the pressure air is additionally directed through a hose to the connection in front of the plant (within the inlet chamber), at the lower edge of which the air is escaping.
- The cellulose plugs in the transparent air hose are used as throttles, so that air comes out of the submerged pipe. Never remove them!
- Depending on the water level, a specific air pressure is required for the air to escape. This air pressure affects the membrane on the pressure switch side, whilst the other side is affected by the ambient air pressure.
- The membrane is supported on one side by a spring.

- When the pressure reaches the preset level (10 cm WG), the membrane touches the double-throw switch and actuates it.

Adjusting the level control system:

- Differential pressure switch: It is adjusted to 10 cm level difference by the supplier. Please do not change this adjustment!
- Mount the air injection pipe in the inlet chamber. The bottom edge of the injection pipe must lie on the feed pipe axis.

The deeper the air injection pipe is installed, the lower the machine will become submerged and switched on earlier, which results in longer machine operating hours.

The data specified above are only basic settings which need to be adjusted to the individual wastewater quality and operational requirements!

NOTICE

Maintenance required after 3 years:
Replacement of the miniature compressor membrane and air filter.

Other single-probe level control systems: (Option, e.g. conductive level probe)
Start-up and adjustment according to the individual manufacturer's operating instructions.

NOTICE

Ultrasonic level control must not be used for this machine!

Requirement: Potential-free contact at the respective water level!
The operator is responsible to prevent metallic dry objects from falling onto the dry plant. In explosive areas plant control must be via level control to prevent dry running of the plant.

2 EC-Declaration of Conformity resp. of Incorporation

The machine complies with the EC standards which prescribe the CE label. The EC Declaration of Conformity confirms that the operable machine fulfils all relevant safety and health requirements. The EC Declaration of Conformity is provided only if the HUBER plant is supplied as a ready-to-operate unit complete with the electrical switchboard and control panel, and if plant installation and commissioning are performed by HUBER SE.

The Declaration of Incorporation is required if the supplied incomplete machine is not operable independently, i.e. if the machine is to be incorporated in other machines for example to obtain an operable complete plant, or if the electrical switchboard and control panel is supplied by a third party. We herewith declare that the design of the incomplete machine as supplied complies with standards, EC directives and DIN EN standards, as far as applicable as delivery does not include the electrical switchboard and control panel. Any modification of the machine without our prior approval will invalidate this declaration. Start-up of the machine is prohibited until the complete plant is in conformity with the quoted directives.

The Declaration of Incorporation is attached in the appendix and additionally included in the table of contents.

3 Safety

3.1 General safety instructions

DANGER

„DANGER“ indicates a hazardous situation which, if not avoided, will result in death or serious injury.

WARNING

„WARNING“ indicates a hazardous situation which, if not avoided, could result in death or serious injury.

CAUTION

„CAUTION“ indicates a hazardous situation which, if not avoided, may result in minor or moderate injury.

NOTICE

„NOTICE“ indicates a property damage message. Follow notices to avoid material damage!

These operating instructions have to remain attached to the machine. It must be made sure that the operating instructions are ready to hand any time for any person that has to perform work on the machine. In addition to these operating instructions, instructions in the sense of the labour protection law and ordinance regulating the use of tools have to be available. As these operating instructions contain fundamental instructions to be observed when installing, operating and servicing the machine, the responsible staff must read the instructions prior to machine installation and start-up. The operating instructions must at any time be available ready to hand at the installation place of the machine/plant. Not only the general safety instructions contained in this chapter have to be observed but also the special safety instructions added under the main items.

3.1.1 Operator's duty of care

The plant has been constructed and manufactured taking into consideration a risk analysis and after careful selection of the applicable harmonized standards and other technical specifications. The machine complies with the state-of-the-art technology and offers a maximum amount of safety.

To achieve such safety in practical operation, it is however necessary to take any measures required therefore. It is the operator's duty of care to plan these measures and control their implementation.

The operator must especially ensure that

- The equipment is applied according to its intended use (see chapter Product Specification)
- The equipment is operated only in a perfect ready-to-operate condition and especially the safety devices are regularly controlled.
- Protective gear for the operating, maintenance and repair personnel is available and used.

- These operating instructions are permanently available on site complete and in a legible condition.
- Only sufficiently qualified and authorized personnel are in charge of machine operation, maintenance and repair.
- Such personnel receive regular briefing concerning all questions of safety and environmental protection and knows these operating instructions, especially the safety instructions contained.
- Any safety and warning symbols attached to the machine remain there in a legible condition.

3.1.2 Definition of safety symbols

	! WARNING
	Occupational safety symbol This symbol will accompany all safety instructions that are associated with risks to life and/or limb. Follow these instructions and proceed carefully! At the same time, follow all applicable laws, general safety and accident prevention regulations.

	! WARNING
	Electric current warning This symbol warns of electric current. Prior to performing any work, switch off mains isolator and make sure that the system is off-circuit. At the same time, follow all applicable laws, general safety and accident prevention regulations.

	! WARNING
	Automatic machine start: Be careful not to get caught when starting up, servicing or repairing the machine!

NOTICE
This symbol is found where special attention is required to ensure compliance with instructions concerning correct operating sequences to prevent damage to the machine or its function.

Instructions directly attached to the machine, e.g.

- Instructions and warning signs
 - Labels for liquid connections
 - Arrow showing the direction of rotation
- must be strictly followed and kept in absolutely legible condition.
 Signs or labels that have become illegible must be replaced immediately.

3.1.3 Qualification and training of personnel

Only well-trained and briefed persons who know these operating instructions and act according to these instructions are authorized to operate the machine. The individual areas of responsibility of operating staff must be defined clearly. The area of authority, responsibility and control of the personnel must be precisely regulated by the operator. The operator must further ensure that the personnel have fully understood these operating instructions.

Personnel being trained must in the beginning work under the supervision of an experienced person. The completed successful training and briefing must be confirmed in writing.

Any electrical control and safety devices must generally be operated by instructed and authorized persons only.

Any person performing work on the machine must read these operating instructions and confirm by signature that the operating instructions have been understood.

3.1.4 Safety instructions for maintenance, inspection, installation

Any maintenance work must be carried out by qualified staff only.

Any inspection and installation work must be carried out by authorized and qualified staff only. Work on electrical equipment may only be carried out by a skilled electrician in accordance with DIN VDE 1000-10.

Work on the plant may only be carried out when the plant is disconnected from the mains.

Dangers for people:

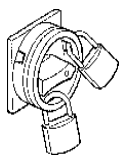
- Toxic gases, or suffocation due to lack of oxygen
- Falling into the tank, or parts falling down.
- Explosion of biogases
- Dangers when entering/leaving the tank
- Drowning inside the tank
- Danger caused by application of electrical devices

Enclosed rooms of wastewater treatment plants that need to be entered for service and maintenance have to be aerated in a way that prevents a dangerous explosive atmosphere, lack of oxygen and presence of harmful concentrations of gas or vapour. The maintenance company or the manufacturer's personnel is therefore only permitted to enter the tank after sufficient prior aeration and use of a gas warning device. In addition, appropriate measures must be taken to protect maintenance and start-up personnel against falling into the tank.

WARNING

Secure the machine against unexpected start in case of open access to the screenings discharge resp. secure the access to this danger area before changing the screenings container or doing maintenance work or disassembling on a downstream machine.

WARNING



Shutdown procedure: Switch off mains isolator and lock it.

Each person who is commissioned to perform maintenance work on the machine must have his own padlock. Starting the machine is only possible when all padlocks on the mains isolator have been removed.

Check if upstream and downstream equipment may cause a danger, switch off power supply for upstream and downstream equipment before performing maintenance or repair.

If the whole plant cannot be disconnected from the mains for operational reasons, individual machines must be disconnected by a skilled electrician in accordance with DIN VDE 1000-10 and secured against unintentional restart acc. to DIN EN 50110-1 VDE 0105-1.

As an option all machine drives can be supplied with a repair switch. For maintenance work the machine or area concerned can be switched off acc. to DIN VDE 0100-200. The repair switches must be switched off and locked as described above.

⚠ WARNING**Attention: Danger of falling into the tank**

Only the maintenance or start-up personnel with appropriate protective equipment is permitted to enter the tank! One supervisory person must stay outside the tank. Use protective gear. (Safety shoes, gloves, etc.). Safeguard the open tank against hazards, secure the open tank against falling into. Enter the tank only on with appropriate rescue equipment, secured by a rope, using a tripod and rope. Rescue from outside in case of an accident.

⚠ WARNING**Attention: How to behave in the event of an accident**

Whenever someone enters a septic tank, a second person must be there to secure him. Never follow unconscious persons into the tank. Immediately call for help instead.

**⚠ CAUTION****Risk of explosion and danger when using electrical devices**

Kapitel 1 Sufficient ventilation required.
Low voltage to be applied (boring machines, etc.) or isolating transformers to be used.

Do not immediately re-start the machine, if the reason why it has stopped is unclear. Somebody could have stopped the machine in order to make a manual adjustment and may have forgotten to secure it against starting. The unexpected start could result in serious injury of personnel.

It is in your own interest to clean the machine prior to working on it to prevent the danger of infections.

⚠ CAUTION

Always protect yourself by means of waterproof protective gear, boots, gloves, and, if possible, also by face protection during cleaning of the machine - especially if a high pressure cleaner is being used – to avoid being hit by waste water, organic material, etc.

Re-attach all safety equipment, covers, grates completely to their original place and assure that they are properly and completely reattached.
Use only tools and means that are expressly intended for such work.

Before starting the machine check the closed position of the turning lock on safety equipment like covers, these locks keep the covers in safe position.

**⚠ Caution**

When working on the machine, always wear safety goggles with integrated side protection (in accordance to standard EN 166) and protective gloves.

Before starting the machine again, check the items mentioned in chapter Start-up.

3.1.5 Residual risk

The residual risk describes the risk of a system that remains after the implementation of planned risk containment measures and in spite of existing safety devices.

The following residual risks may appear:

- Automatic machine start
- Aerosols contaminated by viruses and bacteria caused by manual cleaning
- Reaching into the running machine

3.1.6 Unauthorised rebuilding and production of spare parts

Alterations or changes to the machine:

For safety reasons, it is not permitted to make unauthorized alterations or changes to the machine. This applies also to welding work on bearing components.

Any intended modifications, alterations or changes require the prior written consent of HUBER SE.

Use only original spare parts, original wearing parts and original accessories as these are especially designed for the machine. Components purchased from other sources give no guarantee that they have been designed and manufactured to suit the specific operating and safety requirements.

3.2 Machine identification

Any specifications made in these operating instructions apply to only the type of machine that is named on the title page.

The identification plate is attached to the screenings discharge and specifies the following.

- Name and address of supplier
- CE sign
- Serial name and type, optionally serial number
- Year of manufacture

Always forward the machine type, year of manufacture and order number when inquiring or ordering spare parts to ensure perfect and prompt processing of your queries and orders.

3.3 Incorporated safety systems

The incorporated safety systems are subject to regular checkups (**t** = daily, **w** = weekly, **m** = monthly, **j** = yearly). The following methods are applied:

S = sight inspection, **F** = functional test, **M** = measuring.

Mains isolator

The mains isolator is located on the control panel and disconnects/connects the machine from/with the mains supply.

Padlock the mains isolator after switch-off prior to performing service or repair work.

Checkup	
Interval	Method
j	F

Emergency cutoff circuit

The machine is equipped with an emergency cutoff circuit. Whenever the emergency cutoff switch (option) is operated, the machine or complete plant including incorporated units will be set into a safe operating state.

The emergency cutoff switch can be released by pulling or turning to the right.

Checkup	
Interval	Method
m	F

Motor protection switch

The machine is equipped with an indirect current overload protection with a motor protection switch. The motor will be switched off, if the overload is too much. The thermal delayed overcurrent protective device must be selected that the drive unit will be switched off in time t_E .

Checkup	
Interval	Method
j	F, M

Motor temperature control (option)

The motor is equipped with a temperature sensor, which must be evaluated with an evaluation relay or a frequency converter. If the permissible temperature is exceeded, this must lead to the motor being switched off.

Checkup	
Interval	Method
j	F, M

Machine control

Internal machine control includes a 5-conductor feed system, 3 phase, with separate earth line with GREEN/YELLOW line coating.

Additionally earth the machine casing for potential equalization. Please check earthing connection on corrosion.

Checkup	
Interval	Method
j	S, F, M

Categories acc. to DIN EN ISO 13849-1

The following categories are used on the plant:

Category	System performance	Component	Inspection interval
1	- A fault may cause the loss of safety function, - Some faults may stay unrecognized	Main switch, water shut-off device with protection against restart	1x per year or acc. to instructions of supplier of components
2	- A fault may cause the loss of safety function between the tests. - The loss of safety function is recognized by the test.	Emergency stop (emergency cut-off relays with push-button and cut-off protection)	1x per year or acc. to instructions of supplier of components

	! WARNING
	Control panels can only be opened by using a special key. The special key may only be handed to an authorized person. Take care that doors of control panels are only opened by skilled staff for maintenance work and fault detection, otherwise the doors must stay locked!

NOTICE
These operating instructions are part of the machine and have to be available for the operating staff at any time. The safety instructions contained must be observed.

It is strictly prohibited to override any safety instructions or change the mode of action of safety instructions.

3.4 Safety measures

It is the operator's responsibility to instruct his operating and servicing staff concerning:

- Protective devices on the machine,
- Control of observance of safety measures.

This copy of operating instructions has to be stored to be at hand when needed in future. Observe the intervals for inspection and control measures! In these operating instructions, the work is described so that it can be understood by

- an instructed person (referring to chapter Operation and operation modes
- skilled staff (referring to chapters Transport, Installation, Maintenance, Trouble Shooting and Repair).

The chapters Transport, Installation, Maintenance, Trouble Shooting and Repair are intended for skilled staff only. Any work described under these chapters must be performed by skilled staff only.

Instructed person

An instructed person is a person that has been instructed by a skilled person, and trained if necessary, about the assigned jobs and possible risks arising from improper performance and informed about necessary protective devices and protective measures.

Skilled persons

Skilled persons are persons that are able to evaluate assigned jobs and recognize possible risks, due to their professional skills, expertise and experience and knowledge of corresponding standards.

This definition follows EN 60204-1.

3.5 Operator's duty of care

NOTICE

The valid national version of the framework directive 89/391/EEG and corresponding individual directives, especially 89/655/EEG concerning minimum requirements for safety and health protection of staff when using work equipment, are applicable in EEA countries and must be observed.
For Germany, the occupational safety directive of 2015 is applicable and must be observed.

The operator has to obtain the local operating license and observe the respective requirements.

In addition, the operator has to observe the local laws concerning

- Safety of personnel (accident prevention regulations)
- Safety of work equipment (protective gear and maintenance)
- Product disposal (Waste Management Law)
- Material disposal (Waste Management Law)
- Cleaning (cleaning agent and disposal)
- Environmental compliance

Connections:

The operator has to ensure before start-up of the machine, if installation and start-up are performed by the operator himself, to comply with local standards (such as for electrical connection for instance).

NOTICE**Lighting**

The operator has to provide sufficient and equal lighting in all areas of the plant. The recommended illumination level is 300 lux (value for maintenance; in Germany acc. to ASR).

3.6 Safety tests

Performed by the manufacturer before delivery

1. Airborne sound measurement

- as per rules for machines, appendix 1 (1.7.4/f)

The noise level of the plant lies below 70 dB(A).

2. Test and checkup as per DIN EN 60204-1

- Check of electrical equipment for correspondence with the technical documentation (chapter 18.1)
- Functional tests (chapter 18.1)
Test of functions of electrical devices, especially those relating to safety and protective measures.

4 Handling and transporting

Observe the following points to avoid damage to the machine or persons when handling the equipment:

- Only qualified persons are permitted to perform transport work, observing the safety instructions.
- Lifting and righting of the equipment must be done only by the lifting eyes provided.
- Use only the lifting devices specified hereunder to transport the machine.
- Read also the chapter General Safety Instructions.

4.1 Dimensions and weight

The plant is available with 300 to 700 mm screen basket diameter. The machine length is adjusted to suit specific site requirements (available space, depth, discharge height).

The dimensions are specified in the project-specific installation drawing or general dimension sheet attached to these installation instructions.

The weight of the screen depends on the machine size and length. The weights are specified in the following table. The weight is also specified on the nameplate.

Size	Length	3000	4000	5000	6000	7000	8000	9000
300	242	284	326	368	242	504	546	588
	282	334	386	438	282	584	636	688
	332	399	461	528	332	684	746	811
500		462	504	546		593	634	675
		512	564	616		673	724	775
		611	675	740		810	873	937
700		896	991	1086		1217	1311	1408
		966	1071	1176		1317	1421	1528
		1163	1295	1426		1594	1726	1858
700XL		966	1060	1154		1266	1360	1455
		1036	1140	1244		1366	1470	1575
		1276	1407	1538		1686	1817	1949

All weights (kg) apply to machines without extra equipment for outdoor installation or any other optional equipment. The machine size and length are specified in the data sheet. The machines are packed on pallets for trucking, and in sea-water-proof boxes for sea transport.

4.2 Permitted transport devices and auxiliaries



Have transport and unloading done by experienced experts only.

Lifting vehicles of sufficient lifting capacity to handle the weight and size of the equipment, to be operated by qualified personnel only. Rope slings or straps of required load bearing capacity. Prior to unloading, remove the small parts supplied with the machine, such as supports, from the transport rack.

	⚠ WARNING
	Transport by forklift is forbidden, never insert the fork under the machine!

	⚠ WARNING
	Attachment: Hook shackle, load hook etc. into the lifting eyes on the upper side of the rising pipe. Fixing points are marked with the symbol LIFT HERE (as illustrated here). The ropes of the lifting device must hang freely and must not be attached more than 60° from the vertical. The machine must hang horizontally during unloading. Never lift at the thicker part of the rising pipe prior to the screenings discharge. This is the press liquor collection chamber which is made of thinner plate or plastic and is unable to bear the machine weight. Never wrap a chain around the machine as this may slip and cause the machine to tilt over. In addition, the machine will rust on those points where it gets into touch with black steel if it is not re-pickled.

Inspect all materials for damage before and during unloading. Any transport damage found should be noted on the bill of lading and the forwarder and manufacturer/supplier notified immediately.

NOTICE
Make sure the delivery is complete by carefully checking all received materials against the bill of delivery.

4.3 Machine storage

The machine and components can be stored as supplied. The ambient temperature should be between a minimum of +5°C and a maximum of +40°C. Avoid high temperature fluctuations.

If outdoor storage cannot be avoided, the motors, electrical components and cardboard packaging must be protected against direct exposure to the weather (moisture, direct sunlight, dirt/dust), e.g. by using a breathable cover.

When selecting the storage location, please ensure that there is no risk of damage, e.g. from construction vehicles or negligent work in the vicinity of the machine. Avoid soiling from cement, mortar, concrete and paint splashes. Protect the machine from sparks from angle grinders and other tools to prevent corrosion.

Stable storage must be ensured to prevent the machine from tipping over. The storage location must be vibration-free.

Never store organic solvents at the machine's storage location. Avoid ambient air containing hydrogen sulphide and chloride.

Observe the storage instructions for equipment attached to or supplied with the machine.

NOTICE

Do not lay the machine on its thicker part prior to the screenings discharge (press liquor collection chamber) since this will damage the press liquor collection chamber.

4.4 Transport to the installation place

As the situation and possibilities are different on each individual site, we cannot provide exact installation instructions. This work must be carried out by qualified fitters.

You always need lifting equipment for transporting the machine from the intermediate storage place to the installation place because of the machine weight.

**! WARNING**

The maximum working load of the lifting device must be above the weight of the single machine components.
Observe the stability of the lifting device.

! CAUTION

Wear safety shoes with steel caps to prevent injuries.

**! WARNING**

Always stand clear off a suspended load!

Unpacking:

Do not loosen the clamping bands or steel ropes before the machine is right in its installation place.

5 Installation

Observe the following safety instructions when installing the machine to avoid critical injuries, damage to the machine and other damage.

- Only qualified persons are permitted to perform installation work, observing the safety instructions.
- Check the machine for transport damage prior to starting with any installation work.
- Make sure that only authorized persons have access to the working area and that installation work does not endanger any other persons.
- When laying machine connections, make sure that no one can trip over laid cables, hoses, pipelines, etc.
- Observe the prescribed bending radii when laying cables/hoses/pipelines.
- Observe the instructions for operating media, lubricants, auxiliary material used.
- Read also the chapter General Safety Instructions.

5.1 Acceptable environmental conditions

The machine is manufactured for installation in a building or for outdoor installation as specified by the customer.

	CAUTION Stainless steel parts can get very hot in case of heavy solar radiation, danger of burning.
---	--

Frost protection:

Machines without trace heating and insulation and/or casing must not be operated in winter due to the danger of freezing. Appropriate measures must be taken (e.g. complete emptying) to ensure that the machine cannot freeze.

Protection against dust and water:

As the atmosphere in the buildings or manholes is permanently humid, the equipment is designed to resist these conditions.

All components in contact with water and solids are made of stainless steel which is insusceptible to moisture and wetness.

The gear motors are made for IP 65 and are therefore protected against permeation of dust and spray water from any direction.

The solenoid coils, if fitted to the machine, also provide IP 65 protection.

Protection against explosion (optional):

The plant and its components are available in the following ex-protected versions:

Gear motor:	EEx e II T3
Adjacent control box	EEx ia II T5/T6
Solenoid valves	EEx ed II C T4

Conformity certificates are attached in the appendix.

Lightning protection concept:

The plant must be integrated into the complete concept of the treatment plant for observing the Lightning Protection Standard DIN EN 62305-3.

The state of the art of a functioning overall concept for wastewater treatment plants is particularly described in the supplement 2.

Lighting:

The lighting must make safe working on each part of the plant possible without any risk.

NOTICE**Lighting**

The operator has to provide sufficient and equal lighting in all areas of the plant. The recommended illumination level is 300 lux (value for maintenance; in Germany acc. to ASR).

5.2 Site requirements

Cover all manhole/channel sections that do not need to be accessible for machine installation. Fix a railing along open channel sections. The regulations of DIN EN ISO 13857 "Safe distances preventing upper limbs reaching dangerous areas" and corresponding standards must be observed.

There must be enough space available around the machine for repair and maintenance work. Required minimum distance to the machine (or railing): 1m. An emergency bypass is integrated in the inlet chamber.

A bottom step is essential to ensure the minimum depth of immersion of the screen surface and to keep the switch-on frequency as low as possible.

Static dimensioning of the points where the machine is supported on the channel bottom and wall must be carried out according to the weights specified by the manufacturer.

5.3 General instructions for installation

Installation must be carried out in accordance with these instructions if installation is not part of the supply contract with HUBER SE. If installation is not performed by HUBER SE, HUBER SE cannot accept responsibility for incorrect offloading or installation. Installation must be performed by qualified and experienced personnel.

Prior to installation:

- Completely read these instructions. They contain important information how to prevent damage caused by lack of knowledge.
- Plane approach roads to the installation place must be provided so that the machine can be installed either by means of a crane in case of outdoor installation or by a lifting truck and rollers as well as by lifting chain block or hoist in case of installation in a building.
- Electrical power must be provided to be available on site at the installation date.

Preliminary work:

- If it is an installation in a building, make sure a mounting support is fitted centrally above the manhole.
- Prepare a lockable water supply line exiting from the bottom (see installation drawing).
- Prepare sufficiently sized cable ducts for electrical installation from the control panel to the machine and installation place for level control (for details see installation drawing).

Preparatory work:

- Check all assembly and fixing material making sure it is complete.
- Prepare all material necessary for water connection.
- Prepare all cables according to the cable list (see wiring diagram) and all necessary small components (e.g. air hose for level control).
- Prepare lifting device that is able to lift the load during installation.
- Clean the installation place with a broom before installation in order to prevent injuries caused by slipping.

5.4 Assembly and installation

5.4.1 Mechanical installation

General instructions:

- Fix lifting devices (2 chain hoists or similar) to the prepared installation supports over the installation place. If necessary, prepare points of suspension with required load bearing capacity and arrange in such a way that the plant can hang above the installation area without manual moving.
- Fix safety load hooks to the transport eyes, or rope straps to the machine and lifting device.

NOTICE

Blow the holes for the stainless steel plugs under pressure after drilling (using bellows, air pump, etc.) to ensure a professional durable adhesive joint.

Use grease for all unlockable screws, thus providing a durable mobility of thread.

How to proceed:

- Hang the machine up as vertical as possible.
- Place the machine down into the manhole. The correct lower landing point can be taken from the installation drawing.
- Fix the lower vertical support provisionally. Fix the horizontal support so that the machine stands vertically.
- Then, dowel the supports to the manhole.
- Remove the lifting device.
- Check that the machine runs without vibrations.
- Secure the lower support with a tackweld against slipping.
- Fix the discharge chute or bagging unit with the supplied screws.
- Cover the manhole or mount a circumferential railing that complies with EN 292 standards.

5.4.2 Electrical installation

Electrical installation to be carried out by qualified electricians only.

General:

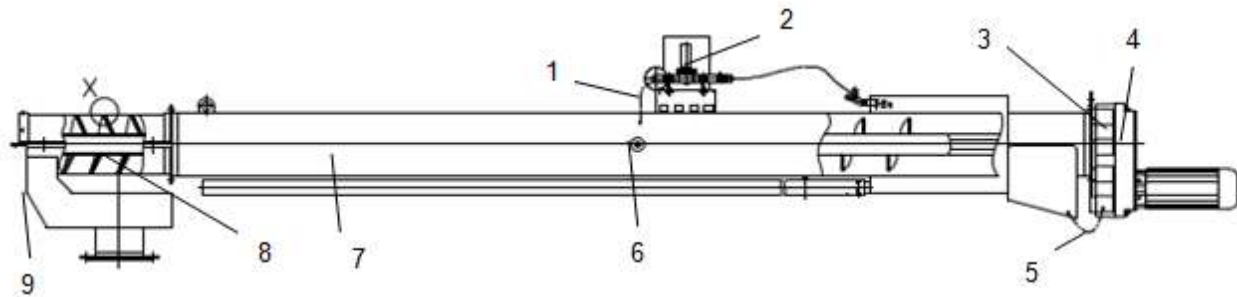
The following instructions are offered for guidance if installation is not included in the supply contract.

If installation is not included in the supply contract, HUBER SE cannot accept responsibility for incorrect installation.

Wiring:

	! WARNING Make sure that power supply is disconnected! Secure this condition by appropriate measures!
---	--

- Prepare earth connection to the plant prior to beginning any other work, and earth the gear motor and solenoid valves (optional). The protection system of the terminal sockets and screw connections must correspond to the protection zone in which the machine has been installed.



- | | |
|---|---|
| 1 Equipotential bonding 6 mm ² | 6 Protective conductor terminal |
| 2 Solenoid valve | 7 Flights, pipe, guide bars, perforated plate |
| 3 Gear motor | 8 Flights, basket, brush |
| 4 Drive pin | 9 Endplate screw, floor plate basket |
| 5 Equipotential bonding 6 mm ² | |

Protective conductor terminal: (M8 x min. 20 mm)

The conductor terminal as connection to the main potential equalisation acc. to DIN EN 60204-1 (VDE 0113-1) is shown by the conductor marking grounding EN 61310 D 20, see pictogram aside. Observe local protective measures as per local standards (DIN, VDE, EN, EeEx-Atex 100a).



- Fix the control panel with bolts in its intended position.
- Fix the adjacent control box with dowels next to / onto the machine.
- Prepare all cables between the machine, control panel and adjacent control box and connect the plant to the power supply according to the specifications in the wiring diagram. The wiring diagram and cable list are attached in the appendix, if the electrical switchboard and control panel is part of the HUBER supply contract.

If there is a pneumatic single-probe water level control system ensure that there are no bends in the air hose going to the air injection pipe. To avoid condensed water inside the air hose, it must be laid to the channel in a slope. Mount the air injection pipe (or customer's probe) in the screen basket area.

NOTICE

If the air hose has to be laid in ducts in the bottom, seal both ends by means of an insulating tape so that they are water-proof, as the ducts may be filled with water if a "bag" has been produced by wrong laying. Water in the air pipe of the inlet (≥ 10 cm water column) may cause continuous operation of the screen in the automatic mode.

Note:

The correct rotation direction of the drive, when topviewed, is clockwise!

Prior to first start resp. prior to recommissioning (e.g. after changing the power supply) check the rotation direction of the motors!

6 Start-up

Observe the following safety instructions for machine start-up to avoid damage to the machine or injuries.

- Only qualified persons are permitted to perform start-up work, observing the safety instructions.
- Check before the first start-up that all tools and foreign objects have been removed.
- Activate all safety devices and emergency cutout switches before start-up.
- Prior to initial start-up close the water supply hose and flush the pipe network. This will prevent that pollution particles can impair the function of or damage (optional) solenoid valves.

Read also the chapter General Safety Instructions.

6.1 Customer-supplied connections

All customer-supplied connections must be installed on the marked positions, or at least as close as possible, according to the manufacturer's instructions respectively installation drawing.

6.1.1 Electrical connection

The electrical connections must be laid to the installation place of the control panel.

Supply of 3 x 400 V (or special voltage) with 50 Hz (special: 60 Hz) PEN. The connecting lead must be defined according to local directives, observe the project specific connecting lead. Provision of multi-polar cables for transmission of operation and trouble signals or any other signals to a central control plant.

Pay attention to separately arranged laying of control and load cables, which means a separate laying of cable ducts. They must be laid from the control panel to the installation site. The position of the cable duct ends can be obtained from the installation drawing. The number of the cable ducts depends on the number of functions.

NOTICE

Pay attention that the cable ducts for an optional pneumatic differential level control system protrude laterally from the channel wall and must be laid in a slope to the channel. This is necessary to enable the condensed water in the air hose to run off.

1x ground, 1x screen drive, 1x level control point, 1x adjacent control box switch, min. 1x additional equipment (see equipment list).

6.1.2 Connection for wash water

Required operational water pressure: min. 5 bar max. 7 bar

Water quality:

Use screened or filtered service or used water, with no particles bigger than 0.2 mm grain size. The water should have an as low as possible chloride and ferric oxide content and preferably a pH > 6.5.

If this is disregarded, the servo bores of the solenoid valves will clog and jet intensity reduced respectively the valves not perfectly close any longer.

Water supply line:

The water supply line must not be sized smaller than 1".

Dimension the supply line one size bigger if the line is longer than 100 m (shock pressure, line loss).

The connection in the press zone chamber area should emerge from an adjacent wall or the floor and should be lockable.

A water connection (not below $\frac{3}{4}$ ") for cleaning work, maintenance etc. is required. For very greasy wastewater a warm water connection or steam jet should be available.

The connection must be made of high-quality rubber fabric hose or a pipe connection.

If fresh water is used, it is necessary to provide a return flow inhibitor complying with EN 1717 to prevent return flow of wastewater into the potable water network.

If solenoid valves are used, a rubber hose is preferably recommended because its elasticity reduces shock pressures when the valve closes.

6.2 Checks prior to initial start-up

Prior to start-up:

Make yourself familiar with the

- operation and control elements of the machine
- machine equipment
- operation principle of the machine
- immediate vicinity of the machine
- safety devices of the machine
- measures to be taken in case of emergency

Perform the following work prior to any start-up:

- Check and make sure that all safety device are attached and in a ready-to-operate condition.
- Check the machine for visible damage and eliminate any damage found immediately or report them to the supervisory staff, as machine operation is only permitted if the machine is in a perfect condition.
- Check and make sure that authorized persons only have access to the operation area of the machine and no other persons are endangered by starting the machine.
- Check and make sure that the protection circuit system is persistently connected.
- Remove any objects or other material from the operation area of the machine, which is not needed for machine operation.

Check that the screw-type cable fitting fits tight and re-tighten the screw, if necessary, to prevent ingress of water into the motor. (See below picture of the screw-type cable fitting.)



Screw-type cable fitting

7 Operation

Observe the following instructions when operating the machine to avoid damage to the machine or injuries.

- Never use the machine for any other purpose than the intended use!
- Inform yourself about the correct behaviour in case of a fault prior to switching the machine on.
- Check prior to switching the machine on that the following units are in a ready-to-operate condition:
 - Protective devices
 - Emergency cutout switch

Read also the chapter General Safety Instructions.

7.1 Control philosophy

The electrical control regulates all automated functions of the plant both in automatic and hand mode. After switching the mains isolator on, the plant works in automatic mode. The control panel must be installed outside the screening room.

If you have the electrical control panel supplied by a third company, pay attention to the control philosophy of HUBER SE. The panel must be completely wired in order to connect the machine. The wiring diagram must lie inside the panel.

7.2 Operation possibilities

7.2.1 Operation by means of adjacent control switch

The control units are displaced from the main control panel into an adjacent control box if the main control panel is not positioned close to the machine. The machine can for example be installed in an Ex protection zone and operated via an adjacent control box with Ex protection while the main control panel is installed in a room without Ex protection.

Equipment:

- Emergency cutoff button
- Lockable key switch (AUTO – HAND)
- Rotation selector switch

Arrangement:

Directly adjacent to the machine, installation options:

- On standard upright dowelled to the floor
- Clamp-mounted on the rising pipe

Rotation direction selector switch positions on adjacent control box:

Position	Action	Condition
Forwards (Stop button)	Machine runs forwards	As long as the lockable key switch remains in this position
0	-	-
Reverse (push key)	Machine reverses continuously	As long as the key remains in this position and the lockable key switch is in HAND position

Lockable key switch positions:

Position	Action	Condition
Remote/AUTO (Turn key)	Machine runs in cycles	When level control or timer actuate
Local/HAND (Turn + press key)	Machine runs forwards in cycles	As long as the rotation selector switch remains in this position

7.2.2 Time-dependent automatic start control (option)

If the inflow rate is low, screenings may accumulate in front of the screen if the screen is not switched on via water level control. To prevent accumulation of screenings, the screen is forced to start automatically after a selectable standstill time and continue to run for a selected run-on time.

Adjustment of the standstill and run-on time on a timer or the text display key. The screen should be active for the set run-on time after a break of 30, 45 or 60 minutes. Standard ex factory setting: plant start after a two hours standstill time

7.2.3 Run-on time

After activation via level control or automatic start control the screen runs until the set run-on time expires to ensure that the screenings are conveyed optimally up to the press zone and prevent thus accumulation of screenings within the rising pipe.

The run-on time to be set at the time relay or in the menu field.

Standard ex factory setting: 10 sec.

8 Trouble shooting and repair

Symptom	Possible cause	Repair
Machine does not run in automatic mode	Local control switch set to "manual"	Check the local control switch.
	PLC is on STOP	Set switch to RUN.
Emergency stop activated	Intended activation	Check the cause and take suitable measures.
	Accidental activation	Release the emergency-stop button and press RESET.
Fault message from drive XY	Motor protection switch has tripped.	Check the motor protection switch (set to nominal current). Check phases and pre-fuse.
	Option: Current relay has tripped.	Check the current relay (see Electrical Control Description).
	Option: True power monitor has tripped.	Check the settings of the true power monitor (see Electrical Control Description).
	Option: Frequency converter has tripped.	Check the frequency converter fault message with the use of the operation manual of the frequency converter. Check the settings of the frequency converter (see Electrical Control Description).
	Option: repair switch	Check the switch position of the repair switch.
	Option: full motor protection (thermistor) has tripped	Check fan wheel on motor; See for sufficient cooling
Bus system fault	Connection between SCADA (superordinate control system) and PLC interrupted	Re-establish the connection
Screen Level upstream of screen	Too much blinding material on the screen basket.	Manually remove material entangled around the screen basket.
		Check the functionality of the cleaning brush.
		Check the settings/function of the level measurement system.
Screen Level downstream of screen	Too much blinding material on the screen basket.	Manually remove material entangled around the screen basket.
		Check the functionality of the cleaning brush.
		Check the settings/function of the level measurement system.

If you are unable to eliminate the fault, contact the HUBER Service specialists.

9 Maintenance and repair

9.1 General information

In order to prevent faults during operation in the long term, the machine must be inspected at regular intervals (visual inspection). Maintenance must be carried out after certain time intervals or if any deviations from the nominal condition are detected during inspection. If defective machine parts are detected during inspection and/or maintenance, these must be replaced with original parts during a maintenance repair.

In order to prevent injuries or damage to the machine, it is important to observe the following points when carrying out maintenance on the machine:

- First cordon off a large space around the area in which maintenance work is to be carried out.
- **Before starting with any maintenance and repair work** all voltage sources must be switched off and secured against unintentional restart.

Inspections (visual inspections) must be carried out

- in automatic mode only and with all protections closed.
- with open protection devices only after machine shutdown. Machine inspection in manual operation mode is only permitted in exceptional cases if machine operation is essential.

CAUTION

There is an increased risk of injury if any guards or separating protections (e.g. covers) are open. Keep therefore a safety distance of at least one metre from any dangerous areas (e.g. moving parts and openings).

- Never use any other than the specified operating media.
- Only use spare parts that are listed in our spare parts lists.
- Read also the chapter *General Safety Instructions*.

CAUTION

Sealed spaces in wastewater treatment plants which need to be accessed for inspection, maintenance or repair must be ventilated in such a way that potentially explosive atmospheres, a lack of oxygen and gas or vapour at levels hazardous to health do not occur.

CAUTION

The chapter *Maintenance and repair* is intended for skilled staff only.
Any inspection, maintenance or repair work must be carried out by qualified personnel only as according to chapter 3.1.3.
Skilled staff has to be equipped with personal protective gear according to chapter 3.1.4 (such as gloves, gas alarm device, etc.).

	⚠ CAUTION Always wear safety goggles and rubber gloves if the machine is being used to process materials that are hazardous to health. Observe the instructions for wearing hearing protection in the operating manual for the compressor (optional).
---	---

	⚠ CAUTION Stainless steel parts may become very hot during operation. Be careful to touch such parts!
---	--

9.2 Overview – regular repair work

Weekly:

- Check the general function of the machine by visual inspection according to chapter 9.3.1.
- Check the function of the level control system according to chapter 9.3.4.
- Washing/inspection of the press zone chamber according to chapter 9.3.3
- Clean the screen basket according to chapter 9.3.5.
- Check the screenings discharge connections according to chapter 9.3.6.

Monthly:

- Clean the plant according to chapter 9.3.2.
- Check the condition of the cleaning brush according to chapter 9.3.7.

Other maintenance and repair work:

CAUTION
Regularly check the potential equalisation as described under chapter 3.3 <i>Safety Systems</i>.

⚠ CAUTION
Under no circumstances must electrical equipment be sprayed with a high-pressure cleaner while the system is being cleaned!

9.3 Inspections and preventive maintenance

Chapter 9.2 describes in more details the procedure for regular repair work specified in chapter 9.1.

9.3.1 General function check and visual inspection

Visually inspect the general function of the machine and pay particular attention to irregularities in operation, such as changes in dewatering performance.

During the following work steps, the plant is operating in automatic mode and the protective equipment (covers) is completely closed.

- Check the fill level of the screenings container and empty it if necessary to avoid back-up of screenings into the screenings discharge unit.

9.3.2 General cleaning work

- Make sure that the feed to the plant is shut off and secured against switching on.

CAUTION

Before carrying out further work, the shutdown procedure described in section 3.1.4 must be observed.

Skilled staff has to be equipped with personal protective gear according to chapter 3.1.4 (such as gloves, gas alarm device, etc.).

CAUTION

Under no circumstances must electrical equipment be sprayed with a high-pressure cleaner while the system is being cleaned!

- Remove any protection equipment (covers) that may be hindering its inspection/cleaning.
- Hose down the complete plant, preferably with a high pressure cleaner.

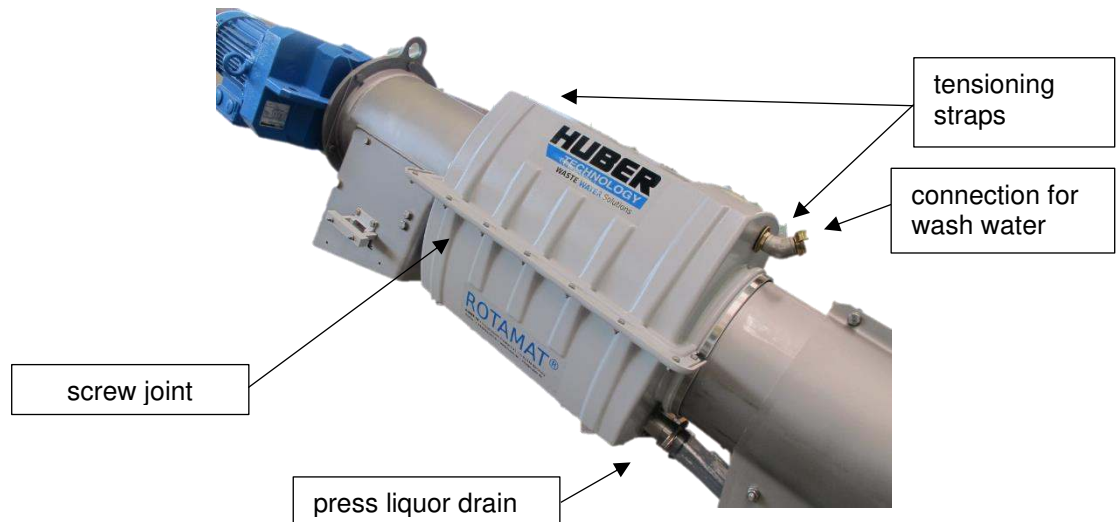
By general cleaning work you avoid lasting deposits and chloride accumulation, which can lead to corrosion and clogging of the system in the long term.

9.3.3 Press zone inspection

The so-called press zone is located just before the screenings discharge on the rising pipe. Two removable plastic half shells form the press zone chamber. In the press zone, a plug of screenings is produced; the press water flows through openings in the stainless steel rising pipe into the press zone chamber and back into the inlet.

The intermittent flushing of the press zone avoids deposits within the press zone chamber, which could lead to a blockage of the openings in the rising pipe or the press water hose.

The inspection of the press zone is required if the screenings are not dewatered sufficiently any longer. Clogged dewatering holes can result in screenings that are too wet.



9.3.3.1 Press zone washing

Irrespective of whether manual or automatic press zone washing is used, the frequency and duration of the washing intervals must be set so that clear water flows out of the press water drain after an interval has ended.

Manual press zone washing

Connect the customer-provided hose to the wash water connection on the press zone and open the on-site valve / water tap.

Automatic press zone washing system

The press zone is washed automatically via a solenoid valve. The basic setting can be taken from the description of the control system and can be adjusted to the operating conditions.

If irregularities occur during the washing process, e.g. no press water flows out of the press zone chamber, inspect the press zone as described in chapter 9.3.8.

9.3.4 Check the function of the level control system

CAUTION

Before carrying out further work, the shutdown procedure described in section 3.1.4 must be observed.

- Remove any protection equipment (covers) that may be hindering its inspection.
- If polluted, hose down the level control equipment with a water hose.
- Measure the water level and check the measurement against the output value on the PLC.
- Reattach the protective equipment.

9.3.5 Screen basket cleaning

CAUTION

Before carrying out further work, the shutdown procedure described in section 3.1.4 must be observed.

- Remove any protection equipment (covers) that may be hindering its inspection.
- If polluted, hose down the screen basket with a high pressure cleaner.
- Reattach the protective equipment.

If you detect any damage in the screen basket area, contact the HUBER Service team.

9.3.6 Check the screenings discharge

CAUTION

Before carrying out further work, the shutdown procedure described in section 3.1.4 must be observed.

- Remove any protection equipment (covers and possibly the discharge chute) that may be hindering its inspection.
- Visual inspection of the screenings discharge at the machine discharge for any tangling material.
- Remove the tangling material if any.

9.3.7 Check the condition of the cleaning brush

NOTICE

A worn brush does not only impair screen basket cleaning!

A worn brush can also result in increased wear on the screw, screen basket, rising pipe and possibly on the intermediate bearing.

- Set the operation mode of the plant from “automatic” to “manual” (all drives OFF).
- Remove the protection equipment (covers) for visual inspection.

CAUTION

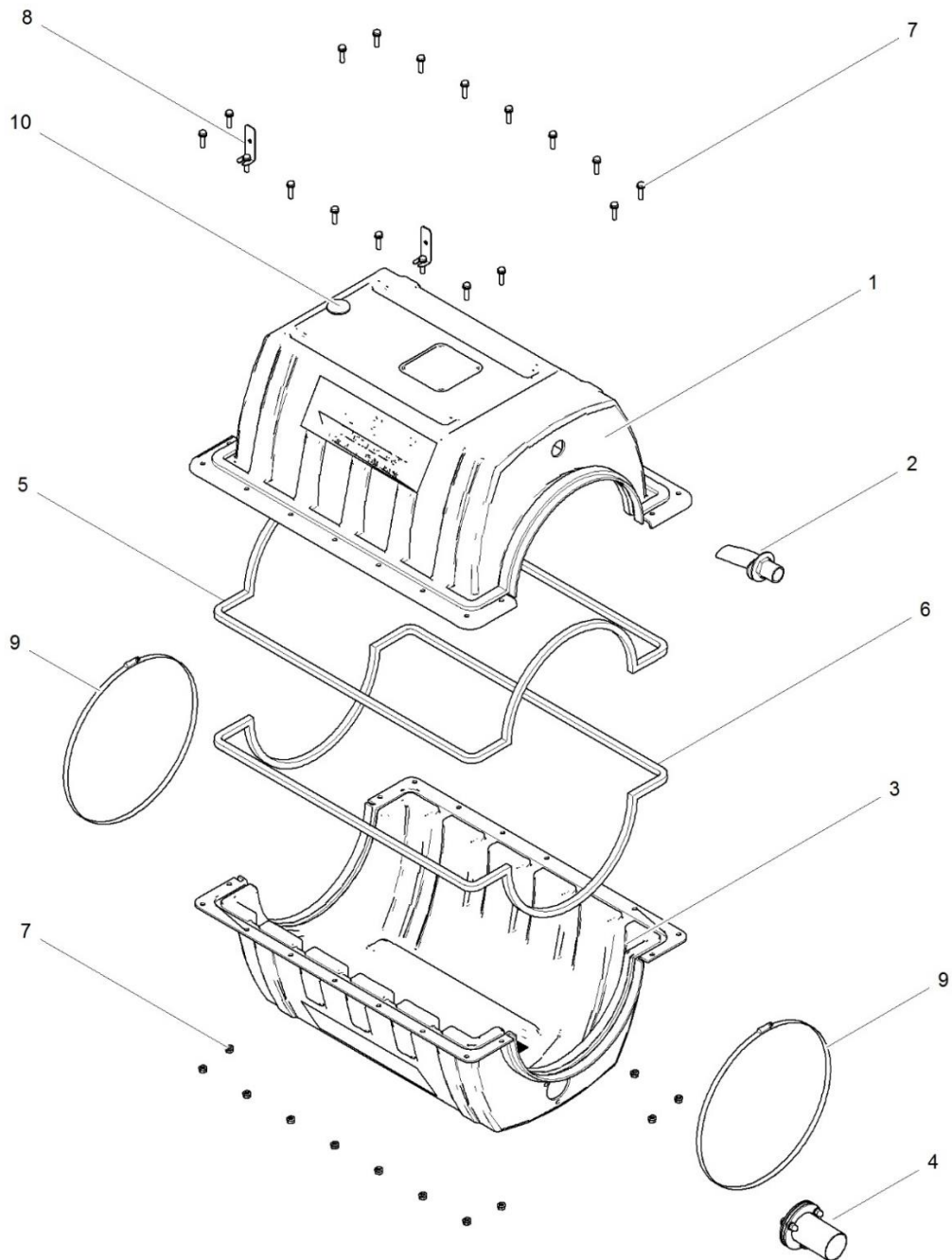
There is an increased risk of injury if any guards or separating protections (e.g. covers) are open. Keep therefore a safety distance of at least one metre from any dangerous areas (e.g. moving parts and openings).

- Switch the plant drive in “manual” operation mode to continuous operation so that the screw shaft is rotating.
- Visually inspect the cleaning brush in the screen area for missing or faulty brush hairs. Special application conditions (e.g. grit, abrasive materials) cause increased wear.

Please contact the HUBER Service Department directly if you notice any irregularities during operating hours or in case of increased clogging of the screen basket.

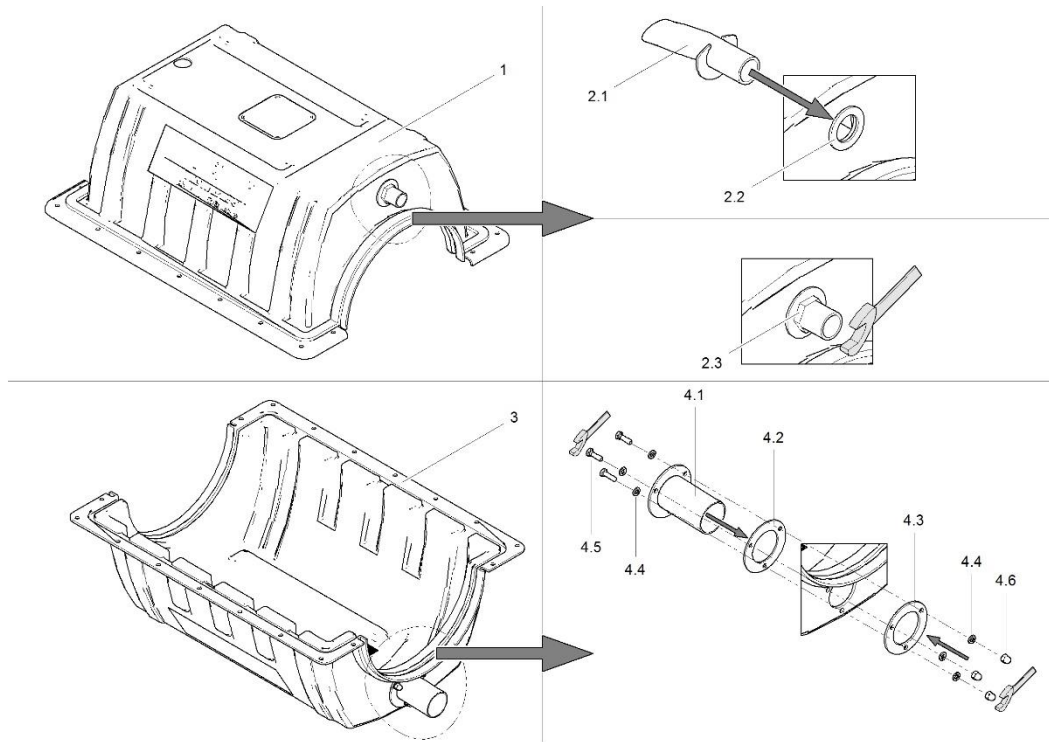
9.3.8 Installation instructions for a plastic dewatering chamber

9.3.8.1 Components overview

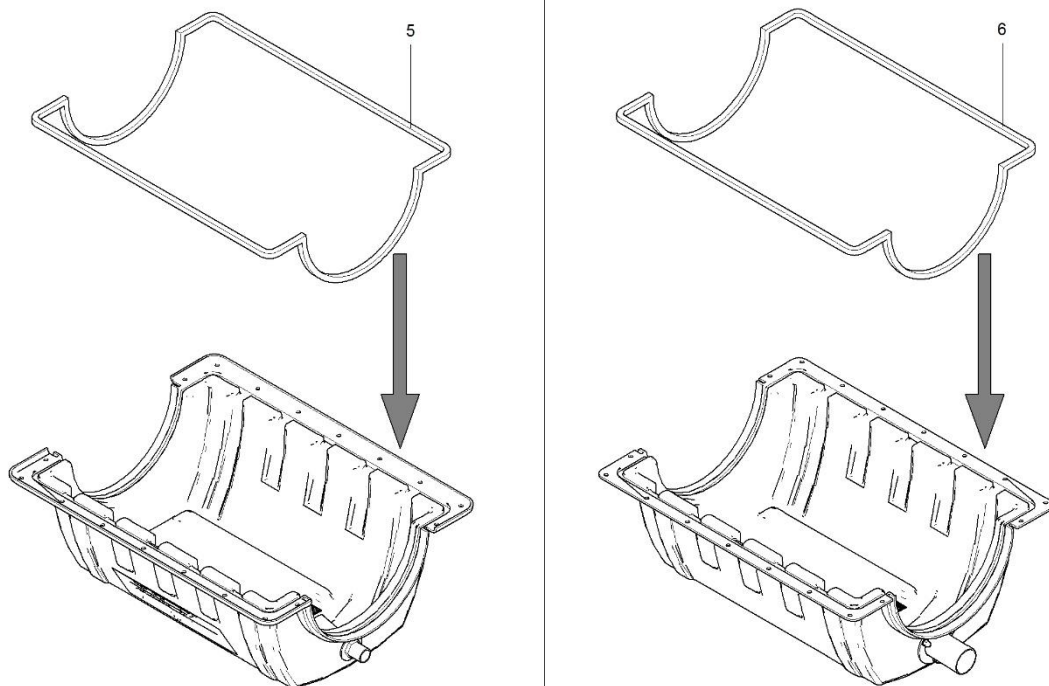


1.	press zone chamber (top)	6.	sealing (bottom)
2.	components for wash water connection	7.	screw fastening for press zone chamber
3.	press zone chamber (bottom)	8.	angle for wiring
4.	components for press liquor drain	9.	tensioning straps
5.	sealing (top)	10.	overpressure plug

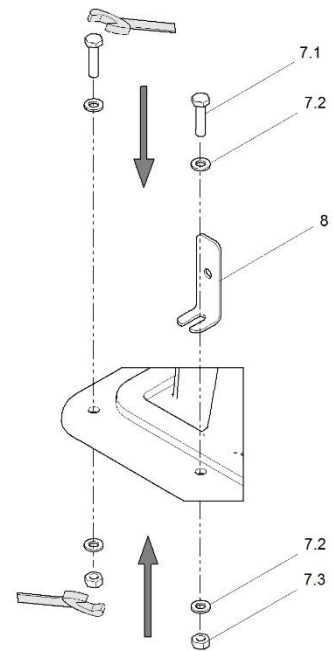
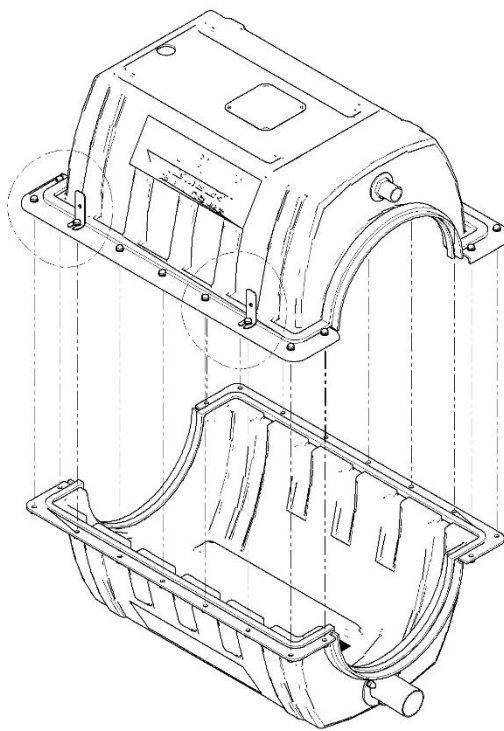
9.3.8.2 Assembly instructions for press zone chamber



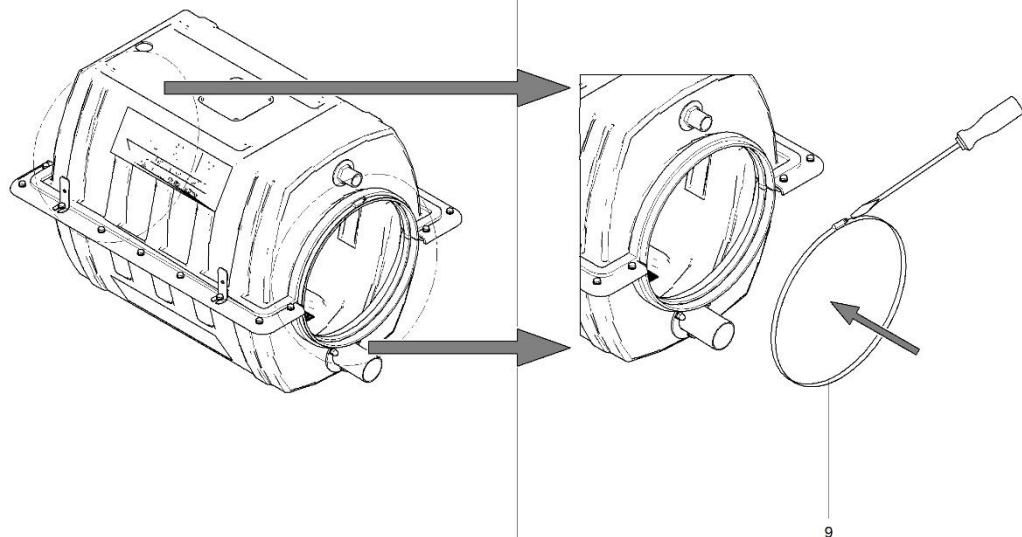
Step 1: Installation of the wash water connection on the press zone chamber (top) and press water drain on the press zone chamber (bottom)



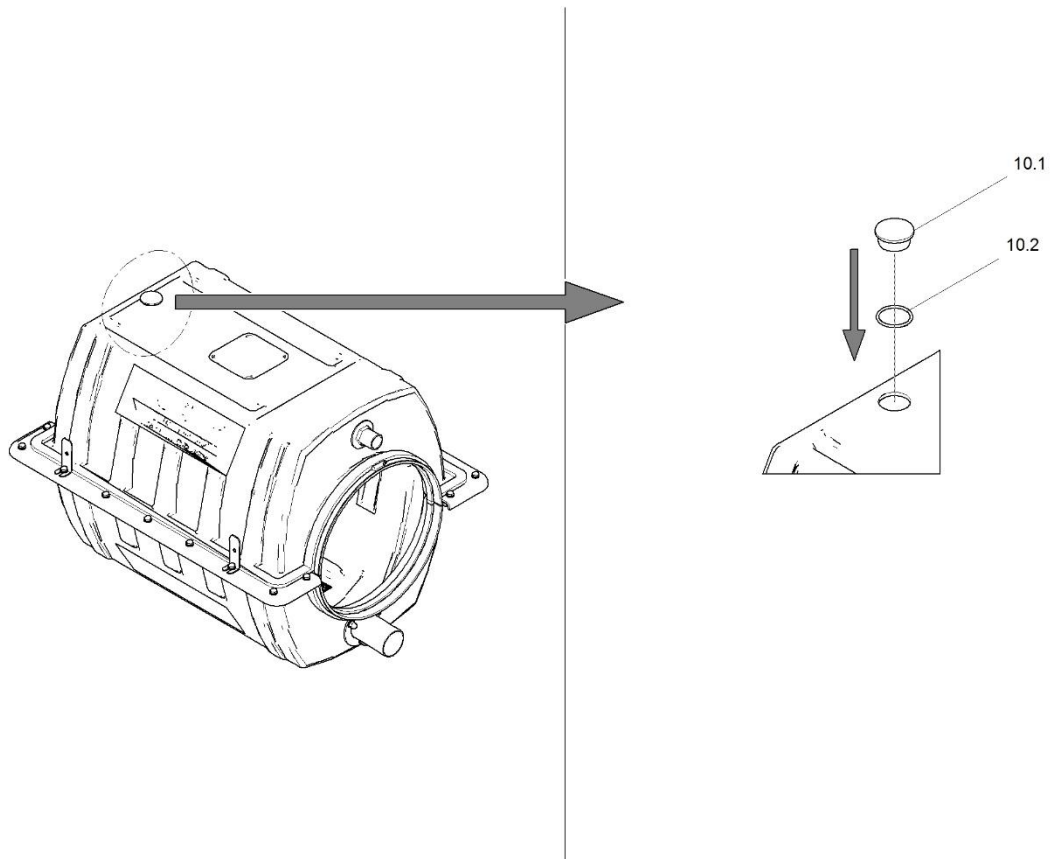
Step 2: Inserting the gaskets into the press zone chamber (top) and the press zone chamber (bottom)



Step 3: Screw the two press zone shells to the rising pipe (incl. angle [8] of the cable fitting).
When fixing the screws, use only grease, do not use a screw lock.



Step 4: Attaching the two tensioning straps to the pre-assembled press zone chamber.



Step 5: Assembly of the overpressure connection

9.3.9 Operating media and lubricant replacement

Gear motor:

The plant is equipped with a gear motor, the technical data and required lubricant amounts of which can be taken from the attached motor data sheet. The required lubricant amount is additionally specified on the identification plate. If two amounts are indicated, the first refers to the main gearbox and the second to the intermediate gear.

An operating manual for the drive motor can be found in the appendix. The approved lubricants are listed in this operating manual.

Example: BP gearbox oil: Energol GR-XP220

Under normal operating conditions it is recommended to replace the lubricant after approx. 15000 operation hours. Independent of the operation time, lubricant replacement is recommended after 2 or 3 years at the latest.

CAUTION

Use a working platform complying with the safety and accident prevention regulations for work to be performed on the gear motor.

9.3 Spares, wear parts

Long life and virtually complete corrosion resistance against municipal and industrial wastewater as all components that are in contact with water are completely made of stainless steel, acid-treated in a pickling bath and passivated. A time frame of 2 - 6 hours/d can be

specified as average operation time. This time period always depends on the specific local site conditions, method of plant feeding, etc.

a) Wear parts

Our guarantee does not include wearing parts that are subject to natural wear during operation. Wear parts are parts that are subject to increased wear due to their function. The level of wear depends greatly on the conditions of use, operating life and maintenance of the system.

These are the main wear parts of the plant:

- Cleaning brushes
- Screen basket
- Wear protection spiral

b) Spare parts

For other spare parts such as gear motor, solenoid valve, etc., see attached spare parts list.

Please provide the following details when ordering spare parts:

- Type of machine
- Size
- Order number = machine number
- Year of manufacture
- Operating voltage of the electrical part
- Order number from the spare parts list (see appendix)
- Required number of units
- Delivery address

9.4 Regular tests of machines subject to ATEX – Recurring tests and permanent control

The regular tests for maintenance of the plant's nominal condition shall include after every 800 respectively 4500 operation hours, at least however after 6 months respectively 3 years, check-up of the components listed below. If there are signs of an unacceptable running condition, replace the affected components.

Operating hours	Interval	Inspection of	Recommended method
800	6 months	Running state: Screw shaft and basket vibrations	Check the brushes and screen basket for wear.
		Wear of screen basket / flights	If necessary, replace the brushes and/or screen basket.
4500	3 years	Running state: screw shaft in rising pipe section L > 6200 mm or size 700	Check bearing for wear.

NOTICE

Additionally observe the separate operating instructions for electro motors, drives and bearings for maintenance.



⚠ CAUTION

Repair or extensive maintenance work on machines with ex protection should be performed in a separate room on the dismantled machine.

10 Shutdown

Observe the following safety instructions for machine shutdowns to avoid damage to the machine, injuries or environmental damage.

- Qualified staff only is authorized to carry out shutdowns.
- Contact the manufacturer for questions concerning disposal of the machine.
- Take care of environmentally sound disposal of operating media, auxiliary material and lubricants (such as gearbox oil). Observe the regulations for eco-friendly waste disposal!
- Lifting and righting of the equipment must be done only by the lifting eyes provided.
- Use only the lifting devices specified hereunder to transport the machine.
- Read also the chapter Transport.
- Read also the chapter General Safety Instructions.

10.1 Temporary shutdown

- Clean the screen trough with a high-pressure unit prior to a longer shutdown.
- Let the screen operate for approx. 10 minutes. This will empty the machine, screenings are discharged. The emptying process can be accelerated by throwing polystyrene into the screen basket while the machine is running until all screenings have been discharged.
- Thoroughly rinse the press zone.
- If the machine is installed outdoors the motor should be covered (in order to avoid ingress of water) but not wrapped. Otherwise, water may ingress into the terminal box.
- It is recommended to change the gearbox oil prior to start-up after a several-year shutdown. Further conservation is not necessary.

10.2 Final shutdown / Disposal

Qualified staff only is authorized to perform electrical and mechanical shutdowns. Prior to a final shutdown, follow the instructions for a temporary shutdown and the following additional instructions:

- Drain the press zone completely and clean the press liquor collection chamber.
- If the machine has the plastic press liquor collection chamber, remove the plastic shell and dispose of to waste recycling.
- Remove then the inner cover on the rising pipe to have access to the press zone for cleaning.
- Drain the gearbox oil. Disposal of gearbox oil: Observe the regulations for eco-friendly waste disposal!

11 Additional Information

If you require more information, please write or phone. We will do our best to support you.

Our headquarters:

HUBER SE
Industriepark Erasbach A1
92334 Berching
Germany

Phone:	+49-8462-201-0
Fax:	+49-8462-201-810
E-mail	info@huber.de

Our customer service:

Phone:	+49-8462-201-455
Fax:	+49-8462-201-459
E-mail	service@huber.de

We will help you to quickly find the right specialist to answer your questions.

Or visit our website <http://www.huber.de> providing up-to-date information concerning our Business Unit Service.

Our service comprises **preventive maintenance, routine servicing, short-term repair**.
Our service hotline is available **24 hours a day, 7 days a week**.

Our qualified team offers a customer-oriented and reliable service. This service includes:

- **Installation and start-up**
- **Expert support, information and briefing of operating staff**
- **Regular servicing**
- **Optimisation of plant operation**
- **Maintenance of machine performance**
- **Repair and standard spare parts within 48 hours**

These additional services guarantee reliable plant operation, which is an important aspect for both municipal and industrial applications, and will help you to meet the requirements within your area of responsibility.

Correct handling of stainless steel products

HUBER stainless steel products are carefully manufactured under precisely the right conditions for the material. The final pickling bath treatment and subsequent passivation ensures that each product receives the best possible protection against corrosion. It is however essential that this protection is not damaged by improper delivery, storage, handling in transit or on site.

What impairs corrosion protection?

Stainless steel must not come into contact with salt solutions (sodium chloride)!

- Avoid contact with de-icing salt during winter transport (provide covers).
- Avoid contact with sea or brackish water.
- Avoid drying of salty liquids (wash off).

Stainless steel must not come into contact with other metals. Do not ...

- store with other metals (storage platforms, scaffolding, tools).
- fix stainless steel with fixings of dissimilar materials (galvanised nuts, brackets).
- attach stainless steel products with other metals directly (pipelines).

Stainless steel must not be treated improperly. Do not ...

- clean stainless steel with carbon steel brushes.
- expose stainless steel to red-hot cutting fountains (from angle grinders).
- use grinding discs containing iron.
- carry out thermal or mechanical treatment without subsequent pickling and passivation.

Do's on site

Avoid 'cold welding', caused by fixing stainless steel of the same grade. Do ...

- use screws made of V2A and nuts made of V4A material.
- use suitable lubricants.
- separate stainless steel from other materials electrolytically through the use of synthetic gaskets, washers and sleeves.

Connections should be preferably bolted. If on-site welding cannot be avoided, it should be carried out correctly. Do ...

- use the correct welding supplements.
- use only TIG welding with filler rod for one-sided butt welding.
- use MIG or electric arc welding only for two-sided butt welding.
- always pickle the surface with a suitable compound after welding.
- allow passivation by flushing the surface with salt-free water after pickling.

When setting in concrete. Do ...

- use stainless steel wall ducts.
- allow for different rates of thermal expansion.
- treat the surface carefully.
- use only suitable cleaning materials and always flush afterwards.
- use a protective covering when burying stainless steel in ground containing iron.

Properly treated stainless steel does not require any further corrosion protection. However, if the surface must be painted for aesthetic reasons, do use a suitable primer.

Only with your cooperation, to ensure correct delivery, storage, handling in transit or on site and installation, can we guarantee you the best from our stainless steel products. In doing so, you will also promise the operator maintenance and trouble-free operation for years.

EC Declaration of Incorporation

in terms of

the EC-Directive Machines **2006/42/EG**
the EC-Directive EMC **2014/30/EU**

We hereby declare, that the machine

TYPE: **HUBER Pumping Station Screen ROTAMAT® RoK4**
Project: **Parque Industrial Taiwan**
Project number: **13018435**

as supplied complies with all relevant terms of the above-mentioned directive and that the conformity assessment procedure was carried out.

The technical documentations were prepared according to above-mentioned directive.

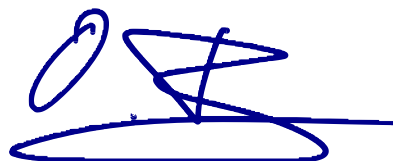
In particular the following harmonised standards (reference: Official Journal of the European Community) were applied:

EN Standards	Title
EN ISO 12100	Safety of machinery - General principles for design - Risk assessment and risk reduction (ISO 12100)
EN 60204-1	Safety of machinery - Electrical equipment of machines - Part 1: General requirements (IEC 60204-1 (modified))

- The incomplete machine may only be put into operation, if it was ascertained, that the complete machine complies with the relevant terms of the directive. According to enclosure.
- The start-up of the machine is prohibited until the conformity of the whole machine was ascertained.
- This declaration relates exclusively to the incomplete machine in the state, in which it was placed on the market.
- The relevant special technical documentation pertaining to the partly completed machinery described in Annex VII, part B has been prepared.
- HUBER SE undertakes to transmit, in response to a reasoned request by the national authorities, relevant information on the partly completed machinery in electronic form.

Responsible for documentation: HUBER SE, Berching

Berching: 2025-10-06



Name: Dr. Oliver Rong, Chief Technical & Production Officer
Grund: Bestätigt / Confirmed
Datum: 09.10.2025 08:11:47 (UTC+02:00:00)

HUBER SE
Industriepark Erasbach A1
92334 Berching

Adresse / address: HUBER SE · Industriepark Erasbach A1 · 92334 Berching · Germany · Telefon / phone: + 49 - 84 62 - 201 - 0 · Fax / fax: + 49 - 84 62 - 201 - 810
e-mail: info@huber.de · Internet: http://www.huber.de

Sitz der Gesellschaft / Headquarters: Berching · AG Nürnberg / Register of companies: HRB 25558
Vorstand / Board: Georg Huber (Vorsitzender / CEO), Dr.-Ing. Oliver Rong (stellvertretender Vorsitzender / Vice CEO), Dr.-Ing. Johann Grienberger, Rainer Köhler
Aufsichtsratsvorsitzender / Chairman of the Supervisory Board: Alois Ponnath

USt (VAT)-IdNr.: DE 812353219
Bank: HypoVereinsbank Nürnberg (BLZ 760 200 70) 5 008 409 · SWIFT-BIC: HYVEDEMM460 · IBAN: DE 30 7602 0070 0005 0084 09



Appendix to the declaration for the installation of the partly completed machinery

Nr.	Essential safety protection- and health requirements	relevant	not relevant	fulfilled	unfulfilled	comment
1.1	GENERAL REMARKS					
1.1.2.	Principles of safety integration	X		X		
1.1.3.	Materials and products	X		X		
1.1.4.	Lighting	X		X		note in the operation manual
1.1.5.	Design of machinery to facilitate its handling	X		X		
1.1.6.	Ergonomics	X		X		
1.1.7.	Operating positions		X			
1.1.8.	Seating		X			
1.2	CONTROL SYSTEMS					
1.2.1.	Safety and reliability of control systems	X		X		
1.2.2.	Control devices	X		X		
1.2.3.	Starting		X			
1.2.4.1.	Normal stop	X		X		
1.2.4.2.	Operational stop		X			
1.2.4.3.	Emergency stop	X		X		
1.2.4.4.	Assembly of machinery		X			
1.2.5.	Selection of control or operating modes	X		X		
1.2.6.	Failure of the power supply	X		X		
1.3	PROTECTION AGAINST MECHANICAL HAZARDS					
1.3.1	Risk of loss of stability	X			X	Correct mounting of the machine and connection to the structure (load bearing capacity, concrete quality, anchor type, correct anchor mounting)
1.3.2.	Risk of break-up during operation	X		X		
1.3.3.	Risks due to falling or ejected objects	X			X	Safety equipment (covers, etc.) is present and securely held in position
1.3.4.	Risks due to surfaces, edges or angles	X		X		
1.3.5.	Risks related to combined machinery		X			
1.3.6.	Risks related to variations in operating conditions	X		X		note in the operation manual
1.3.7.	Risks related to moving parts	X			X	Secure dangerous areas (e.g. screenings discharge H: < 2500 mm)
1.3.8.	Choice of protection against risks arising from moving parts	X			X	See chapter 1.3.7

Adresse / address: HUBER SE · Industriepark Erasbach A1 · 92334 Berching · Germany · Telefon / phone: + 49 - 84 62 - 201 - 0 · Fax / fax: + 49 - 84 62 - 201 - 810
e-mail: info@huber.de · Internet: http://www.huber.de

Sitz der Gesellschaft / Headquarters: Berching · AG Nürnberg / Register of companies: HRB 25558
Vorstand / Board: Georg Huber (Vorsitzender / CEO), Dr.-Ing. Oliver Rong (stellvertretender Vorsitzender / Vice CEO), Dr.-Ing. Johann Grienberger, Rainer Köhler
Aufsichtsratsvorsitzender / Chairman of the Supervisory Board: Alois Ponnath

USt (VAT)-IdNr.: DE 812353219

Bank: HypoVereinsbank Nürnberg (BLZ 760 200 70) 5 008 409 · SWIFT-BIC: HYVEDEMM460 · IBAN: DE 30 7602 0070 0005 0084 09



Nr.	Essential safety protection- and health requirements	relevant	not relevant	fulfilled	unfulfilled	comment
1.3.8.1.	Moving transmission parts	X		X		
1.3.8.2.	Moving parts involved in the process	X		X		
1.3.9.	Risks of uncontrolled movements	X		X		
1.4	REQUIRED CHARACTERISTICS OF GUARDS AND PROTECTIVE DEVICES					
1.4.1.	General requirements	X		X		
1.4.2.1.	Special requirements for fixed guards	X		X		
1.4.2.2.	Special requirements for Interlocking movable guards		X			
1.4.2.3.	Special requirements for Adjustable guards restricting access		X			
1.4.3.	Special requirements for protective devices		X			
1.5	RISKS DUE TO OTHER HAZARDS					
1.5.1.	Electricity supply		X			
1.5.2.	Static electricity	X			X	Correct electrical installation (main equipotential bonding, connection and testing of metal components and electrical equipment)
1.5.3.	Energy supply other than electricity		X			
1.5.4.	Errors of fitting	X		X		
1.5.5.	Extreme temperatures	X		X		note in the operation manual
1.5.6.	Fire		X			
1.5.7.	Explosion		X			if relevant, note EU-Declaration of Conformity 2014/34/EU ATEX
1.5.8.	Noise	X		X		
1.5.9.	Vibrations		X			
1.5.10.	Radiation		X			
1.5.11.	External radiation		X			
1.5.12.	Laser radiation		X			
1.5.13.	Emissions of hazardous materials and substances		X			
1.5.14.	Risk of being trapped in a machine		X			
1.5.15.	Risk of slipping, tripping or falling		X			
1.5.16.	Lightning	X			X	Take into account the lightning protection concept on site and correct electrical installation
1.6	MAINTENANCE					
1.6.1.	Machinery maintenance	X		X		
1.6.2.	Access to operating positions and servicing points	X			X	If necessary: provide security-

Adresse / address: HUBER SE · Industriepark Erasbach A1 · 92334 Berching · Germany · Telefon / phone: + 49 - 84 62 - 201 - 0 · Fax / fax: + 49 - 84 62 - 201 - 810
e-mail: info@huber.de · Internet: http://www.huber.de

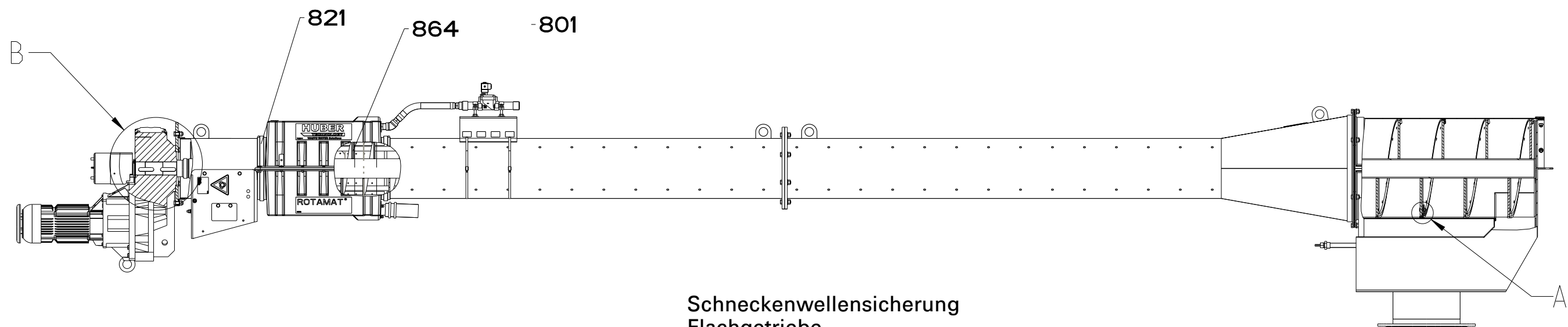
Sitz der Gesellschaft / Headquarters: Berching · AG Nürnberg / Register of companies: HRB 25558
Vorstand / Board: Georg Huber (Vorsitzender / CEO), Dr.-Ing. Oliver Rong (stellvertretender Vorsitzender / Vice CEO), Dr.-Ing. Johann Grienberger, Rainer Köhler
Aufsichtsratsvorsitzender / Chairman of the Supervisory Board: Alois Ponnath

USt (VAT)-IdNr.: DE 812353219

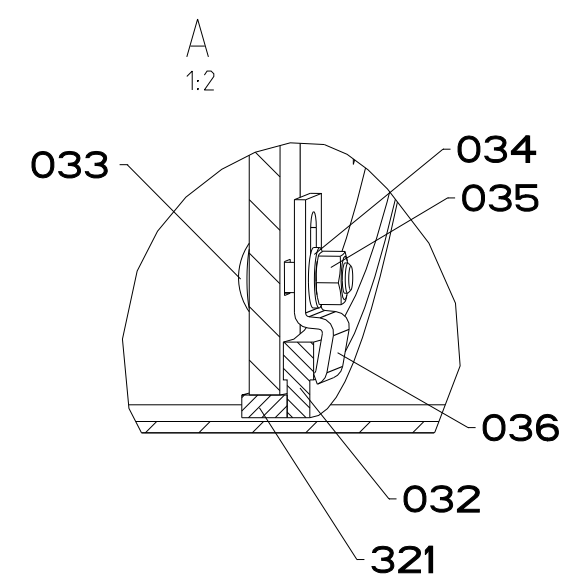
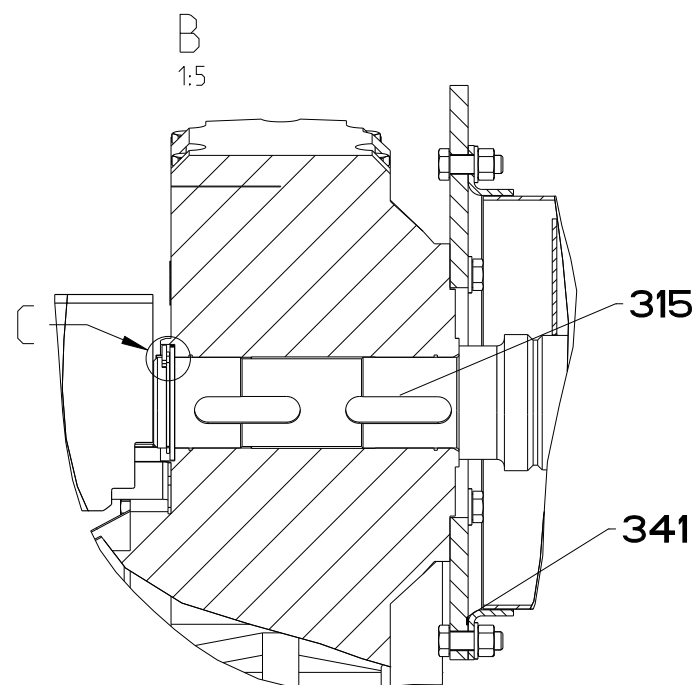
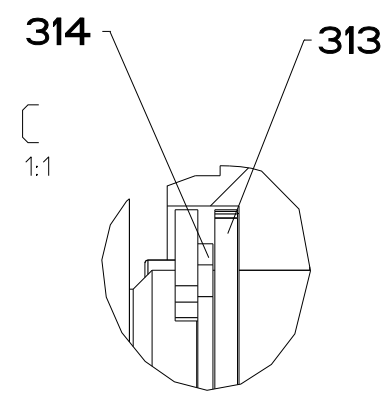
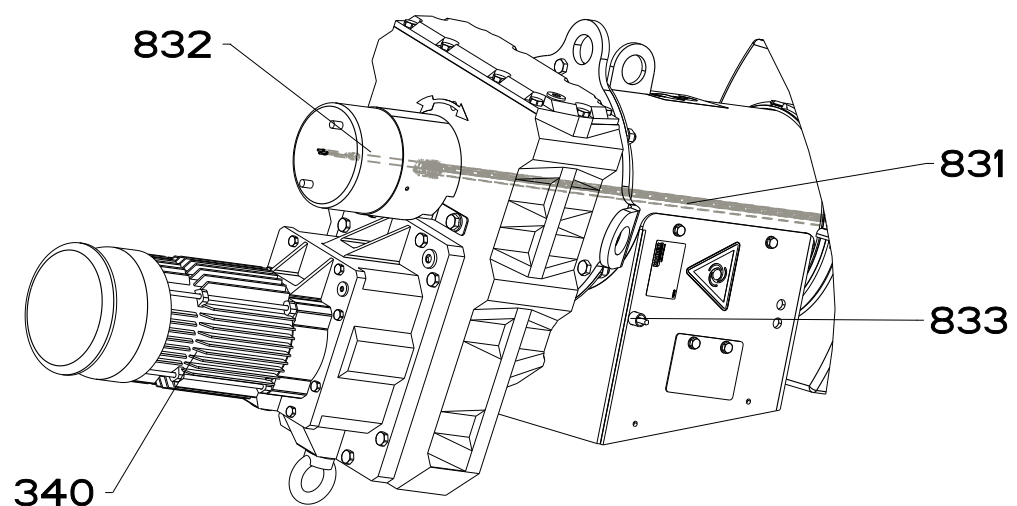
Bank: HypoVereinsbank Nürnberg (BLZ 760 200 70) 5 008 409 · SWIFT-BIC: HYVEDEMM460 · IBAN: DE 30 7602 0070 0005 0084 09



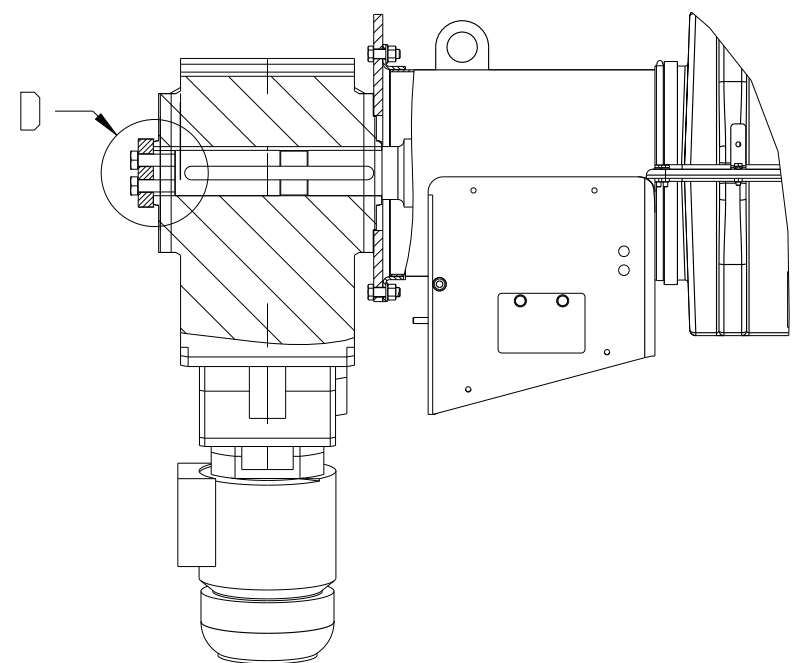
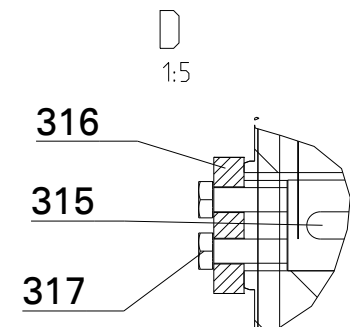
						relevant equipment (railings, covers, etc.)
Nr.	Essential safety protection- and health requirements	relevant	not relevant	fulfilled	unfulfilled	comment
1.6.3.	Isolation of energy sources		X			
1.6.4.	Operator intervention	X		X		
1.6.5.	Cleaning of internal parts	X		X		
1.7	INFORMATION					
1.7.1.	Information and warnings on the machinery	X		X		
1.7.1.1.	Information and information devices	X		X		
1.7.1.2.	Warning devices		X			
1.7.2.	Warning of residual risks	X		X		
1.7.3.	Marking of machinery	X		X		
1.7.4.	Instructions	X		X		
1.7.4.1.	General principles for the drafting of instructions	X		X		
1.7.4.2.	Contents of the instructions	X		X		



Schneckenwellensicherung
Flachgetriebe



Schneckenwellensicherung
Kegelradgetriebe



Diese Zeichnung ist geistiges Eigentum der Fa. HUBER SE und damit urheberrechtlich geschützt. Zuwiderhandlungen verpflichten zum Schadensersatz.
This is a copyrighted drawing which is the intellectual property of HUBER SE.
Any contravening offender will be held liable for payment of damages.

Technische Änderungen vorbehalten / Subject to change					ISO 2768-mK			
				Datum/Date	Name	HUBER Schachtsiebanlage		
				Bear./Rev.	04.12.2024		ab	
c	Änd.3852	04.12.24	ab	Gepr./Appr.	II		II	
Revision	Änderung Modification	Tag Date	Name	Norm.			HUBER Pumping Stations Screen	
							ROTAMAT® RoK4	
Projekt project				Attachment part		Massstab scale	Nummer number	Blatt sheet
						1:20		

HUBER
TECHNOLOGY
WASTE WATER Solutions
D-92334 Berching / Tel.: +49-8462-201-0

011_000021
1/1

Page 55 of 149

Ersatzteilstückliste

Projekt:	13018435	Parque Industrial Taiwan, PY	Ersatzteilzeichnung:
Artikel:	13018435_10546731_HUBER	HUBER Schachtsiebanlage ROTAMAT® RoK4 300/6	011_000021C1
	SCHACHTSIEBANLAGE ROTAMAT®	6500	

Z-Pos.	Bezeichnung	Menge	Eh.	Artikel
032	Streifenbürste D 262 d 210 d0,60 W 8 h10 BL16 H26	4,00	wdg	706604
033	Flachrundschraube DIN 603 M 8x 20	24,00	pcs	702548
034	Scheibe DIN 125 A 8,4	24,00	pcs	702424
035	Sechskantmutter DIN 934 M 8	24,00	pcs	702837
036	Klemmblech 49/25/3 1xsh9x22	24,00	pcs	503235
313	Stützscheibe 45,2x 58,5x3,0 2xM4	1,00	pcs	510837
314	Sicherungsring DIN 471 45x1,75	1,00	pcs	710537
315	Passfeder DIN 6885 A14x 9x125	1,00	pcs	711053
321	Spirale R 126 10/5 1xturn	4,00	wdg	560016
340	Motor BF40-74W 1,10kW 13,0U 400/50 800Nm IE3 noEx	1,00	pcs	51718583
341	Streifendichtung klebend 10/ 3	1,20	m	703505
821	Dichtungsprofil 700x441 11x10,5 ET-N	2,00	pcs	51348420
864	Schneckenwendel FR 2431001 d 89,5 D243 P100 t10 end	1,00	pcs	503786

Assembly and Operating Instruction BA 200 EN - Edition 11/22

Geared Motors / Asynchronous and PMSM-Motors



TRANSLATION

The operating instructions are an integral part of the product. It contains important information for your safety. Make sure that the operating instructions are always available in a legible condition and complete at the assembly or installation site. Read the operating instructions carefully and observe their contents. If you have any questions, please contact Bauer Gear Motor before putting the drive into operation. Further documentation can be found on our homepage.

TABLE OF CONTENTS

EU-Declaration of Conformity	4
Safety information for the operation of geared motors	6
Geared motors with three-phase motor	8
Lubricants	19
Lubrication quantity series BG	19
Lubrication quantity for BG20-01R	20
Lubrication quantity series BF	21
Lubrication quantity series BK	22
Lubrication quantity series BK (HiflexDRIVE)	23
Lubrication quantity series BS	24
Lubrication quantity for gear design with free running input shaft	25
Lubrication quantity for gear design with integral motor	26
Lubrication quantity for gear design with free running input shaft	27
Lubrication quantity for coupling lantern	28
Lubrication quantity for pre-stage	29
Lubrication quantity for intermediate gear	30
Brakes	31
Spring-loaded brakes with direct current solenoid release Models E003B and E004B	31
Spring-loaded brakes with DC solenoid release	
Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A	36
Electrical brake connection	53
Brake connection: special rectifier ESG 1.460A	57
Brake connection: DC voltage supply	59
Brake connection: special rectifier MSG...I	60
Brake connection: special rectifier MSG...U	62
Brake connection: standard rectifier SG 3.575B	63
Rectifier on motor terminal block or cage clamp	67
Manual release for spring loaded brake with DC solenoid	
Model E003B and E004B	68
Manual release for spring loaded brake with DC solenoid	
Model ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A	69
Geared Motors	72
Gear units designed with torque arms and rubber buffers for series BF	72
Gear units designed with torque arms and rubber buffers for series BK	73
Gear units designed with torque arms made of stainless steelder for series BK	74
Gear units designed with torque arms and rubber buffers for series BS	76
Geared motors with backstop	77
Assembly of standard motor with C-Adapter (IEC and NEMA)	78
Assembly and disassembly of the shrink-disc	79
Installation of oil expansion tank	80
Information on the storage of geared motors with cage rotors	83

Low Voltage Directive 2014/35/EU
Ecodesign Directive 2009/125/EC

Bauer Gear Motor GmbH

Postfach 10 02 08
73726 Esslingen (Germany)
Eberhard-Bauer-Str. 37
73734 Esslingen (Germany)
Telephone: +49 (0)711 35 18 0
Fax: +49 (0)711 35 18 381
Email: info@bauergears.com
Website: www.bauergears.com

B 010.0800-02 Version: 10/2021

Bauer Gear Motor GmbH

Eberhard-Bauer-Str. 37, 73734 Esslingen (Germany)

declares on its sole responsibility the conformity of the following products:

Asynchronous motors from the series

**D..04, D..05, D..06, D..07, D..08, D..09, D..11, D..13, D..16, D..18, D..20, D..22, D..25, D..28
E..04, E..05, E..06, E..07, E..08, E..09**

where necessary, in conjunction with

gears from the series: BG.., BF.., BK.., BS.., BM..

with the requirements of the European Directives

**DIRECTIVE 2014/35/EU OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 26 February 2014
on the harmonisation of the laws of the Member States relating to the making available on the
market of electrical equipment designed for use within certain voltage limits.**

Published on 29 March 2014 in the Official Journal of the EU L96/357.

**DIRECTIVE 2009/125/EC¹⁾ OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 21 October 2009
establishing a framework for the setting of ecodesign requirements for energy-related products.**

Published on 31 October 2009 in the Official Journal of the EU L285/10.

¹⁾ Where the products fall under the scope of this Directive, the requirements of COMMISSION REGULATION EU 2019/1781 - laying down ecodesign requirements for electric motors and variable speed drives – of 1 October 2019 are met. Additional product labels HE; PE (IE2; IE3 according to EN 60034-30-1).

**The object of declaration described above is in conformity with the Union's relevant harmonization
legislation, proven by the compliance with the following harmonised standards:**

EN 60034-1:2010 + Cor.:2010	EN 60034-2-1:2014	EN 60034-5:2001/A1:2007
EN 60034-8:2007/A1:2014	EN 60034-30-1:2014	EN 60529:1991/A1:2000/A2:2013

Additional information:

Three-phase motors do not fall under the scope of the EMC Directive, since as passive elements they are neither susceptible for the purpose of interference resistance, nor do they affect the environment by emitting high-frequency interference. Operation of these motors in the case of power supply through switch-mode supplies (converter) and the associated EMC-relevant aspects are the responsibility of the user of this PDS (Power Drive System). The instructions in the converter's product documentation must be observed. The installer is responsible for the final EMC properties of the device, system or installation.

Esslingen 27 October 2021

Bauer Gear Motor GmbH



N. Halmuschi
(Managing Director)



P. Cagan
(Quality Director)

This certificate does not contain any guarantee of features in terms of product liability. The technical documentation is produced and administered by Bauer Gear Motor GmbH.

Low Voltage Directive 2014/35/EU

Bauer Gear Motor GmbH

Postfach 10 02 08
73726 Esslingen (Germany)
Eberhard-Bauer-Str. 37
73734 Esslingen (Germany)
Telephone: +49 (0)711 35 18 0
Fax: +49 (0)711 35 18 381
Email: info@bauergears.com
Website: www.bauergears.com

B 010.0800-03 Version: 07/2020

Bauer Gear Motor GmbH

Eberhard-Bauer-Str. 37, 73734 Esslingen (Germany)

declares on its sole responsibility the conformity of the following products:

**Permanent magnet three-phase synchronous motors in the series
S..04, S..05, S..06, S..07, S..08, S..09, S..11, S..13, S..16, S..18**

where necessary, in conjunction with
gears from the series: BG.., BF.., BK.., BS.., BM..

with the requirements of the European Directives

**DIRECTIVE 2014/35/EU OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 26 February 2014
on the harmonisation of the laws of the Member States relating to the making available on the
market of electrical equipment designed for use within certain voltage limits.**

Published on 29 March 2014 in the Official Journal of the EU L96/357

**The object of declaration described above is in conformity with the Union's relevant harmonisation legislation,
proven by the compliance with the following harmonised standards:**

EN 60034-1:2010/AC:2010

EN 60034-5:2001/A1:2007

EN 60034-8:2007/A1:2014

EN 60529:1991/A1:2000/A2:2013

Additional information:

Three-phase motors do not fall under the scope of the EMC Directive, since as passive elements they are neither susceptible for the purpose of interference resistance, nor do they affect the environment by emitting high-frequency interference. Operation of these motors in the case of power supply through switch-mode supplies (converter) and the associated EMC-relevant aspects are the responsibility of the user of this PDS (Power Drive System). The instructions in the converter's product documentation must be observed. The installer is responsible for the final EMC properties of the device, system or installation.

Esslingen 01 July 2020

Bauer Gear Motor GmbH



N. Halmuschi
(Managing Director)



P. Cagan
(Quality Director)

This certificate does not contain any guarantee of features in terms of product liability. The technical documentation is produced and administered by Bauer Gear Motor GmbH.

Safety information for the operation of geared motors

(in accordance with the Low Voltage Directive 2014/35/EU)

General

This safety information applies in addition to the relevant product-specific operating instructions and for safety reasons must be taken into particular consideration in every case.

This safety information is intended to protect persons and objects from injury and hazards which can arise from improper use, incorrect operation, inadequate maintenance or other incorrect handling of electric drive units in industrial installations. Low-voltage machines have rotating parts and may have parts that are live, even when the machine is at rest, and surfaces that may become hot in operation. Warning signs and information signs on the machine are to be observed without exception. Details may be found in our detailed operating instructions. They are provided with the machine when it is supplied and can be requested separately as required by stating the motor model.

1 Personnel

All necessary work on electric drive units, in particular also planning work, transport, assembly, installation, commissioning, maintenance, repair, may only be performed by adequately qualified personnel (e.g. electrical engineers as specified in draft EN 50 110-1/ DIN VDE 0105), who have the operating instructions provided and other product documentation available during any corresponding work and who are obliged to abide by the instructions contained therein. This work is to be monitored by a specialist supervisor. Qualified personnel are persons who are authorised due to training, experience and instruction as well as their knowledge of relevant standards, rules, accident prevention regulations and operating conditions by the person responsible for the safety of the installation to perform the activities required in each case and who are able to recognise and avoid possible hazard.

Knowledge of first-aid measures and of the available lifesaving equipment is also required.

Non-qualified personnel shall be forbidden to work on the geared motors.

2 Intended use taking into account the relevant technical regulations

These machines are intended for commercial installations, unless otherwise expressly agreed. They comply with the standards of the series EN 60034/DIN VDE 0530. Use in a potentially explosive atmosphere is forbidden, if not expressly intended for this purpose (refer to additional information). If in a special case --use in non-commercial installations-- Increased safety precautions are required (e.g. protection against access by children's fingers), these conditions are to be ensured when setting up the installation. The machines are designed for ambient temperatures between -20 °C to +40 °C as well as for installation heights up to 1000 m above sea level. Data for Units designed for differing ambient temperatures on the rating plate. Any deviations found on the rating plate must be taken into consideration.

The conditions at the place of work must correspond to all rating plate data.

Low-voltage machines are components for installation in machines in the sense of the Machinery Directive 2006/42/EC.

It is forbidden to use the machine until conformity of the final product with this directive is established (consult EN 60204-01).

3 Transportation, storage



Only the attachment points provided by the factory may be used for transporting and installing the geared motors.

When the electric drive units are being transported, the eye bolts -- where provided in the design -- must be firmly tightened down their bearing surface. They may be used only for transporting the drive unit and not for lifting both the drive unit and the driven machine. Damage sustained after delivery must be reported to the haulage company immediately. Commissioning may have to be suspended.

If drive units are to be stored, ensure a dry, dust free and low vibration (veff < 0,2 mm/s) environment (damage sustained during storage). The life of the lubricants and seals is reduced with longer storage times.

There is a risk of fracture at very low temperatures (under approximately - 20 °C). If the transport eye bolts are replaced, drop forged eye bolts as specified in DIN 580 are to be used.

4 Mounting arrangement, assembly

The drive unit is to be fastened by its flange or foot if an IM.. mounting arrangement is intended. Gear units with hollow shafts are to be attached on the driven shaft using the means provided.

Caution! Depending on the reduction ratio, geared motors develop substantially higher torques and forces than high-speed motors of similar power.

Mounts, substructure and torque restraint are to be rated for the high forces to be anticipated during operation and secured sufficiently against loosening. The output shaft(s) and any second motor shaft extension present as well as the transmission elements mounted on it (couplings, chain wheels etc.) are to be covered so that they cannot be touched.

5 Connection

All work shall only be carried out by qualified technical personnel on a stationary machine which has been protected against re-starting. This applies also to auxiliary circuits (e.g. stationary heating). Remove any transportation blocks before start-up.

Safety information for the operation of geared motors

Check to ensure safe isolation from the supply!

The terminal box may only be opened once it has been ensured that the power is switched off. The information on voltage and frequency on the rating plate must correspond with the mains voltage under observance of the terminal circuit. Exceeding the tolerances as in EN 60034 / DIN VDE 0530, i.e. voltages $\pm 5\%$, frequency $\pm 2\%$, cam form, symmetry, increases heating and reduces service life.

Accompanying connection diagrams, particularly for special equipment (e.g. pole-changing, thermistor protection etc.), are to be observed. Type and cross-section of the main conductors as well the protective conductors and any potential equalization which may become necessary must correspond to the general and local installation regulations. With switching duty, the starting current is to be taken into account. The drive unit is to be protected against overloading and in dangerous situations against automatic restarting due to inadvertent starting.

The terminal box is to be locked again to protect against contact with live components.

6 Commissioning

Before commissioning, protective films are to be removed, the mechanical connection to the driven machine disconnected as far as possible and the direction of rotation examined in the no-load state. Feather keys are to be removed or secured in such a way that they cannot be ejected as this is done. Ensure that the current draw in the loaded condition does not exceed the rated current indicated on the rating plate for any length of time. Observe the drive unit after first commissioning for at least one hour for any unusual heat or noise.

7 Operation

With certain layouts (e.g. unventilated machines), relatively high temperatures can occur on the motor frame, which are however within the limits specified in the standard. If these drive units are located in a place where they are subject to intensive contact, measures must be taken by the installer or operator to provide protective shielding.

8 Spring-loaded brakes

Spring-loaded brakes are safety brakes which continue to work in the event of power failure or usual wear. With a spring loaded brake with manual release option, the operator must ensure that the manual release of the brake cannot be actuated accidentally. If the operator removes the manual release lever for this purpose, then, in the case that the brake is built in under the fan cowl, he must allow for suitable contact protection at the emerging opening of the sheet metal fan cowl. Since other components could also fail, suitable safety precautions are to be taken to avoid any injury to persons or damage to objects caused by un-braked operation.

9 Maintenance

In order to prevent breakdowns, danger and damage, the drive units must be examined at regular intervals depending on the operating conditions. The lubrication intervals for bearings and gear units specified in the respective operating instructions are to be observed. Worn or damaged parts are to be replaced using original spare parts or standard parts. In the event of heavy dust accumulation, clean airways regularly. For all inspection and maintenance work, observe Section 5 and the information provided in the detailed operating instructions.

10 Operating instructions

For reasons of clarity, the operating instructions and safety information do not contain all information relating to all geared motors types and cannot take into account every conceivable case of installation, operation or maintenance. The information is essentially limited to that which is required for qualified personnel in normal working situations. Any unclear points can be clarified by contacting Bauer.

11 Faults

Changes in relation to normal operation, such as higher temperatures, vibrations, noises etc. tend to indicate that the function is impaired. To avoid faults which could lead directly or indirectly to injury to persons or damage to property, the maintenance staff responsible must be informed. If in any doubt, the geared motors are to be switched off immediately.

12 Electromagnetic compatibility

The operation of the low-voltage machine in its intended application must meet the protection requirements of the EMC (electromagnetic compatibility) Directive 2014/30/EU.

Correct installation (e.g. screened cables) is the responsibility of the system's installers. Precise information can be taken from the operation instructions. For systems with frequency inverters and rectifiers, the manufacturer's electromagnetic compatibility information is also to be taken into consideration. The electromagnetic compatibility directive in accordance with EN 61000-6-2 and EN 61000-6-4 is complied with given proper use and installation of BAUER geared motors. This is also true in combination with frequency inverters and rectifiers. The additional information provided in the operation instructions is to be taken into consideration when using the motors in the residential, commercial and trade sectors, as well as in small businesses in accordance with EN 61000-6-1 and EN 61000-6-3.

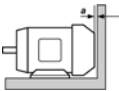
13 Warranty and liability

The warranty obligations of Bauer arise out of the relevant supply contract, which is neither expanded nor restricted by this safety information or other instructions.

This safety information is to be kept in a safe place.

Geared motors with three-phase motor

In the standard version the drives are designed for an ambient temperature range of -20 °C to +40 °C and for installation elevations up to 1000 m above sea level; differing conditions are marked on the nameplate. Exposure to dirt, moisture and the usual outdoor conditions may not exceed the level corresponding to the IP protection rating. The air inlet and outlet must be kept free of obstruction (for example, due to an acoustic cover).

Motor sizes	Minimum clearance a at air inlet		
Up to D .. 16		35 mm	
D .. 18 to D. 22		85 mm	
D .. 25 and above		125 mm	

General Notes

The operating instructions are an intergral element of the product and must be available in a readable form at all times. Employees responsible for the plant and operations together with all who have to work with the unit must have read and understood the operating instructions completely.

Disclaimer

Observation of the operating instructions is a prerequisite for the safe use and the attainability of the product performance of the geared motor. Bauer does not accept liability for personal, material or economical damages caused by non-observation of the operating instructions. Liability for defects are excluded in such cases.

Geared motors in degree of protection IP65

(Motor models D/E06... to D.28...) complying with EN 60529 and IEC 34-5/529 are totally enclosed and dust-tight as well as hose proof.

For outdoor installation, the geared motor must be coated with several layers of durable paint to protect against corrosion. The condition of the paint must be checked and repaired at regular intervals, depending on ambient influences. The paint finish must be compatible with the other components. Paints with a synthetic resin base have proved well suited to this purpose.

Geared motors in degree of protection IP54

(Motor models D/E04... and D/E05...) complying with EN 60034, part 5 and IEC 34-5 are protected against dust and occasional splashing water. Installation outdoors or in wet areas is not permissible without special protective measures.

Geared motor in stainless steel design

Caution:
Risk of injury on hot surfaces
Slight to moderate injuries
Work only on cooled drives

Geared motors with three-phase motor

Mounting arrangement

It is recommended that drinking water, food, textiles etc. beneath the geared motor be covered.

The drive unit should be installed as free from vibration as possible.

Special instructions are to be observed in installation locations with abnormal operating conditions (e.g. prolonged exposure to dripping water, high ambient temperatures above 40 °C, explosion hazards). The fresh air intake must not be restricted by unsuitable installation or by fouling.

Flexible couplings with zero play, if possible, are recommended for direct power transmission from the gear unit to the driven machine and commercially available slip clutches are recommended if there is a risk of blocking.

Care must be taken when fitting transmission elements onto the output shaft of the gear unit, which is finished to ISO k 6 or m 6, and the tapped end hole intended for this purpose according to DIN 332 should be used if possible. Warming the machine part to be fitted onto the shaft to approximately 100 °C has proved to be advantageous. The bore must be dimensioned in accordance with following table and must thus exhibit the following tolerances:

Nominal size of bore (in mm)	k 6 or m 6 output shaft Bore H7 with tolerances (in $1/_{1000}$ mm)
over 6 to 10	0 to + 15
over 10 to 18	0 to + 18
over 18 to 30	0 to + 21
over 30 to 50	0 to + 25
over 50 to 80	0 to + 30
over 80 to 120	0 to + 40

Where the gear units have a hollow shaft and keyway for high profile feather keys as specified in DIN 6885, Part 1 and hollow shaft for shrink-disc connection, the shafts intended to form the counterpart must be dimensioned to ISO h 6. They must, therefore, exhibit the following tolerances:

Shaft diameter (in mm)	Nominal allowance (in $1/_{1000}$ mm)
over 18 to 30	0 to - 13
over 30 to 50	0 to - 16
over 50 to 80	0 to - 19
over 80 to 120	0 to - 22
over 120 to 140	0 to - 25

Geared motors with three-phase motor

In all cases, particular care shall be taken to ensure that any burring, swarf etc. is carefully removed before assembly. The keyways should be lightly greased to prevent seizing. Hollow shafts to be fitted with shrink-disc connections must not be greased. The following installation instructions are to be noted here.

The eye bolt is to be retightened firmly if it has worked loose during transportation.

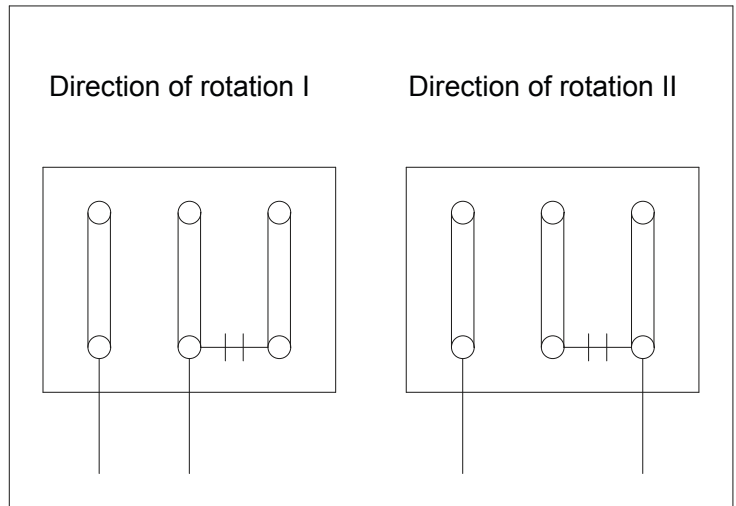
Electrical connection

When connecting the motor, take note of the rating plate information and the connection diagram as well as the relevant safety regulations and rules for the prevention of accidents.

Unless a special design is concerned, the rating data refer to $\pm 5\%$ voltage tolerance, -20 to $40\text{ }^{\circ}\text{C}$ ambient temperature and altitudes up to 1000 m above sea level.

Small motors can be connected directly (the regulations of the local electricity supply companies are to be observed). The permissible switching frequency depends on the design of the motors, the load torque and the mass moment of inertia.

The direction of rotation of single phase motors may generally only be changed after the motor has stopped and in accordance with the following diagram:



Geared motors with three-phase motor

Unless otherwise stated, the three phase motor is connected for the higher of the two rated voltages specified. The motor may have to be connected from star to delta at the terminal board, if necessary, to ensure that it complies with the mains voltage.

Specially designed motors (e.g. for two rated voltages in the ratio 1:2 or with pole-changing windings) are to be connected in accordance with the relevant circuit diagram.

Two mains leads must be swapped in the event of incorrect direction of rotation. When closing the terminal box, particular attention must be given to obtaining a perfect seal. With motors in sizes D/E 04 to D/E 09 with moulded terminal boxes, two connection openings each on side A and C are possible.

The cable entry openings required depending on the installation position should carefully pierced with a suitable tool. Take care not to damage the terminal board.

Two lock nuts and seals are provided in the terminal box for the cable glands (metric). Plugs must be screwed into unused cable entry holes. The maximum spanner opening for the cable glands of the D04 is 24 mm and of the D05 to D09 is 29 mm.

To guarantee electromagnetic compatibility (EMC) as defined in EMC Directive 2014/30/EU, all signal lines must use shielded cables. The cable sheath is to be earthed at both ends. The frequency inverter operating instructions will indicate whether a shielded cable is necessary for the motor supply line. A shielded motor cable is not required when connecting to the low-voltage network or to a frequency inverter with an output filter. Signal cables and power cables should not be laid parallel over long distances.

Overload protection A motor protection switch must be used to protect the winding against overloading and against the consequences of operating on only 2 mains leads (e.g. when only one fuse blows or in the event of a wire breakage)

Example	Motor winding for 230/400 V; Rated currents Setting of the motor protection switch at Connection for 230 V (delta): Connection for 400 V (star):	5.7/3.3 A 5.7 A 3.3 A
---------	--	---------------------------------

The overcurrent relay of the motor protection switch is to be set to the correct rated current intensity for the rated voltage concerned (see rating plate). Take note of the relevant circuit diagram for motors with thermally activated winding protection (e.g. thermostats or thermistors).

Automatic restarting after the winding has cooled must be avoided in most applications.

The output of the motors is normally adequately rated, particularly in connection with four and multistage gear units. The rated current does not represent a measure of gear unit utilization in these cases and cannot be used as overload protection for the gear unit. In some cases, the way in which the driven machine is loaded can exclude any overloading as a matter of course. In other cases it is prudent to protect the gear unit by mechanical means (e.g. slip clutch, sliding hub etc.). The maximum permissible limit torque M_2 in continuous running duty specified on the rating plate is decisive here.

Permanent-magnet synchronous motor (PMSM)

The rotors on the PMSM machines are fitted with embedded permanent magnets.

Caution: The magnetic field which is created may be harmful to health.

For this reason, it is imperative to observe the accident prevention regulations of the respective country in workplaces where people are exposed to magnetic fields. In Germany, the accident prevention regulations BVG B11 (VBG25) – Electromagnetic Fields must be observed.

Please note that additional electromagnetic fields occur in operation.

Warning: Magnetic fields emanating from the permanent magnets exert a very strong pull on magnetisable materials.

Dismantlement of the motor must therefore only be carried out with special tools and auxiliary equipment. Once engine components, tools or other magnetisable materials are attracted to the rotor, it takes great effort to separate them again.

In the event of dismantlement, the rotor must be removed in a guided and insulated manner.

Caution: The removed rotor must be protected from contamination such as metal shavings. Before reinsertion, the rotor must be thoroughly cleaned.

Hazard: The attraction of other objects may lead to severe bruising and injuries from attracted tools, such as screwdrivers, wrenches etc.

Commissioning

Caution: Permanent-magnet synchronous motors can only be operated with a frequency converter. Direct connection to the mains is not possible.

The motor data listed on the nameplate for the motor parameterisation must be used for commissioning.

The values of the torque limit, limiting currents and limit speed on the nameplate must be observed.

Caution: Exceeding these limit values may lead to motor damage as a result of heating, centrifugal force and demagnetisation of the permanent magnets, to gear damage resulting from overstressing and to system damage.

In the event of application-related overloading, Bauer Gear Motor must be consulted.

In generator operation, a PMSM works like a dynamo and generates tensioning on the open motor clamps through the movement of the rotor shaft or the drive components.

Warning: Electric shock with slight personal injuries as a result of generator operation at the open motor clamps.

Lubricant changes

The gear units are supplied with lubricant ready for operation.

In normal operating conditions and with a lubricant temperature of approximately 80° C, the oil should be replaced after approximately 15000 operating hours when using CLP 220, or after 25000 operating hours when using PGLP 220/PGLP 460. The lubrication interval must be reduced at higher temperatures (halve it for each 10 K increase in the lubricant temperature).

The lubricant must be changed after 2 or 3 years at the latest whatever the operating hours.

The medium and larger gear units have filling plugs and drain plugs. In the standard designs, these make it possible to change the lubricant without disassembly.

With smaller gear units, the interior is accessed by unscrewing the connecting bolts. Alignment pins and centrigs secure the precise assembly.

Worm-gear units are sliding gear units whose tooth flanks, contrary to rolled gear units, only become smooth once run in. They should therefore initially be run in under partial loading (about $\frac{2}{3}$ of the rated load) until the full load capacity of the flanks and the optimal efficiency is achieved. After approximately 200 operating hours, the lubricant should be changed and the gear unit enclosure thoroughly flushed, so that the minimal, but inevitable amount of material removed by smoothing abrasion is cleared.

It is also necessary to flush the gear unit enclosure if the lubricant grade or lubricant type is changed.

If the motor is only used briefly it is sufficient to drain off the original oil and use the original lubricant type to refill the maximum possible amount for the gear unit as defined in the lubricant volume table. Then operate the drive unit briefly under no load, drain this oil off again and refill with the new lubricant as defined on the rating plate. In special cases, refill up to the oil level mark. If necessary, drain off the original lubricant and flush out the gear unit with petroleum until all traces have been washed out. Then perform the procedure described above for short-term operation twice before filling with the specified volume of new lubricant in accordance with the rating plate, in special cases up to the oil level mark.

It is advisable to inspect and if necessary replace the wear parts (bearings and seals) when changing the lubricant.

Lubricant grade

Oils CLP 220, PGLP 220 and PGLP 460 complying with DIN 51502 and DIN 51517 are suitable for lubricating the gear unit, or in special cases use soft flow grease GLP 00f with good EP properties.

The lubricant must permit low-friction, virtually wear-free continuous operation. The damage load level on the FZG test as specified in DIN 51354 shall be in excess of load level 12, and the specific wear below 0.27 mg/kWh. The lubricant should not foam, should protect against corrosion and should not attack the interior paint, the rolling contact bearings, gearwheels and seals.

Lubricants of different types may not be mixed, as otherwise the lubrication characteristics may be impaired. A long service life is only ensured by the use of a lubricant listed below or which is demonstrably equivalent.

Storage

Should geared motors need to be stored for a longer period of time before installation, please observe the chapter „Information on the storage of geared motors with cage rotors“

Wear-protecting EP gear lubricant oils as listed in the lubricant table below have proved particularly suitable.

Geared motors with three-phase motor

Manufacturer	Lubricant type				
	Mineral Oil	Synthetic Oil			USDA H1 Oil
	ISO VG 220	ISO VG 68	ISO VG 220	ISO VG 460	ISO VG 220
	Standard oil for gearboxes in the series BF06-BF90 BG04-BG100 BK60-BK90	Low temperature oil for gearboxes in the series BF06-BF90 BG04-BG100	BK06-BK90 BM09-BM40 BS02-BS40 High temperature oil for gearboxes in the series BS02-BS10 BK06-BK10 BF06-BF90 BG04-BG100 BK60-BK90 BM09-BM10	Standard oil for gearboxes in the series BS20-BS40 BK17-BK50 BM20-BM40 High temperature oil for gearboxes in the series BS20-BS40 BK17-BK50 BM20-BM40	Food and Beverage Industry Oil for gearboxes in the series BF06-BF90 BG04-BG100 BK06-BK90 BM09-BM40 BS02-BS40
AGIP	BLASIA 220 [13 02 08]	—	—	BLASIA S 220 [13 02 06]	BLASIA S 460 [13 02 06]
BECHER RHUS	STAROIL G 220 [13 02 08]	—	BERUSYNTH EP 68 [13 02 06]	BERUSYNTH EP 220 [13 02 06]	BERUSYNTH EP 460 [13 02 06]
CASTROL	ALPHA EP 220 [13 02 08] ALPHA SP 220 [13 02 08] OPTIGEAR EP 220 [13 02 08] OPTIGEAR 1100/220 [13 02 08]	Alphasyn T68 [13 02 06]	—	ALPHASYN PG 220 [13 02 06] OPTIGEAR 800/220 [13 02 06] OPTIGEAR 1300/220 [13 02 06] ALPHASYN GS 220 [13 02 06]	ALPHASYN PG 460 [13 02 06] OPTIGEAR 800/460 [13 02 06] OPTIGEAR 1300/460 [13 02 06] ALPHASYN GS 460 [13 02 06]
CHEVRON	Meropa 220 [13 02 08] GEARTEX EP-A SAE 85W-90 [13 02 06]	—	Meropa Synlu-be WS 68 [13 02 06]	Meropa Synlube WS 220 [13 02 06]	Meropa Synlube WS 460 [13 02 06]
FUCHS	RENOLIN CLP 220 [13 02 08] RENOLIN CLPF 220 SUPER [13 02 08] RENOLIN CLP 220 PLUS [13 02 08]	RENOLIN UNI-SYN CLP 68 [13 02 06]	RENOLIN PG 68 [13 02 06]	RENOLIN PG 220 [13 02 06]	RENOLIN PG 460 [13 02 06]
KLÜBER	KLÜBEROIL GEM 1-220 N [13 02 08]	—	KLÜBER-SYNTH GH 6-80 [13 02 06]	KLÜBERSYNTH GH 6-220 [13 02 06]	KLÜBERSYNTH GH 6-460 [13 02 06]
MOBIL	MOBILGEAR 600 XP 220 [13 02 08]	MOBIL SHC 626 [13 02 06]	—	MOBIL SHC Gear 220 [13 02 06] MOBIL SHC 630 [13 02 06]	MOBIL SHC Gear460 [13 02 06] MOBIL SHC 634 [13 02 06]
OEST	Gearoil 220 [13 02 06]	—	—	—	—
SHELL	OMALA S2 GX220 [13 02 08]	—	—	OMALA S4 WE 220 [13 02 06]	OMALA S4 WE 460 [13 02 06]
TOTAL	CARTER EP 220 [13 02 08] CARTER KEP 220 [13 02 06]	—	—	CARTER SY 220 [13 02 06]	CARTER SY 460 [13 02 06]
WINTERSHALL	SRS ERSOLAN 220 [13 02 08]	—	—	—	—

[...] European Waste Catalogue Code (Decision 2001/118/CE)



ATTENTION!

Synthetic gear oils with a polyglycol base (e.g. PGLP etc) must be kept separate from mineral oils and disposed of as special waste.

As long as the ambient temperature does not fall below -20 °C, ISO viscosity grade VG 220 (SAE 90) is recommended in accordance with the international definition of viscosity grades at 40 °C in accordance with ISO 3448 and DIN 51519, and AGMA 5 EP in North America.

For lower ambient temperatures, oils of a lower nominal viscosity, with correspondingly better starting characteristics should be used, such as PGLP with a nominal viscosity of VG 68 (SAE 80) or AGMA 2 EP. These grades may also be required at temperatures around the freezing point if the drive unit's breakaway torque has been reduced with a view to achieving soft starting or if the motor has a relatively low power output.

Lubricant volume

The recommended lubricant quantity for the particular style is indicated on the rating plate of the motor (symbol ). When filling, make certain that the upper gear unit components, depending upon the installation position, are also well lubricated. The oil level mark should be taken into consideration in special cases. Information about the lubricant volume required for other styles construction can be obtained from the works.

Disposal

The metallic parts of the gear unit and the geared motor can be disposed of as scrap, segregated into steel, iron, aluminium and copper. The lubricants used are to be disposed of as waste oil, and the synthetic oils are to be disposed of as special waste. Information on this can be found on the lubrication chart or the rating plate.

Bearing lubrication

Gearbox assemblies

Gearbox bearings are usually open bearings. Open bearings in standard BAUER gearboxes are lubricated together with the gearing lubrication. Their maintenance therefore takes place as a part of the routine gearbox lubrication change.

The bearings in special designs (Ex) can be in encapsulated design and equipped with their own grease reservoirs. In this case, the grease lubrication is renewed by changing the bearings themselves in the course of a regular gearbox lubrication change. Cleaning and re-greasing the bearings is not recommended due to the risk of impurifying the lubrication.

Input assemblies

Input assemblies of BAUER geared motors

- with integrated motors, all gearbox sizes with all combinable motors
- with free running input shaft (-SN) for the gearbox sizes 06 to 70 together with 10 to 100 with pre-stage
- free running input shaft (-SN) the gearbox size BF80
- for assembly with standard motors in sizes up to and including IEC180 or NEMA286

include factory-installed greased encapsulated bearings.

When operated at an input speed of 1500 rpm, a grease change is required after 10,000 hours of operation. In specific cases a maximum allowable input rpm of 3600 is allowable. If the input rpm is doubled, the grease change interval is halved.

If the gearbox has encapsulated bearings in the input components, the grease change takes place by replacing the bearings while maintaining/inspecting the radial shaft seals. Cleaning and re-greasing the bearings is not recommended due to the risk of impurifying the lubrication..

Input assemblies of BAUER geared motors

- free running input shaft (-SN) the gearbox sizes 80 and 90, however not for BF80
- for assembly with standard motors in sizes as from and including IEC200 or NEMA324

are, in contrast to the encapsulated bearings described above, equipped with open, regreasable input bearings, an individual lubrication point or nipple is installed for each regreasable bearing.

The maximum speed is 1800 rpm. The grease change interval is 2500 hours of operation however, a maximum of 6 months. The grease filling in the bearing is to be renewed after every 800 hours of operation by topping up with fresh grease.

The entire grease filling is to be replaced after two grease top-ups at the latest. Approx. 40 g of additional grease is required to top up free running input shaft assemblies (-SN) and standard motor installation and three times as much (120 g) being required for a complete grease change. For integrated motor assemblies, 60 g of grease is required to top up and 180 g for a complete grease change.

The topping up or exchange of the grease filling should be performed while the motor shaft is rotating in order to effect an optimal distribution of the grease in the bearing.

When removing the grease, excess and used-up grease is to be removed from the grease discharge chamber. KLÜBER PETAMO GHY 133 N grease is to be used as lubricant.

The grease type for encapsulated and regreasable bearings may vary from the standard for special-purposes lubrication (food-compatible, biodegradable, etc.). Ask the geared motor manufacturer about the correct lubrication for specific special applications.

Geared motors with three-phase motor

Operating problems: Gear unit

Failure	Possible causes	Remedy
Oil leakage at: • output shaft sealing • motor shaft sealing • Gearbox cover • Motor flange	• Shaft seal defective • Excess pressure in gear unit	• Change shaft sealing • Excess pressure function check breather valve • Contact Bauer Service
• Gear box output shaft does not turn although motor is switched on.	• Interruption of movement chain	• Contact Bauer Service

Operating problems: Motor

Failure	Possible causes	Remedy
• Output shaft is turning in the wrong direction	• Motor connection wrong	• Change 2 phases with each other
• Motor gets too hot	• Bad ventilation of the motor • Motor runs against a closed brake	• Check ventilation system • Clean surface of motor (packing of dust) • Check brake function • Contact Bauer Service
• Brake failure	• Brake is not lifting • Friction partner worn out	• Check brake lift voltage • Change friction partner • Contact Bauer Service

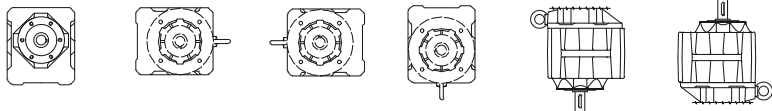
Gear-housing with flange or foot

Flange (Code-2./Code-3./Code-4./Code-7.)

Foot with threaded holes (Code -6.)

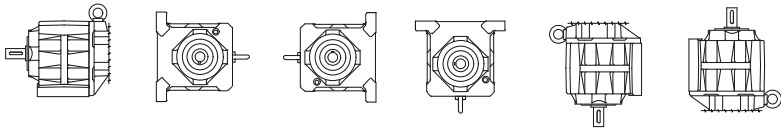
Foot with clearance holes (Code-9.)

Completely machined (Code -8.)



Foot housing

cast foot with clearance holes (Code -1.)



Gearbox type		H4	H1	H2	H3	V1	V2
BG04	*	0.03	0.03	0.03	0.03	0.05	0.05
	**	0.05	0.05	0.05	0.05	0.1	0.05
BG05	*	0.05	0.05	0.05	0.05	0.08	0.08
	**	0.08	0.08	0.08	0.08	0.16	0.8
BG06	*	0.08	0.08	0.08	0.08	0.15	0.15
	**	0.12	0.12	0.12	0.12	0.24	0.15
BG10	*	0.65	0.65	0.65	0.85	1.05	0.85
	**	0.45	0.45	0.45	0.6	0.75	0.6
BG15	**	0.4	0.4	0.4	0.35	0.62	0.55
	*	0.8	0.8	0.8	1.1	1.4	1.1
BG20	**	0.6	0.6	0.6	1.0	1.15	0.9
	*	1.0	1.0	1.0	1.7	2.4	1.6
BG30	**	1.0	1.0	1.0	1.7	2.3	1.7
	*	1.7	1.7	1.7	2.5	3.5	2.1
BG40	**	1.7	1.7	1.7	2.5	3.5	2.1
	*	3.0	3.0	3.0	4.5	5.5	3.3
BG50	**	3.0	3.0	3.0	4.5	5.5	3.3
	*	5.5	5.5	5.5	7.0	10.9	6.4
BG60	**	5.5	5.5	5.5	7.0	10.9	6.4
	*	6.5	6.5	6.5	8.0	13.5	9.0
BG70		6.5	6.5	6.5	8.0	13.5	9.0
BG80		11.0	11.0	11.0	11.0	22.5	15.0
BG90		19.0	19.0	19.0	19.0	40.0	26.0
BG100		35.0	35.0	55.0	50.0	66.0	50.0

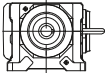
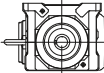
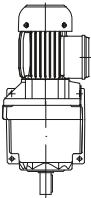
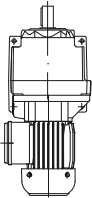
* Flange Housing

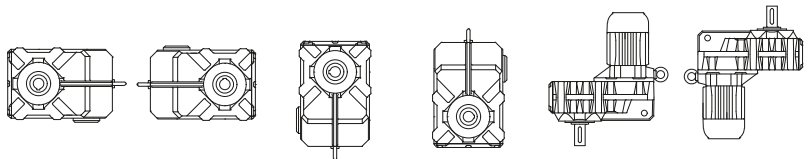
** Foot Housing

Lubrication quantity in litre

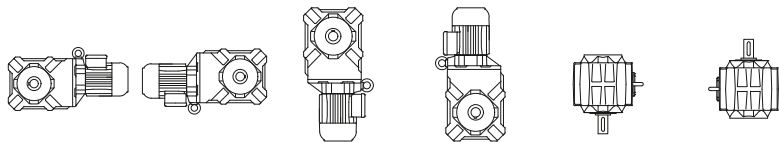
Lubricants

Lubrication quantity for BG20-01R

						
Gear type	H4	H1	H2	H3	V1	V2
BG20R	0.8	1.0	0.8	1.4	1.65	1.0
Lubrication quantity in litre						



Gear type	H1	H2	H3	H4	V1	V2
BF06	0.25	0.25	0.25	0.37	0.35	0.3
BF10	0.85	0.85	0.85	1.1	1.45	1.5
BF20	1.3	1.3	1.3	1.7	2.2	2.25
BF30	1.7	1.7	1.7	2.2	3.2	3.0
BF40	2.7	2.7	2.7	3.5	4.9	4.8
BF50	3.8	3.8	3.8	5.0	6.7	6.7
BF60	6.7	6.7	6.7	9.0	12.3	12.0
BF70	12.2	12.2	12.2	16.0	24.2	21.8
BF80	17.0	17.0	17.0	21.0	32.2	27.5
BF90	32.0	32.0	32.0	41.0	62.0	53.0
	Lubrication quantity in litre					



Gear type	H1	H2	H3	H4	V1	V2
BK06	0.15	0.23	0.29	0.31	0.18	0.23
BK10	0.83	0.83	0.92	1.75	0.92	0.92
BK17	1.0	1.7	1.8	2.6	1.3	1.8
BK20	1.5	1.5	1.6	2.9	1.65	1.65
BK30	2.2	2.2	2.3	4.4	2.4	2.4
BK40	3.5	3.5	3.5	7.0	3.7	3.7
BK50	5.8	5.8	5.8	11.5	6.0	6.0
BK60	6.0	8.7	6.9	12.0	8.6	8.6
BK70	10.2	15.0	11.5	20.5	13.5	14.5
BK80	18.0	25.5	19.0	37.0	23.5	25.5
BK90	33.0	48.0	36.0	69.0	45.0	48.0
Lubrication quantity in litre						

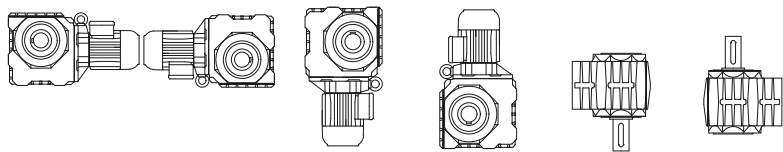
Lubricants

Lubrication quantity series BK (HiflexDRIVE)

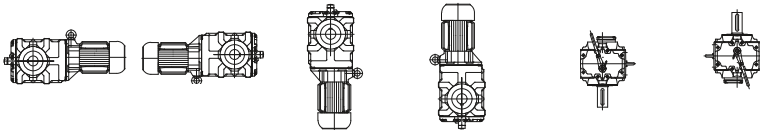


Gear type		H1	H2	H3	H4	V1	V2
BK04	AI	-	-	-	on request	-	-
	VA	0.37	0.75	0.75		0.56	0.7
BK08	AI	0.7	1.05	1.1		0.85	0.9
	VA	0.65	0.96	1.03		0.78	0.83
BK17	AI	1.0	1.7	1.8		1.3	1.8
	VA	1.0	1.7	1.8		1.3	1.8
Lubrication quantity in litre							

Lubricants
Lubrication quantity series BS



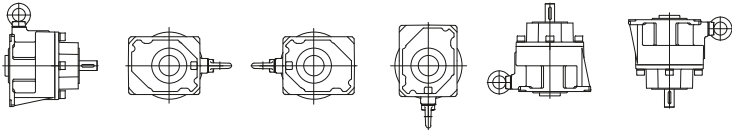
Gear type	H1	H2	H3	H4	V1	V2
BS02	0.06	0.06	0.06	0.06	0.06	0.06
BS03	0.17	0.17	0.17	0.17	0.17	0.17
BS04	0.11	0.17	0.11	0.2	0.11	0.11
BS06	0.24	0.36	0.24	0.45	0.24	0.24
BS10	0.9	1.3	0.9	1.6	0.9	0.9
BS20	1.5	2.1	1.5	2.7	1.5	1.5
BS30	2.2	3.0	2.2	3.8	2.2	2.2
BS40	3.5	4.7	3.5	6.0	3.5	3.5
Lubrication quantity in litre						

											
Gear type	H1	H2	H3	H4	V1	V2					
BM09	0.5										
BM10	0.65										
BM20	0.7										
BM30	1.2										
	1.8*										
BM30/S1	1.2										
	1.8*										
BM30/S2	1.3										
	1.9*										
BM40	2.5										
	3.2*										
BM40/S1	2.5										
	3.2*										
BM40/S2	2.6										
	3.3*										
*: with BM30Z/BM40Z the prepress lubricant is filled via the main gearbox.											
Lubrication quantity in litre											



BF	H4	H1	H2	H3	V1	V2
BG	H4 B3/B5	H1 B6	H2 B7	H3 B8	V1 V5/H5	V2 V6/H6
BK	H1	V1	V2	H2	H4	H3
BS	H1	V1	V2	H2	H4	H3
Motor size						
D..04; E..04			not regreasable			
D..05; E..05						
D..06; E..06						
D..07						
D..08; S..08						
D..09; S..09						
D..11; S..11						
D..13						
D..16						
D..18						
D..20; D..22						

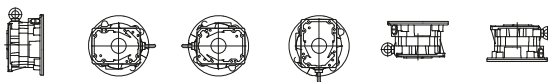
Lubrication quantity in litre



BF	H4	H1	H2	H3	V1	V2
BG	H4 B3/B5	H1 B6	H2 B7	H3 B8	V1 V5/H5	V3 V6/H6
BK	H1	V1	V2	H2	H4	H3
BS	H1	V1	V2	H2	H4	H3
Gear type						
BK06-SN / BS06SN						
BG10-BG10Z						
BF10-BF10Z-SN						
BK10-BK10Z-SN						
BS10-BS10Z-SN						
BG20-BG20Z-SN						
BF20-BF20Z-SN						
BK20-BK20Z-SN						
BS20-BS20Z-SN						
BG30-BG30Z-SN						
BF30-BF30Z-SN						
BK30-BK30Z-SN						
BS30-BS30Z-SN						
BG40-BG40Z-SN						
BF40-BF40Z-SN						
BK40-BK40Z-SN						
BS40-BS40Z-SN						
BG50-BG50Z-SN						
BF50-BF50Z-SN						
BK50-BK50Z-SN						
BG60-BG60Z-SN						
BF60-BF60Z-SN						
BK60-BK60Z-SN						
BG70Z-SN / BF70Z-SN / BK70Z-SN						
BG80Z-SN / BF80Z-SN / BK80Z-SN						
BG90Z-SN / BF90Z-SN / BK90Z-SN						
BG100Z-SN /	2-Z-bearing grease lubricated, sealed for life not regreasable					
BG70-SN						
BK70-SN						
BF70-SN						
BG80-SN						
BK80-SN						
BG90-SN						
BK90-SN						
BF90-SN						
BG100-SN						
grease lubrication for subsequent lubrication regreasable: (PETAMO GHY133N)						
Lubrication quantity in litre						

Lubricants

Lubrication quantity for coupling lantern



BF			H4	H1	H2	H3	V1	V2
BG			H4 BS3/BS5	H1 B6	H2 B7	H3 B8	V1 V5/H5	V2 V6/H6
BK			H1	V1	V2	H2	H4	H3
BS			H1	V1	V2	H2	H4	H3
Gear type								
BK06-C / BS06-C		up to IEC180 or up to Nema284/286TC						
BG10-BG10Z-C	BG20-BG20Z-C							
BF10-BF10Z-C	BF20-BF20Z-C							
BK10-BK10Z-C	BK20-BK20Z-C							
BS10-BS10Z-C	BS20-BS20Z-C							
BG30-BG30Z-C	BG40-BG40Z-C							
BF30-BF30Z-C	BF40-BF40Z-C							
BK30-BK30Z-C	BK40-BK40Z-C							
BS30-BS30Z-C	BS40-BS40Z-C							
BG50-BG50Z-C	BG60-BG60Z-C							
BF50-BF50Z-C	BF60-BF60Z-C							
BK50-BK50Z-C	BK60-BK60Z-C							
BG70-C	BG80-C							
BF70-C	BF80-C							
BK70-C	BK80-C							
BG90-BK90Z-C BF90 BK90-BK90Z-C	BG100-C							
BG70Z-C BF70Z-C BK70Z-C BG80Z-C BF80Z-C BK80Z-C BG100Z-C BF90Z-C								
BG70-C BK70-C BF70-C	only from IEC200 up only from Nema324/326TC up	regreasable lubricant to be used: KLÜBER Petamo GHY133N Amount of grease for regreasing: approx. 40g (-> BA..) Amount of grease for renewal: approx 120g (-> BA..)						
BG80-C BK80-C BF80-C								
BG90-BK90Z-C BK90-BK90Z-C BF90-C								
BG100-C								
Lubrication quantity in litre								

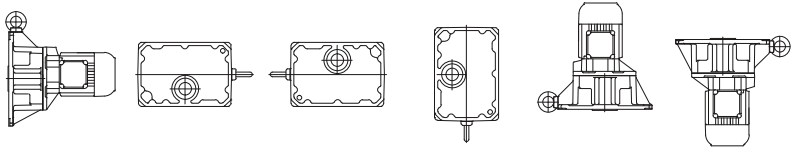
not regreasable

regreasable

lubricant to be used: KLÜBER Petamo GHY133N
Amount of grease for regreasing: approx. 40g (-> BA.)
Amount of grease for renewal: approx 120g (-> BA.)

Lubricants

Lubrication quantity for pre-stage



BF	H4	H1	H2	H3	V1	V2
BG						
BK	H1	V1	V2	H2	H4	H3
BS						
Gear type						
BF10Z BF10Z	0.10	0.05	0.12	0.07	0.16	0.07
BK10Z BS10Z						
BG20Z BF20Z	0.15	0.07	0.19	0.17	0.27	0.10
BK20Z BS20Z						
BG30Z BF30Z	0.2*	0.10	0.35	0.22	0.35	0.19
BK30Z BS30Z						
BM30Z BM30Z						
BG40Z BF40Z						
BK40Z BS40Z	0.32*	0.17	0.50	0.37	0.6	0.32
BM40Z						
BG50Z BF50Z	0.5	0.3	0.92	0.7	1.15	0.5
BK50Z						
BG60Z BF60Z	0.9	0.5	1.55	1.1	2.0	0.7
BK60Z						
BG70Z BF70Z	1.2	0.6	1.8	1.6	2.4	1.4
BK70Z BS70Z						
BG80Z BF80Z	3.1	1.3	4.0	2.6	5.2	2.0
BK80Z BS80Z						
BG90Z	4.2	1.5	5.4	3.5	7.7	3.0
BK90Z						
*: with BM30Z/BM40Z the pre-stage lubricant is filled via the main gearbox.						
Lubrication quantity in litre						

Lubricants

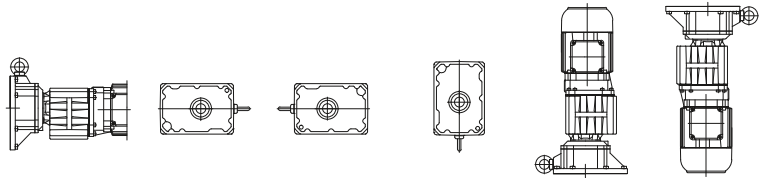
Lubrication quantity for intermediate gear

Definition of the terminal box position

Terminal box position for intermediate gear is similar to the main gearbox that means

Main gearbox BG, BF terminal box pos. I -> intermediate gearbox terminal box pos. I

Main gearbox BK, BS terminal box pos. II -> intermediate gearbox terminal box pos. II



Mounting position of main gearbox	BF	H4	H1	H2	H3	V1	V2
	BG						
	BK	H1	V1	V2	H2	H4	H3
	BS						

Type designation of double gearbox combination

BG06G04	BS06G04	0.03	0.03	0.03	0.03	0.05	0.05
BK06G04							
BG10G06	BF10G06	0.08	0.08	0.08	0.08	0.15	0.15
BK10G06	BS10G06						
BG20G06	BF20G06	0.08	0.08	0.08	0.08	0.15	0.15
BK20G06	BS20G06						
BG30G06	BF30G06	0.08	0.08	0.08	0.08	0.15	0.15
BK30G06	BS30G06						
BG40G10	BF40G10	0.65	0.65	0.65	0.85	1.05	0.85
BK40G10	BS40G10						
BG50G10	BF50G10	0.65	0.65	0.65	0.85	1.05	0.85
BK50G10							
BG60G20	BF60G20	0.8	0.8	0.8	1.1	1.4	1.1
BK60G20							
BG70G20	BF70G20	0.8	0.8	0.8	1.1	1.4	1.1
BK70G20							
BG80G40	BF80G40	1.7	1.7	1.7	2.5	3.3	2.1
BK80G40							
BG90G50	BF90G50	3.0	3.0	3.0	4.5	5.5	3.3
BK90G50	BG100G50						

Lubrication quantity in litre

Brakes

Spring-loaded brakes with direct current solenoid release Models E003B and E004B

Safety information

Connection, adjusting and maintenance work may only be carried out taking into account the safety information given on pages 4/5.



Caution:

Brakes are safety-relevant components; therefore, only qualified personnel with product-specific training may perform work on them. To find your nearest service partner, visit www.bauergears.com.

These spring loaded brakes are working brakes. The brakes perform frictional work in standard duty e.g. a braking function is performed.

In addition to holding loads in the idle state, the spring-loaded brake slows rotating and linear moving masses, thus reducing unwanted overtravel distances and times.

The brake is released electromagnetically. Under zero-load conditions, braking force is applied by spring pressure. Because braking is still effective even if an accidental power failure occurs, it can be considered a safety brake within the context of accident prevention regulations.

During the braking process, the kinetic energy of the mass moments of inertia is converted into heat via the brake disc. The brake disc, which consists of high-quality, asbestos-free material, is highly resistant to wear and heat. A certain amount of wear is unavoidable, however. For this reason, the limit values specified in paragraph MAINTENANCE regarding the working capacity and the minimum lining thickness are to be strictly observed.

Operating principle

The operating principle is described in Figure 1.

Brakes

The brake disc (1) is pressed axially through the retaining plate (2) against the friction plate (4) by springs (3). Radial movement of the retaining plate is prevented by the fillister screws (5). The braking torque is transferred to the rotor via gear teeth connecting the brake disc and the carrier (6) fixed to the shaft. The braking torque and the number of springs can be changed in stages (see paragraph 6).

Brake release

Supplying the coil (7) with the correct DC voltage causes the retaining disc to be attracted by the magnetic field generated in the magnet housing (8) against the spring force. This relieves the brake disc and as a result allows the rotor to move freely.

The increased air gap s_L caused by the wear to the brake discs can be overcome thanks to the generous dimensioning of the electromagnets. No adjustment facility is hence provided.

All brakes can be optionally fitted with either a latching or non-latching manual release, which may be used to release the brake manually e.g. in the event of a power failure.

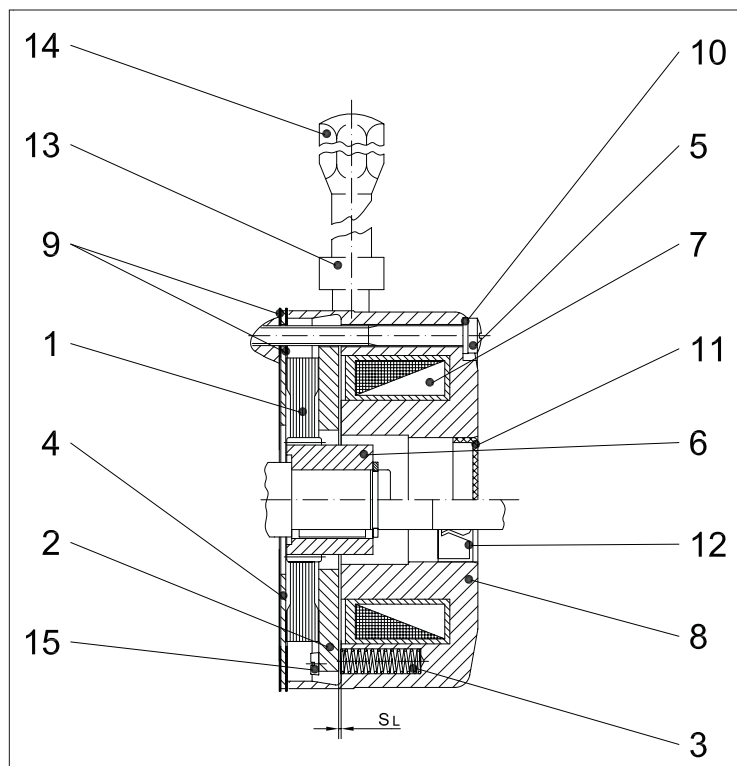


Figure 1: Spring-loaded brake from the series E003B and E004B

Assembly

Generally, the spring-loaded brakes are mounted ready for operation on the motor. Proceed as follows for retrofitting (see Figure 1):

- Fit carrier (6) to the shaft, pay attention to the total supporting length of the keys and fix axially with a retaining ring.
- Push friction plate (4) with both seals (9) and brake disc (1) onto the carrier manually. Ensure that the gearing moves easily.

Do not damage !

Observe the correct installation position of the friction plate (4): Side with engraved marking "Reibseite" (friction side) facing toward brake disc (1).

- Secure the brake (4) using the fillister screws (5) and the USIT rings (10) over the friction plate and both seals (9) on the end shield of the motor. Observe starting torque, $M_A = 2.5 \text{ Nm}$.
- For motor types without a second shaft end, fit a closure cap (11) and for motor types with a second shaft end, fit a shaft sealing ring (12).

The brake is ready for operation once the electrical connection has been made.

Setting of the brake torque



ATTENTION!

Bauer assumes no liability when the springs are changed.

Different braking torques can be obtained with a different spring configuration in the magnet housing (see paragraph MAINTENANCE).

Request the relevant set of springs from the factory, specifying the brake type and the required braking torque setting.

Procedure for changing the spring configuration (see Figure 1):

- 6.1 Remove brake from the motor end shield.
- 6.2 Remove fastening screws (5).
- 6.3 Unscrew the shoulder screws (15) from the magnet housing (8) and remove the retaining plate (2).



ATTENTION!

The springs (3) press against the retaining plate. To remove the shoulder screws, the retaining plate must be pressed against the magnet housing to avoid releasing the springs too quickly. Observe the installation position of the retaining plate and make sure that no springs fall out.

- 6.4 Insert springs (3) according to desired braking torque (see paragraph MAINTENANCE).



ATTENTION!

The springs should be arranged **symmetrically**.

- 6.5 Place the retaining plate (2) on the magnet housing (8) or springs (3) (observe installation position, if necessary use fastening screws (5) as centring assistance), press the retaining plate down against the spring force and screw in the shoulder screws (15) to the stop.
- 6.6 Secure the brake using the fastening screws (5) and USIT rings (10) above the friction plate (4) and both seals (9) on the end shield of the motor. Observe starting torque, $M_A = 2.5 \text{ Nm}$.

Maintenance

The E003B and E004B brakes are to a large extent maintenance-free, since a very long service life is obtained by the durable and wear resistant brake discs.

However, if the brake disc becomes worn due to high total friction and the function of the brake is therefore no longer guaranteed, replacing the brake disc will restore the brake to its original condition.

The state of wear of the brake disc should be checked regularly by measuring the brake disc thickness. This must not fall below the limit value.

Procedure for checking the state of wear and for replacing the brake disc (see Figure 1):

Brakes

Spring-loaded brakes with direct current solenoid release Models E003B and E004B

- Remove brake from the motor end shield.
- Remove fastening screws (5).
- Clean brake. Remove abrasion material using compressed air.
- Remove brake disc (1) from the carrier (6).
- Measure the thickness of the brake disc. At the latest, the brake disc is to be replaced when it reaches the minimum thickness.
- Check retaining plate (2) for wear and parallelism (there should be no significant grooving). Replace retaining plate if necessary.
- Push brake disc (1) onto carrier (6) and check for radial play. If there is increased play in the gear teeth between the carrier and brake disc, the carrier must be removed from the shaft and replaced.
- Secure the brake using the fastening screws (5) and USIT rings (10) over the friction plate (4) and both seals (9) on the end shield of the motor. Observe starting torque $M_A = 2.5 \text{ Nm}$.

Technical data

Type	M_{Br} [Nm]	NS	W_{max} [*10 ³ J]	W_{th} [*10 ³ J]	W_L [*10 ⁶ J]	t_A [ms]	t_{AC} [ms]	t_{DC} [ms]	d_{min} [mm]	P_{el} [W]
E003B9	3	4	1.5	36	55	35	150	15	5.85	20
E003B7	2.2	3	1.8	36	90	28	210	20	5.75	20
E003B4	1.5	2	2.1	36	140	21	275	30	5.6	20
E004B9	5	4x red	2.5	60	50	37	125	15	5.87	30
E004B8	4	4x grey	3	60	100	30	160	18	5.75	30
E004B6	2.8	4x yellow	3.6	60	180	23	230	26	5.55	30
E004B4	2	2x grey	4.1	60	235	18	290	37	5.4	30
E004B2	1.4	2x yellow	4.8	60	310	15	340	47	5.2	30

Explanation of abbreviations

M_{Br}	Nominal braking torque. Brake torque tolerance: -10 / +30 %
NS	Number of springs Because different springs can be used for the E004B, the colour of the relevant springs must also be specified here.
W_{max}	Maximum permissible switching energy for a single braking operation
W_{th}	Maximum permissible switching energy per hour
W_L	Maximum permissible switching until replacement of the brake disc

The values given for W_L are reference values which can vary considerably depending on the specific application. It is recommended to monitor the brake disc thickness regularly.

t_A	Response time when releasing with normal excitation. Overexcitation by the MSG special rectifier results in response times that are approximately half as long.
t_{AC}	Response time when braking with alternating current isolation, i.e. by interruption of the power supply of a separately fed standard rectifier. When the voltage supply of the rectifier results from the motor connection block, considerably higher response times are expected due to the residual magnetism of the motor - dependent on motor size and winding design.
t_{DC}	Response time when braking with direct current interruption by mechanical circuit breaker. Electronic direct-current interruption by a special rectifier (type ESG or MSG) results in response times that are approximately twice as long.

Dependent on the operating temperature and the state of wear of the brake disc, the actual response times (t_A , t_{AC} , t_{DC}) can deviate from the guide values indicated here.

d_{min}	Minimum permissible thickness of the brake disc
P_{el}	Electrical power consumption of the solenoid at 20 °C Depending on the voltage design of the coil, the actual power can differ from the reference value given.



ATTENTION!

The max. permissible frictional work and the min. permissible brake disc thickness are not valid for brake motors used in potentially explosive areas. Please refer here to the special corresponding documentation for explosion proof motors.

Brakes

Spring-loaded brakes with DC solenoid release

Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A

Safety information

Connection, adjusting and maintenance work may only be carried out taking into account the safety information on pages 4/5.



Caution:

Brakes are safety-relevant components; therefore, only qualified personnel with product-specific training may perform work on them. To find your nearest service partner, visit www.bauergears.com.

General information

The brakes ES010A ... ES250A, ZS300A and ZS500A are holding brakes which do not perform frictional work in standard duty but are used principally for holding a specifically reached position. In emergency situations a braking function may also be performed.

The brakes ESX010A ... ESX250A, ZSX300A and ZSX500A are working brakes which perform frictional work in standard duty e.g. a braking function is performed.

In addition to holding loads in the idle state, the spring-loaded brake slows rotating and linear moving masses, thus reducing unwanted overtravel distances and times.

The brake is released electromagnetically. In the de-energized state, braking force is applied by spring pressure. Because the retardation effect is still effective even if an unforeseen power failure occurs, it can be considered to be a safety brake within the context of accident prevention regulations.

During the braking process, the kinetic energy of the mass moments of inertia is transformed into heat via the brake discs. The brake discs consist of high-quality, asbestos-free material and are particularly resistant to abrasion and heat. A certain amount of wear is, however, unavoidable. For this reason, the limit values regarding the working capacity and the minimum lining thickness are to be strictly observed.

Brake mounting

ES and ESX: Brake mounting is under the fan cover

EH and EHx: Brake mounting is on the fan cover

Operating principle

Single disc brakes ES(X)...

The springs (Pos. 3) press the axial movable pressure plate (Pos. 6) and the brake disc (Pos. 2) which is mechanically connected to the rotor shaft against the friction disc or motor end shield. The brake torque is created.

Brakes

Spring-loaded brakes with DC solenoid release

Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A

When a DC voltage is placed on the coil in the magnet housing (Pos. 1), a magnetic force is created whereby the pressure plate (Pos. 6) is pulled against the spring force against the magnet housing (Pos. 1). The brake disc is free to run and the brake is opened.

Caution!

The air gap of the brake is not adjustable due to the construction principal of the magnet housing. When the wear limit or the max. permissible air gap is reached, the brake disc must be replaced. The momentarily existing air gap of the brake can be determined by means of a feeler gauge after removing the plug (Pos. 13). The plug must be replaced and sealed with locking varnish.

Brakes

Spring-loaded brakes with DC solenoid release

Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A

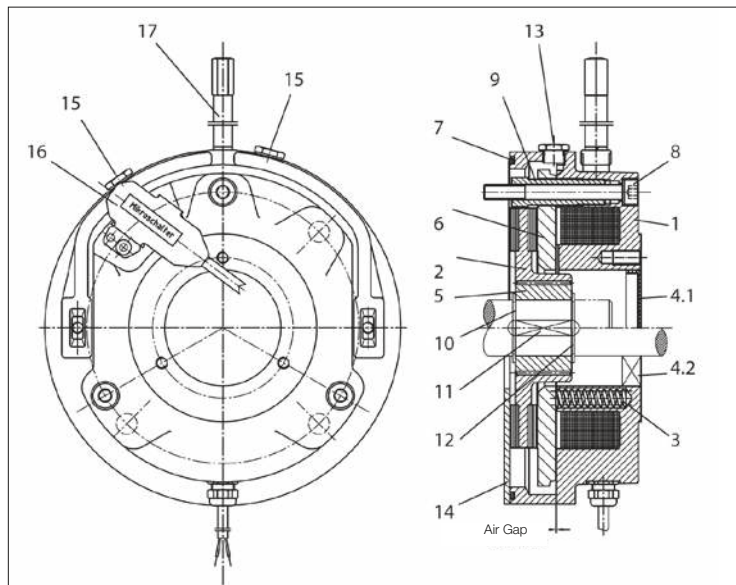


Figure 1: Brake Models ES(X)010A ... ES(X)250A.

- 1 Magnet housing
- 2 Brake disc
- 3 Spring
- 4.1 Sealing cap for encapsulated brakes
- 4.2 Shaft seal for 2nd shaft extension
- 5 Carrier
- 6 Pressure Plate
- 7 O-Ring
- 8 Fastening bolts with copper washer
- 9 Hollow bolt
- 10 Circlip
- 11 Key
- 12 Circlip
- 13 Plug for monitoring of the air gap
- 14 Friction plate - only in motors with aluminium end shield
- 15 Plug for monitoring of the micro switch setting
- 16 Microswitch (Optional)
- 17 Manual release (Optional)

Brakes

Spring-loaded brakes with DC solenoid release

Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A

Double disc brakes ZS(X)...

The springs (Pos. 3) press the axial movable pressure plate (Pos. 6) and the brake disc (Pos. 2.1 and 2.2) which are mechanically connected to the rotor shaft against the motor end shield. The brake torque is created.

When a DC voltage is placed on the coil in the magnet housing (Pos. 1), a magnetic force is created whereby the pressure plate (Pos. 6) is pulled against the spring force against the magnet housing (Pos. 1). The brake discs (Pos. 2.1 and 2.2) are free to run and the brake is opened.

When the wear limit or the max. permissible air gap is reached, the brake can be adjusted. When worn, the brake can be adjusted. The momentarily existing air gap can be monitored after removal of the sealing ring (Pos. 13).

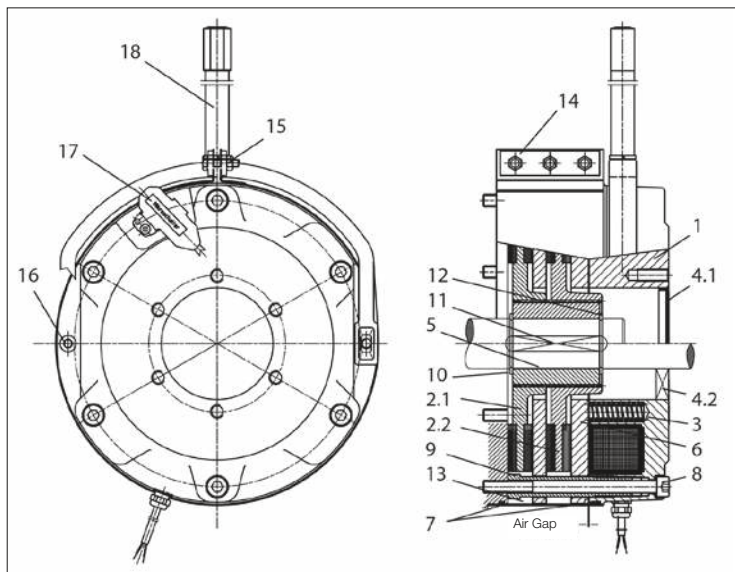


Figure 2: Brake models ZS(X)300A, ZS(X)500A.

Brakes

Spring-loaded brakes with DC solenoid release

Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A

- | | |
|-----|-------------------------------------|
| 1 | Magnet housing |
| 2 | Brake disc 2.1 and 2.2 |
| 3 | Spring |
| 4.1 | Sealing cap for encapsulated brakes |
| 4.2 | Shaft seal for 2nd shaft extension |
| 5 | Carrier |
| 6 | Pressure Plate |
| 7 | O-Ring |
| 8 | Fastening bolts with copper washer |
| 9 | Hollow bolt |
| 10 | Circlip |
| 11 | Key |
| 12 | Circlip |
| 13 | Cover |
| 14 | Fastening bolts |
| 15 | Plate |
| 16 | Assembly screw/Assembly help |
| 17 | Microswitch (Optional) |
| 18 | Manual release (Optional) |

Assembly



ATTENTION!

- Brake disc and brake surface must be free from oil and grease
- The carrier must be positioned so that the spline of the brake disc is in full contact.
- The contacting surface of the key should be along the full length of the carrier.
- Cleaning agents containing solvents may not be used as these corrode the friction material.

The brakes are assembled ready-for-use on the motor.

The assembly of the single disc brakes results as follows:

- Insert the circlip (Pos. 10) in the shaft groove.
- Insert the key (Pos. 11) in the rotor shaft.
- Position the carrier (Pos. 5) on the rotor shaft and secure with the circlip (Pos. 12).
- Assemble friction plate (Pos. 14) (only motor sizes D08 and D09).
- Push the brake disc (Pos. 2) onto the carrier (Pos. 5).
- Assemble the magnet housing with 3 fastening bolts (Pos. 8). Tightening torques refer to Brake Technical Data.

Caution!

Copper washers are found under the bolt heads. These may only be used once.

- The air gap of the brake is defined by the brake design.

Brakes

Spring-loaded brakes with DC solenoid release

Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A

The assembly of the double disc brakes results as follows:

- Insert circlip (Pos. 10) in the shaft groove.
- Insert key (Pos. 11) in the motor shaft.
- Position carrier (Pos. 5) on the shaft and secure with circlip (Pos. 12).
- Push brake disc (Pos. 2.1) onto the carrier (Pos. 5).
- Push the magnet housing with the intermediate flange and the brake disc (Pos. 2.2.) onto the carrier.
- Tighten the fastening bolts with the given torque. Tightening torque refer to Brake Technical Data.
- The brake air gap must be monitored. Values refer to Brake Technical Data

Caution!

Should the air gap not correspond to the given values, it must be set according to section Air Gap.

- Place the O-Rings (Pos. 7) in the corresponding grooves.
- Place the cover (Pos. 13) around the brake.
- Place the plate (Pos. 15) on the lugs of the covering plate.
- Press the cover plate with screws (Pos. 14) so far so that the cover plate lies radially on the magnet housing and the motor end shield.

Caution!

Copper washers are found under the bolt heads. These may only be used once.

Caution!

The brakes without manual release are supplied with an assembly aid which is to be removed after assembly.

The open holes in the magnet housing must be sealed with the plastic plugs supplied along with the brake.

Air Gap

General

The air gap must be monitored regularly. When the max. permissible air gap is reached (see technical data) the brake disc must be replaced or the air gap adjusted. Adjustment of the air gap is only possible for double disc brakes.

Monitoring the air gap of single disc brakes

The brake air gap (max. permissible air gap refer to Brake Technical Data) can be measured by means of a feeler gauge by removing the plug (Pos. 13). The plug must be replaced and sealed with locking varnish.

Caution!

The max. tightening torque of the plug is 10 Nm.

Monitoring of the air gap of double disc brakes

After removal of the cover (Pos. 13), the air gap can be measured by means of a feeler gauge.

Brakes

Spring-loaded brakes with DC solenoid release

Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A

Procedure:

- When looking at the brake, the fastening bolts (Pos. 8) must be released by half a turn.
- Now, the hollow bolts (Pos. 9), which enclose the fastening bolts (Pos. 8) can be screwed into the magnet housing (Pos. 1) by turning anticlockwise.
- Turning the 3 fastening screws (Pos. 8), positioned 120° from another, clockwise, the magnet housing can be moved in the direction of the pressure plate (Pos. 6) so that the nominal air gap measured with a feeler gauge is reached. Nominal air gap refer to Brake Technical Data.
- The 3 hollow bolts are now screwed out clockwise until they sit against the magnet housing (Pos. 1).
- Finally the fastening screws (Pos. 8) are tightened with the defined tightening torque given in the Brake Technical Data.
- The rest of the hollow bolts of the adjacent friction surface must be carefully tightened with the defined tightening torque.
- The brake air gap must now be rechecked. Values refer to Brake Technical Data.

Caution!

The manual release setting may not be changed.

Setting of the brake torque



ATTENTION!

Bauer assumes no liability when the springs are changed.

Single disc brakes ES(X)...

Different brake torques (see Brake Technical Data) can be reached by means of differing spring placement (see figure 7.1) in the magnet housing (Pos. 1).

When the conversion of the brake must result through the user, the required springs (Pos. 3) are to be requested giving the brake size and the brake setting.



ATTENTION!

The brake must be disassembled from the motor and under no load before the springs are changed.

Disassembly

- Release the fastening screws (Pos. 8) when present.
- Remove the manual release (optional).
- Unscrew the hollow bolts (Pos. 9) out of the magnet housing (Pos. 1).



ATTENTION!

The springs (Pos. 3) press against the pressure plate (Pos. 6). The pressure plate (Pos. 6) must be pressed against the magnet housing (Pos. 1) to avoid an abrupt relief of the springs (Pos. 3).

Brakes

Spring-loaded brakes with DC solenoid release

Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A

Assembly

- Remove the springs (Pos. 3).
- Insert the new spring set– positioning see figure 7.1 and figure 7.2

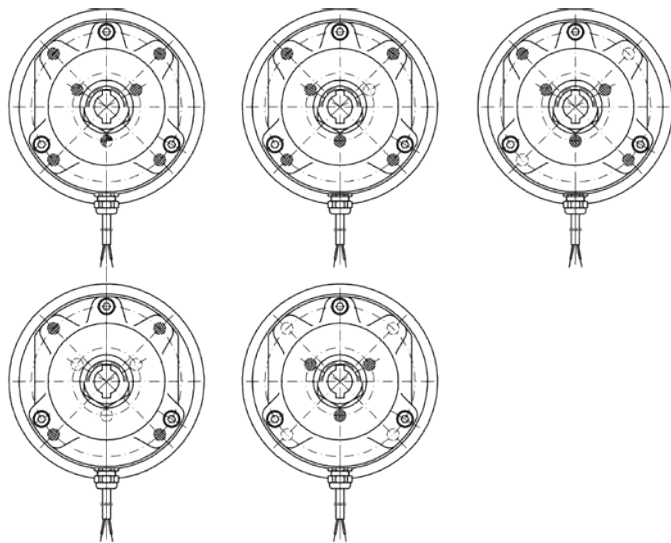


Figure 7.1 Positioning of the springs – Brakes ES(X)010A...ES(X)200A

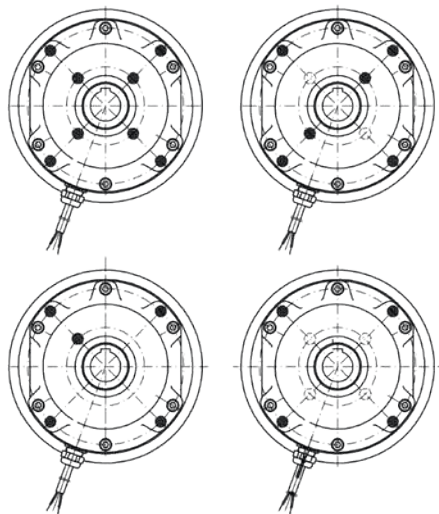


Figure 7.2 Positioning of the springs - Brake ES(X)250A

Brakes

Spring-loaded brakes with DC solenoid release

Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A



ATTENTION!

If a manual release was present, proceed with the assembly and setting according to the section Manual Release.

- Place the pressure plate (Pos. 6) on the springs (Pos. 3).



ATTENTION!

Observe the position of the bore for the manual release.

- Screw in the hollow bolts (Pos. 9) against the spring force until block.
- Assemble the brake on the motor end shield.



ATTENTION!

Refer to Brake Technical Data for the tightening torque of the fastening screws (Pos. 8).

Brake check

Checking the air gap

- Remove plug (Pos. 13).
- Check the air gap by means of a feeler gauge. Air gap refer to Brake Technical Data.
- Replace the plug (Pos. 13).

Double disc brakes ZS(X)...

The brakes ZS(X)300 and ZS(X)500 cannot be modified to other brake torques.

Replacing the brake disc

- Remove the motor and brake from the mains supply. Disconnect the brake supply cable.
- Loosen the fastening screws (Pos. 8) and remove the brake from the motor end shield.
- Clean the brake. Remove brake dust by means of compressed air.
- Remove the brake disc (Pos. 2) from the carrier (Pos. 5).
- Place the new brake disc on the carrier and re-assemble the brake.

Brakes

Spring-loaded brakes with DC solenoid release

Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A

Technical Data of the Holding Brakes with Emergency Characteristics Models ES.../ZS...

Frictional Work, Response Time, Power

Type	M _{Br}	W _{max}	W _{th}	W _L	t _A	t _{AC}	t _{DC}	P _{el}
	[Nm]	[10 ³ J]	[10 ³ J]	[10 ⁶ J]	[ms]	[ms]	[ms]	[W]
ES010AX	15*	3	-	-	110	-	30	35
ES010A9	10	3	-	-	60	100	15	
ES010A8	8	3	-	-	55	150	20	
ES010A5	5	3	-	-	45	220	20	
ES010A4	4	3	-	-	30	250	20	
ES010A2	2.5	3	-	-	25	350	25	50
ES027AX	32*	2.5	-	-	80	-	30	
ES/EH027A9	27	2.5	-	-	120	100	15	
ES/EH027A7	20	2.5	-	-	100	130	20	
ES/EH027A6	16	2.5	-	-	80	170	25	
ES/EH040A9	40	3.5	-	-	100	100	20	65
ES/EH040A8	34	3.5	-	-	80	200	25	
ES/EH040A7	27	3.5	-	-	70	250	30	
ES070AX	90*	3.5	-	-	120	-	40	85
ES070A9	70	3.5	-	-	120	150	18	
ES070A8	63	3.5	-	-	120	200	20	
ES070A7	50	3.5	-	-	90	220	25	
ES/EH125A9	125	4.5	-	-	170	220	25	105
ES/EH125A8	105	4.5	-	-	150	320	28	
ES/EH125A7	85	4.5	-	-	135	350	30	
ES/EH125A6	70	4.5	-	-	120	440	35	
ES125A5	57	4.5	-	-	100	600	40	
ES125A3	42	4.5	-	-	90	700	45	
ES/EH200A9	200	8	-	-	400	150	22	105
ES/EH200A8	150	8	-	-	280	250	35	
ES/EH200A7	140	8	-	-	200	320	35	
ES250A9	250	9	-	-	300	500	45	135
ES250A8	200	9	-	-	200	960	60	
ES250A6	150	9	-	-	160	1100	60	
ES250A5	125	9	-	-	150	1500	90	
ES250A4	105	9	-	-	130	1800	110	
ZS300A9	300	8	-	-	280	220	35	75
ZS300A8	250	8	-	-	210	380	45	
EH400A9	400	10	-	-	300	600	60	180
EH400A7	300	10	-	-	200	850	75	
EH400A5	200	10	-	-	150	1400	85	
ZS500A9	500	9	-	-	320	320	50	100
ZS500A8	400	9	-	-	260	600	60	
ZS800A9	800	20	-	-	400	550	65	140
ZS800A7	600	20	-	-	320	920	80	
ZS800A5	400	20	-	-	250	1450	100	

* only permissible with MSG-Rectifier, over excitation is required

Brakes

Spring-loaded brakes with DC solenoid release

Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A

Explanation of abbreviations

M_{Br}	Nominal braking torque. Brake torque tolerance: -10 / +30 %
W_{max}	Maximum permissible switching energy for a single braking operation
W_{th}	Maximum permissible switching energy per hour
W_L	Maximum permissible switching until replacement of the brake disc

No values given for W_{th} and W_L as the holding brakes in standard duty do not perform frictional work.

t_A	Response time when releasing with normal excitation. Overexcitation by the MSG special rectifier results in response times that are approximately half as long.
t_{AC}	Response time when braking with alternating current isolation, i.e. by interruption of the power supply of a separately fed standard rectifier. When the voltage supply of the rectifier results from the motor connection block, considerably higher response times are expected due to the residual magnetism of the motor - dependent on motor size and winding design.
t_{DC}	Response time when braking with direct current interruption by mechanical circuit breaker. Electronic direct-current interruption by a special rectifier (type ESG or MSG) results in response times that are approximately twice as long.

For the brake torque designs marked with *, which can only be reached by use of the MSG rectifier, the values for t_A and t_{DC} are already valid for use with the MSG rectifier i.e. t_A for over excitation and t_{DC} for electrical DC switching.

Depending on the operating temperature and the production tolerances, the actual response times can differ from the guidance values given here.

P_{el}	Electrical power consumption of the solenoid at 20 °C Depending on the voltage design of the coil, the actual power can differ from the reference value given.
----------	---

Brakes

Spring-loaded brakes with DC solenoid release

Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A

Spring placement, Air Gap, Tightening torque of the screws

Type	M _{Br} [Nm]	Number and colour of the springs		S _{LN} [mm]	S _{Lmax} [mm]		M _A [Nm]
		outside	inside		w/o HL	with HL	
ES010AX	15 ^{*1)}	Special		0.2	0.6	0.6	6
ES010A9	10	4 x yellow	3 x turquoise	0.2	0.6	0.6	
ES010A8	8	4 x yellow	3 x yellow	0.2	0.7	0.7	
ES010A5	5	2 x yellow	3 x yellow	0.2	1.0	1.0	
ES010A4	4	4 x yellow	-	0.2	1.2	1.0	
ES010A2	2.5	4 x silver	3 x silver	0.2	1.5	1.0	10
ES027AX	32 ^{*1)}	Special		0.3	0.6	0.6	
ES/EH027A9	27	4 x turquoise	3 x turquoise	0.3	0.6	0.6	
ES/EH027A7	20	4 x yellow	3 x turquoise	0.3	0.9	0.9	
ES/EH027A6	16	4 x turquoise	-	0.3	1.0	1.0	
ES/EH040A9	40	4 x silver	3 x silver	0.3	0.9	0.9	10
ES/EH040A8	34	4 x silver	2 x silver	0.3	1.1	1.0	
ES/EH040A7	27	2 x silver	3 x silver	0.3	1.3	1.0	
ES070AX	90*	4 x turquoise	3 x turquoise	0.3	1.0	1.0	25
ES070A9	70	2 x turquoise	3 x turquoise	0.3	0.8	0.8	
ES070A8	63	4 x yellow	3 x yellow	0.3	1.1	1.0	
ES070A7	50	4 x yellow	2 x yellow	0.3	1.5	1.0	
ES/EH125A9	125	4 x yellow	3 x turquoise	0.4	0.7	0.7	
ES/EH125A8	105	2 x yellow	3 x turquoise	0.4	1.2	1.2	25
ES/EH125A7	85	4 x yellow	2 x yellow	0.4	1.3	1.2	
ES/EH125A6	70	2 x yellow	3 x yellow	0.4	1.7	1.2	
ES125A5	57	4 x yellow	-	0.4	2.0	1.2	
ES125A3	42	-	3 x yellow	0.4	2.0	1.2	
ES/EH200A9	200 ¹⁾	Special		0.4	0.6	0.6	25
ES/EH200A8	150	4 x yellow	3 x yellow	0.4	0.9	0.9	
ES/EH200A7	140	4 x yellow	2 x yellow	0.4	1.2	1.2	
ES250A9	250	4 x yellow	4 x yellow	0.5	1.0	1.0	
ES250A8	200	4 x yellow	2 x yellow	0.5	1.7	1.5	
ES250A6	150	4 x yellow	1 x yellow	0.5	2.0	1.5	50
ES250A5	125	4 x yellow	-	0.5	2.4	1.5	
ES250A4	105	4 x silver	4 x silver	0.5	2.4	1.5	
ZS300A9	300	4 x yellow	3 x yellow	0.5	0.9	0.9	25
ZS300A8	250	2 x yellow	3 x yellow	0.5	1.1	1.1	
EH400A9	400	4 x yellow	4 x yellow	0.5	1.2	1.2	
EH400A7	300	4 x yellow	2 x yellow	0.5	1.5	1.5	50
EH400A5	200	4 x yellow	-	0.5	1.7	1.5	
ZS500A9	500	4 x yellow	4 x yellow	0.5	1.0	1.0	50
ZS500A8	400	4 x yellow	2 x yellow	0.5	1.2	1.2	
ZS800A9	800	4 x yellow	4 x yellow	0.6	1.2	1.2	
ZS800A7	600	4 x yellow	2 x yellow	0.6	1.2	1.2	50
ZS800A5	400	4 x yellow	-	0.6	1.2	1.2	

* only permissible with MSG-Rectifier, over excitation is required

¹⁾ Brake with special spring bore. Reworking to another brake torque is not possible.

Brakes

Spring-loaded brakes with DC solenoid release

Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A

Explanation of abbreviations

M_{Br}	Nominal braking torque. Brake torque tolerance: -10 / +30 %
S_{LN}	Nominal air gap when new. Tolerance +0,15 mm, except at EH400 and ZS800 Tolerance: +0,20 mm
S_{Lmax}	Maximum permissible air gap
HL	Manual Release
M_A	Tightening torque of the fastening screws



ATTENTION!

The max. permissible frictional work and the min. permissible brake disc thickness are not valid for brake motors used in potentially explosive areas. Please refer here to the special corresponding documentation for explosion proof motors.

Brakes

Spring-loaded brakes with DC solenoid release

Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A

Technical Data of the Working Brakes Models ES(X).../ZS(X)...

Frictional Work, Response Time, Power

Type	M _{Br} [Nm]	W _{max} [10 ³ J]	W _{th} [10 ³ J]	W _L [10 ⁶ J]		t _A [ms]	t _{ac} [ms]	t _{bc} [ms]	P _{el} [W]
				without HL	with HL				
ESX010AX	15*	3	250	120	120	110	-	30	35
ESX010A9	10	3	250	120	120	60	100	15	
ESX010A8	8	3	250	150	150	55	150	20	
ESX010A5	5	3	250	240	240	45	220	20	
ESX010A4	4	3	250	300	240	30	250	20	
ESX010A2	2.5	3	250	390	240	25	350	25	50
ESX027AX	27*	10	350	150	150	80	-	30	
ESX/EHX027A9	22	10	350	150	150	120	100	15	
ESX/EHX027A7	16	10	350	300	300	100	130	20	
ESX/EHX027A6	13	10	350	350	350	80	170	25	
ESX/EHX040A9	32	20	450	420	420	100	100	20	65
ESX/EHX040A8	27	20	450	560	490	80	200	25	
ESX/EHX040A7	22	20	450	700	490	70	250	30	
ESX070AX	72*	28	550	700	700	120	-	40	
ESX070A9	58	28	550	500	500	120	150	18	85
ESX070A8	50	28	550	800	700	120	200	20	
ESX070A7	40	28	550	1200	700	90	220	25	
ESX/EHX125AX	100*	40	700	1900	1900	100	-	70	
ESX/EHX125A9	85	40	700	1700	1700	150	320	28	105
ESX/EHX125A8	70	40	700	1900	1700	135	350	30	
ESX/EHX125A7	58	40	700	2700	1700	120	440	35	
ESX125A5	45	40	700	3300	1700	100	600	40	
ESX125A3	34	40	700	3300	1700	90	700	45	
ESX/EHX200AX	160*	60	850	2000	2000	105	-	70	105
ESX/EHX200A9	120	60	850	1700	1700	280	250	35	
ESX/EHX200A8	110	60	850	2600	2600	200	320	35	
ESX250A9	200	84	1000	2800	2800	300	500	45	135
ESX250A8	160	84	1000	6800	5700	200	960	60	
ESX250A6	120	84	1000	8500	5700	160	1100	60	
ESX250A5	100	84	1000	11000	5700	150	1500	90	
ESX250A4	85	84	1000	11000	5700	130	1800	110	
ZSX300A9	250	60	850	1300	1300	280	220	35	75
ZSX300A8	200	60	850	2000	2000	210	380	45	
EHX400A9	320	120	1100	3000	3000	300	600	60	180
EHX400A7	240	120	1100	4800	4800	200	850	75	
EHX400A5	160	120	1100	6000	4800	150	1400	85	
ZSX500A9	400	84	1000	2800	2800	320	320	50	100
ZSX500A8	320	84	1000	4000	4000	260	600	60	
ZSX800A9	640	120	1150	1550	1550	400	550	65	
ZSX800A7	480	120	1150	1550	1550	320	920	80	140
ZSX800A5	320	120	1150	1550	1550	250	1450	100	

* only permissible with MSG-Rectifier, over excitation is required

Brakes

Spring-loaded brakes with DC solenoid release

Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A

Explanation of abbreviations

M_{Br}	Nominal braking torque. Brake torque tolerance: -20 / +30 % when worn in. In new state up to -30 % possible
W_{max}	Maximum permissible switching energy for a single braking operation
W_{th}	Maximum permissible switching energy per hour
W_L	Maximum permissible switching until servicing i.e. replacement of the brake disc or air gap adjustment. Adjustment of the air gap only possible for ZS(X) brakes possible
HL	Manual release

The values given for W_L are reference values which can vary considerably depending on the specific application. It is recommended to monitor the brake disc thickness regularly.



ATTENTION!

The max. permissible frictional work and the min. permissible brake disc thickness are not valid for brake motors used in potentially explosive areas. Please refer here to the special corresponding documentation for explosion proof motors.

t_A	Response time when releasing with normal excitation. Overexcitation by the MSG special rectifier results in response times that are approximately half as long.
t_{AC}	Response time when braking with alternating current isolation, i.e. by interruption of the power supply of a separately fed standard rectifier. When the voltage supply of the rectifier results from the motor connection block, considerably higher response times are expected due to the residual magnetism of the motor - dependent on motor size and winding design.
t_{DC}	Response time when braking with direct current interruption by mechanical circuit breaker. Electronic direct-current interruption by a special rectifier (type ESG or MSG) results in response times that are approximately twice as long.

For the brake torque designs marked with *, which can only be reached by use of the MSG rectifier, the values for t_A and t_{DC} are already valid for use with the MSG rectifier i.e. t_A for over excitation and t_{DC} for electronical DC switching.

Depending on the operating temperature, wear of the brake disc and the production tolerances, the actual response times can differ from the guidance values given here.

P_{el}	Electrical power consumption of the solenoid at 20 °C Depending on the voltage design of the coil, the actual power can differ from the reference value given.
----------	---

Brakes

Spring-loaded brakes with DC solenoid release

Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A

Spring placement, Air Gap, Tightening torque of the screws

Type	M _{Br} [Nm]	Number and colour of the springs		S _{LN} [mm]	S _{Lmax} [mm]		M _A [Nm]
		outside	inside		w/o HL	with HL	
ESX010AX	15* ¹⁾	Special		0.2	0.6	0.6	6
ESX010A9	10	4 x yellow	3 x turquoise	0.2	0.6	0.6	
ESX010A8	8	4 x yellow	3 x yellow	0.2	0.7	0.7	
ESX010A5	5	2 x yellow	3 x yellow	0.2	1.0	1.0	
ESX010A4	4	4 x yellow	-	0.2	1.2	1.0	
ESX010A2	2.5	4 x silver	3x silver	0.2	1.5	1.0	10
ESX027AX	27* ¹⁾	Special		0.3	0.6	0.6	
ESX/EHX027A9	22	4 x turquoise	3 x turquoise	0.3	0.6	0.6	
ESX/EHX027A7	16	4 x yellow	3 x turquoise	0.3	0.9	0.9	
ESX/EHX027A6	13	4 x turquoise	-	0.3	1.0	1.0	
ESX/EHX040A9	32	4 x silver	3 x silver	0.3	0.9	0.9	10
ESX/EHX040A8	27	4 x silver	2 x silver	0.3	1.1	1.0	
ESX/EHX040A7	22	2 x silver	3 x silver	0.3	1.3	1.0	
ESX070AX	72*	4 x turquoise	3 x turquoise	0.3	1.0	1.0	25
ESX070A9	58	2 x turquoise	3 x turquoise	0.3	0.8	0.8	
ESX070A8	50	4 x yellow	3 x yellow	0.3	1.1	1.0	
ESX070A7	40	4 x yellow	2 x yellow	0.3	1.5	1.0	
ESX/EHX125AX	100*	4 x yellow	3 x turquoise	0.4	1.3	1.3	25
ESX/EHX125A9	85	2 x yellow	3 x turquoise	0.4	1.2	1.2	
ESX/EHX125A8	70	4 x yellow	2 x yellow	0.4	1.3	1.2	
ESX/EHX125A7	58	2 x yellow	3 x yellow	0.4	1.7	1.2	
ESX125A5	45	4 x yellow	-	0.4	2.0	1.2	
ESX125A3	34	-	3x yellow	0.4	2.0	1.2	
ESX/EHX200AX	160* ¹⁾	Special		0.4	1.0	1.0	25
ESX/EHX200A9	120	4 x yellow	3 x yellow	0.4	0.9	0.9	
ESX/EHX200A8	110	4 x yellow	2 x yellow	0.4	1.2	1.2	
ESX250A9	200	4 x yellow	4 x yellow	0.5	1.0	1.0	
ESX250A8	160	4 x yellow	2 x yellow	0.5	1.7	1.5	50
ESX250A6	120	4 x yellow	1 x yellow	0.5	2.0	1.5	
ESX250A5	100	4 x yellow	-	0.5	2.4	1.5	
ESX250A4	85	4 x silver	4 x silver	0.5	2.4	1.5	
ZSX300A9	250	4 x yellow	3 x yellow	0.5	0.9	0.9	25
ZSX300A8	200	2 x yellow	3 x yellow	0.5	1.1	1.1	
EHX400A9	320	4 x yellow	4 x yellow	0.5	1.2	1.2	
EHX400A7	240	4 x yellow	2 x yellow	0.5	1.5	1.5	50
EHX400A5	160	4 x yellow	-	0.5	1.7	1.5	
ZSX500A9	400	4 x yellow	4 x yellow	0.5	1.0	1.0	50
ZSX500A8	320	4 x yellow	2 x yellow	0.5	1.2	1.2	
ZSX800A9	640	4 x yellow	4 x yellow	0.6	1.2	1.2	
ZSX800A7	480	4 x yellow	2 x yellow	0.6	1.2	1.2	50
ZSX800A5	320	4 x yellow	-	0.6	1.2	1.2	

* only permissible with MSG-Rectifier, over excitation is required

¹⁾ Brake with special spring bore. Reworking to another brake torque is not possible.

Brakes

Spring-loaded brakes with DC solenoid release

Models ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A, EH(X)027A ...EH(X)400A

Explanation of abbreviations

M_{Br}	Nominal braking torque. Brake torque tolerance: -20 / +30 % when worn in. In new state up to -30 % possible
S_{LN}	Nominal air gap when new. Tolerance +0,15 mm, except at EH(X)400 and ZS(X)800 Tolerance: +0,20 mm
S_{Lmax}	Maximum permissible air gap
HL	Manual Release
M_A	Tightening torque of the fastening screws



ATTENTION!

The max. permissible frictional work and the min. permissible brake disc thickness are not valid for brake motors used in potentially explosive areas. Please refer here to the special corresponding documentation for explosion proof motors.

General

There are 2 different options for the voltage supply of the DC solenoid:

1. Externally from an existing DC control power supply, or from a rectifier in the control cabinet.
2. From a rectifier built into the motor terminal box or brake terminal box. In this instance, the rectifier can be supplied either directly from the motor terminal board or from the mains.

In the following instances, the rectifier must not however be connected to the motor terminal board:

- Pole-changing motors and wide range motors
- Operation with frequency converter
- Other designs in which the motor voltage is not constant, e.g. operation on smooth-start equipment, starting transformers, ...

Brake Release

If the solenoid is energised using nominal voltage, the coil current and the magnetic field build up follow an exponential function. Only when the power has achieved a specific value (I_{Rel}) is the spring force overcome and the brake begins to release.

Two different cases can occur during response time t_A , assuming the power is supplied to the motor and the brake at the same time:

- Motor is blocked - condition: $M_A < M_L + M_{Br}$
The motor takes the inrush current and is thereby subjected to additional thermal loading.
This scenario is shown in Figure 2.
- Motor tears away from brake - condition: $M_A > M_L + M_{Br}$
At the start, the brake is subject to thermal loading and wears more rapidly.

M_A : starting torque of the motor, M_L : load torque, M_{Br} : braking torque

In both scenarios, the motor and brake are therefore subjected to additional loading. The response time becomes noticeably longer as the size of the brakes increases. A reduction in response time is therefore especially recommended for medium and large-sized brakes as well as with a high frequency of braking operations. A relatively simple electrical solution is possible using the principle of 'overexcitation'. In this instance, the solenoid is briefly supplied with twice the nominal voltage when switched on.

The response time is decreased to approximately half by comparison with 'normal excitation' as a result of the associated steeper rise of the current. This overexcitation function is integrated in the MSG special rectifier (see section brake connection).

As the air gap gets larger, the release current and therefore the response time increase. As soon as the release current exceeds the nominal coil current, the brake no longer releases during normal excitation and the brake disc wear limit is reached.

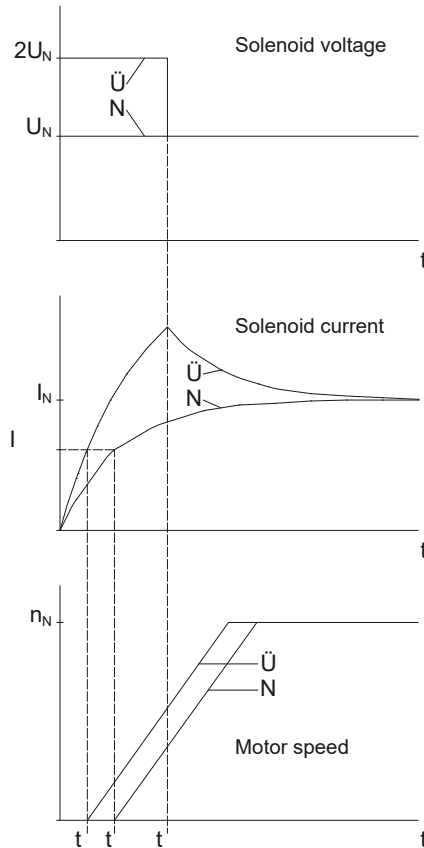


Figure 2: Solenoid voltage, solenoid current and motor speed development at normal excitation (N) and overexcitation (Ü).

t_0 : overexcitation period; t_{AN} , t_{AU} : response times at normal excitation and overexcitation

Braking

The braking torque is not effective immediately after the power supply to the solenoid is switched off. Firstly, the magnetic energy has to reduce until the spring force can overcome the magnetic force. This occurs at holding amperage I_{Hold} which is far smaller than the release current. Dependent on the circuit design, different response times result.

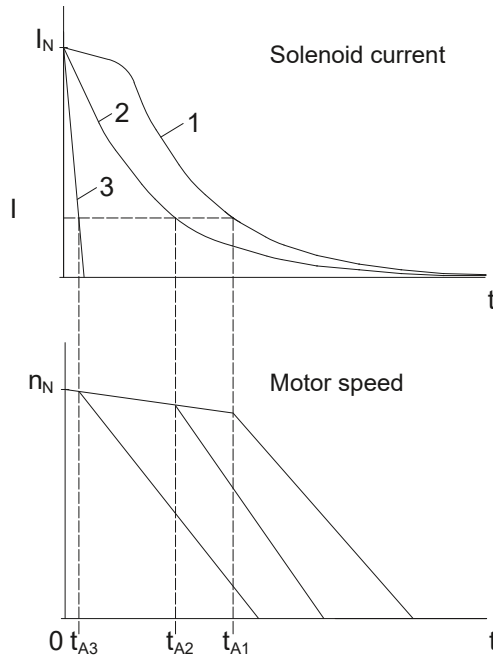
Switching off the AC supply of the standard rectifier SG

- a) Rectifier supply from the motor terminal board (Figure 3, graph 1)
Response time t_{A1} : very long
Cause: after the motor voltage is switched off, the remanence of the motor induces a slowly decaying voltage which continues to supply the rectifier and thus the brake. The magnetic energy of the brake solenoid declines relatively slowly through the freewheeling circuit of the rectifier.
- b) Separate rectifier supply (Figure 3, graph 2)
Response time t_{A2} : long
Cause: after the rectifier voltage is switched off, the magnetic energy of the brake solenoid declines relatively slowly through the freewheeling circuit of the rectifier.

No significant shut-off voltages arise on the solenoid during an AC interruption.

Interruption in the DC switching circuit of the solenoid (Figure 3, graph 3)

- a) By mechanical switch
- for separate supply from a DC control network or
- at the DC switching contacts (A2, A3) of the standard rectifier
Response time t_{A3} : very short
Cause: The magnetic energy of the brake solenoid is rapidly reduced by the arc developing at the switch.
- b) Electronically
By use of a special rectifier, type ESG or MSG Response time t_{A3} : short
Cause: the magnetic energy of the brake solenoid is reduced rapidly by a varistor integrated in the rectifier.



With a DC interruption, voltage spikes u_q are induced through the solenoid the height of which depends on the following relationship between the self-inductance L of the coil and the cut-off speed di/dt :

$$u_q = L \cdot \frac{di}{dt}$$

As a consequence of the winding design, inductivity L increases as the solenoid rated voltage increases. With higher solenoid voltages, the cut-off voltage spikes may therefore be dangerously high. All brakes for voltages in excess of 24 V are therefore connected with a varistor.

The varistor serves only to protect the solenoid and not as protection for the surrounding electronic components and devices against electromagnetic compatibility interference.

On request, brakes for voltages of less than or equal to 24 V can also be produced with varistors.

If the direct current interruption is produced by a mechanical switch, high levels of burn down are caused by the arc produced on the switch contacts. Only special DC direct current contactors or adapted AC alternating current contactors can be used with contacts of usage category AC3 as defined in EN 60947-4-1.

Rectifier technical data

Operating principle	Half-wave rectifier with electronic direct current interruption
Supply voltage U_1	220 - 460 V AC $\pm 5\%$, 50/60 Hz
Output voltage	$0.45 \cdot U_1$ V DC
Maximum output current	1 A DC
Ambient temperature	-20 °C to 40 °C
Possible conductor cross-sections	maximum 1.5 mm ²

The blue conductor routed out of the casing must be connected to PE to activate the integrated high-speed switch-off function.

As this conductor is coupled to the supply voltage with high impedance, leakage currents of up to a maximum of 2 mA may flow, depending on the voltage level.

If the motor undergoes high-voltage testing, the blue wire of the rectifier must be disconnected from the PE beforehand.

When operating on unearthed networks, the blue conductor is to be connected with the right alternating current voltage contact (N) of the ESG. If the rectifier is supplied from the motor terminal board in this case, an increase in the response time on shut-down is to be anticipated.

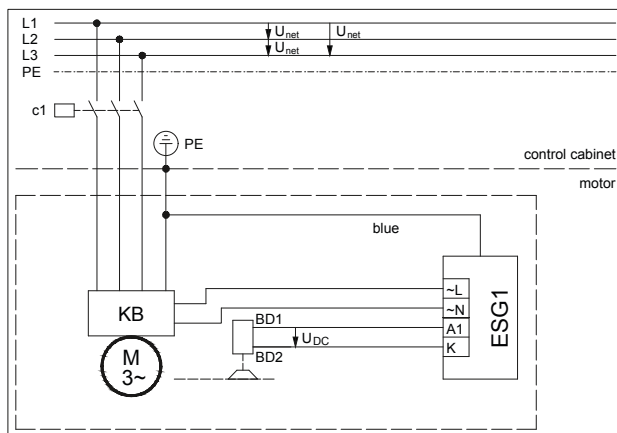
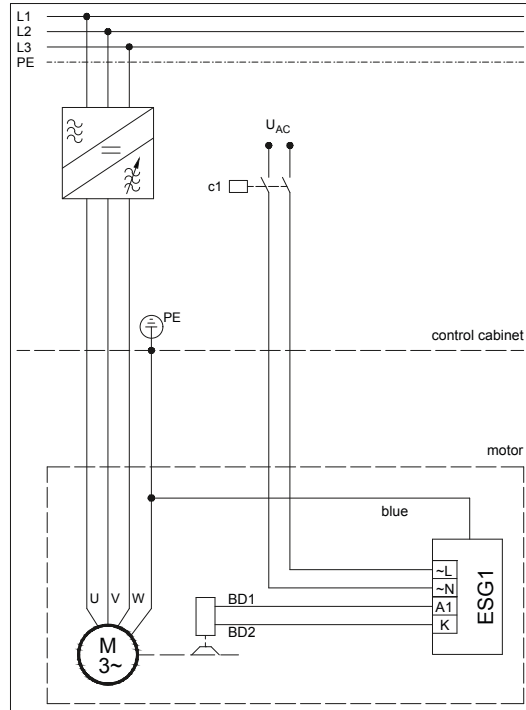


Figure 8: Rectifier voltage supply from the motor terminal board Voltage connection for the rectifier from the motor terminal block or cage clamp (see Rectifier Connection on Motor Terminal Block or Cage Clamp)

Brakes

Brake connection: special rectifier ESG 1.460A



Picture 8a: Separate voltage supply of the rectifier, e.g. for usage on frequency inverter

If the brake is supplied directly from a DC control power supply.

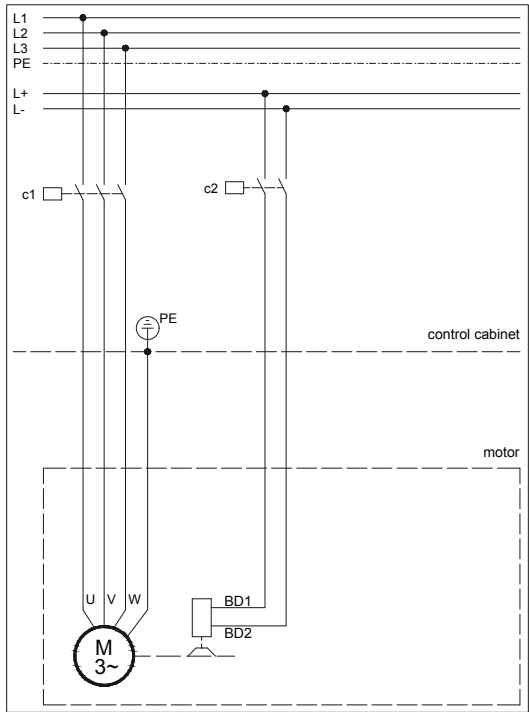


Figure 4: Direct DC power supply from a control network

Rectifier technical data MSG 1.5.480I

Operating principle	Half-wave rectifier with time-limited overexcitation and electronic direct current interruption Quick switch-off due to absence of motor current in a phase.
Operating voltage U_1	220 - 480 V AC +6/-10 %, 50/60 Hz
Output voltage	$0.9 \cdot U_1$ V DC during overexcitation $0.45 \cdot U_1$ V DC after overexcitation
Overexcitation period	0.3 s
Maximum output current	1,5 A DC
Ambient temperature	-20 °C to 40 °C
Possible conductor cross-sections	maximum 1.5 mm ²

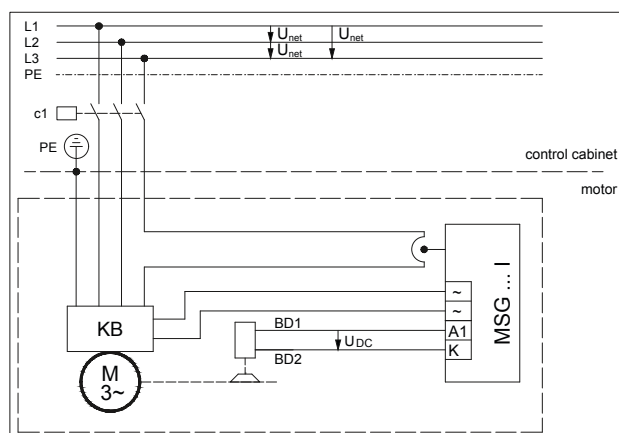
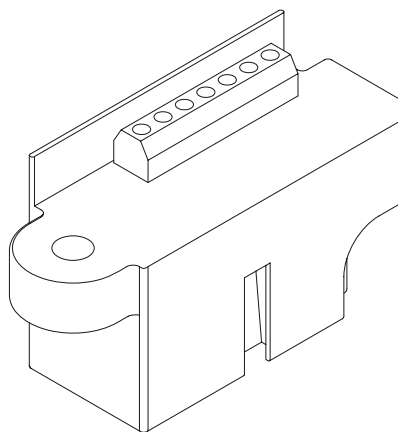


Figure 10: Voltage connection for the rectifier from the motor terminal block or cage clamp (see Rectifier Connection on Motor Terminal Block or Cage Clamp)

To enable the current monitoring, one of the leads of the supply cable must be passed through the current sensor on the side of the rectifier. As the current sensing has a lower threshold, the lead must be passed through the sensor twice if the motor no-load current is lower than 0,4A. In this case a sticker with the number "2" is on the rectifier underneath the sensor itself. The maximum constant current of the sensor is 64A.

**Caution:**

It is imperative for the correct functioning of the rectifier that a motor lead is passed through the sensor. If not, the rectifier will not switch on and in the worst case can be permanently damaged.



The diameter of the sensor bore for passing the leads through is 7 mm. The diameter of the leads of the motor connection cable must therefore not exceed the following values:

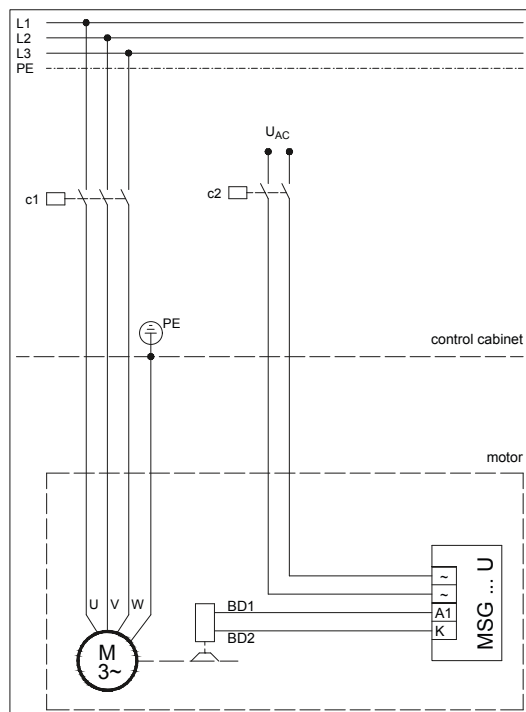
max. lead diameter:

6,7 mm for 1 passthrough

3,2 mm for 2 passthroughs

Rectifier technical data MSG 1.5.500U

Operating principle	Half-wave rectifier with time-limited overexcitation and electronic direct current interruption Quick shut-down due to absence of input voltage.
Operating voltage U_1	220 - 500 V AC $\pm 10\%$, 50/60 Hz
Output voltage	$0.9 \cdot U_1$ V DC during overexcitation $0.45 \cdot U_1$ V DC after overexcitation
Overexcitation period	0.3 s
Maximum output current	1,5 A DC
Ambient temperature	-20 °C to 40 °C
Possible conductor cross-sections	maximum 1.5 mm ²



Picture 9: Separate voltage supply of the rectifier

Rectifier technical data

Operating principle	Half-wave rectifier
Supply voltage U_1	maximum 575 V AC +5 %, 50/60 Hz
Output voltage	$0.45 \cdot U_1$ V DC
Maximum output current	2 A DC
Ambient temperature	-40 °C to 40 °C
Possible conductor cross-sections	max. 1.5 mm ² without wire end ferrule max. 1.0 mm ² with wire end ferrule

1 Rectifier voltage supply from the motor terminal board Voltage connection for the rectifier from the motor terminal block or cage clamp (see Rectifier Connection on Motor Terminal Block or Cage Clamp)

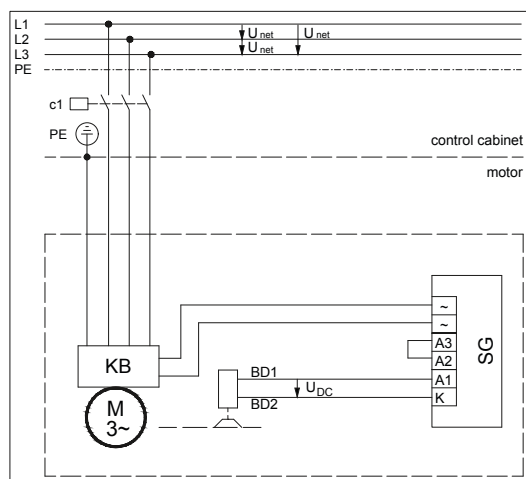


Figure 5: AC disconnection → Terminal A2 and A3 bridged

Brakes

Brake connection: standard rectifier SG 3.575B

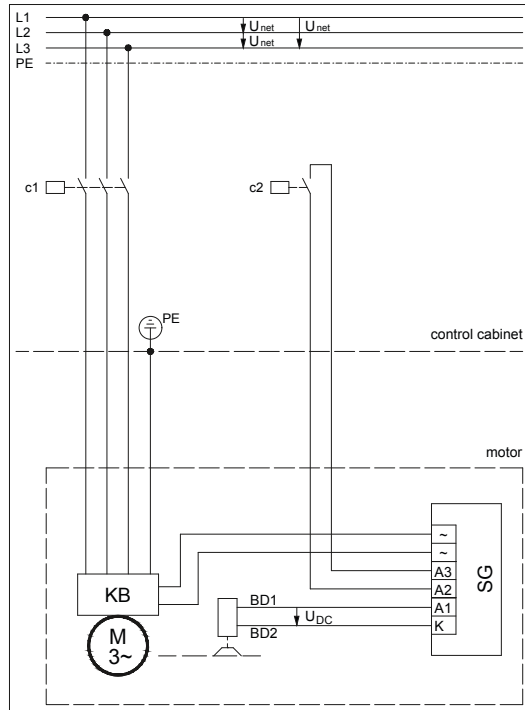
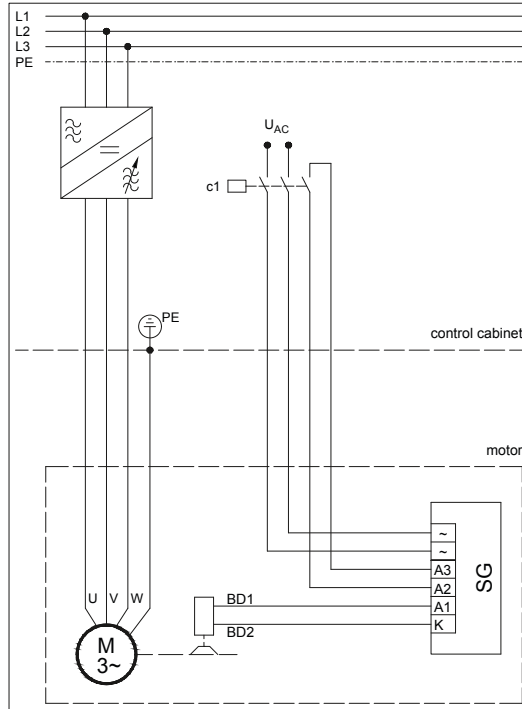


Figure 6: DC disconnection at terminals A2 and A3 e.g. via direction of rotation contactor via relay

Brakes

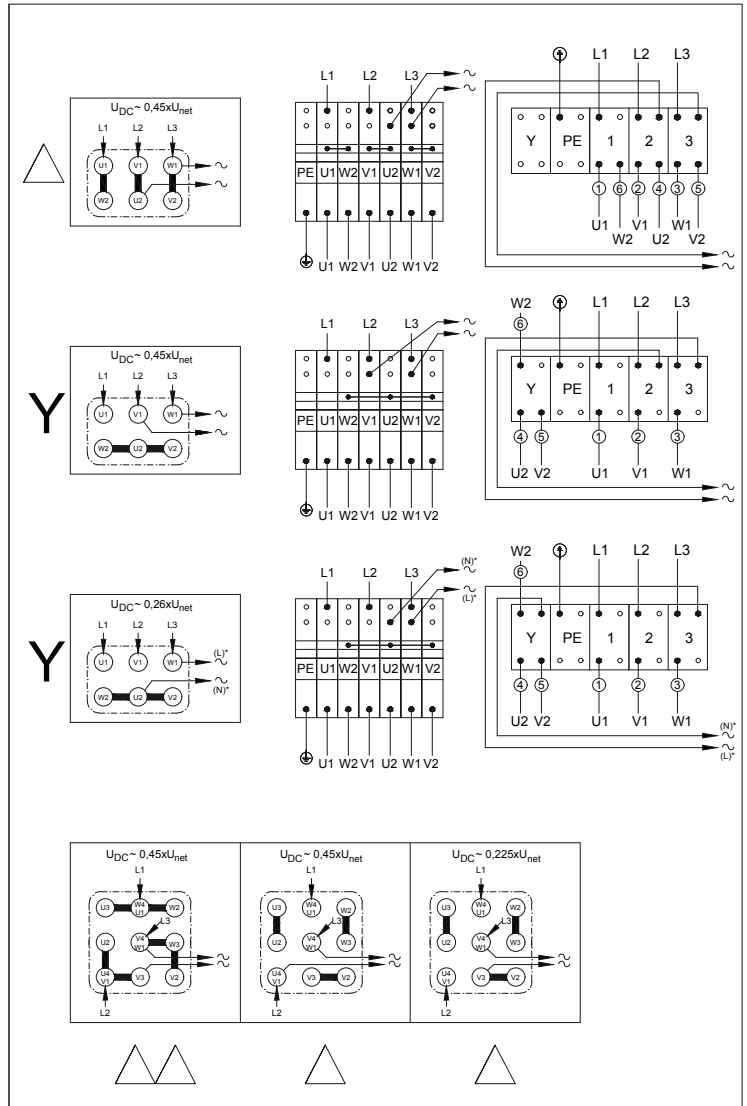
Brake connection: standard rectifier SG 3.575B



Picture 7a: Separate voltage supply of the rectifier. DC side switching on the terminals A2 and A3 via relay.

Brakes

Rectifier on motor terminal block or cage clamp



* Connection at the rectifier ESG1.460A

Brakes

Manual release for spring loaded brake with DC solenoid Model E003B and E004B

Assembly

The manual release can only be assembled with the brake removed.

Procedure (see Figures 1 and 12 in section spring loaded E003B and E004B):

- 1.1 Remove brake from the motor end shield.
- 1.2 Remove stopper plugs from the manual-release holes in the magnet housing (8).
- 1.3 Push compression springs (16) onto the manual-release bolts (17).
- 1.4 Push manual-release bolts (17) with compression springs (16) into the manual-release holes on the magnet housing (8) from the inside (in the direction of the coil (7)).
- 1.5 Push the O-rings (18) onto manual-release bolts (17) and push into the countersinks on the magnet housing (8).
- 1.6 Push spacer plates (19) onto the manual-release bolts (17).
- 1.7 Locate manual release bracket (13), push on washer (20) and screw on self-locking nut (21) loosely.
- 1.8 Tighten both lock nuts (21) until the retaining plate (2) is flush with the magnet housing (8).
- 1.9 With non-lockable manual release:
Unscrew both lock nuts (21) by 1.5 turns, thereby creating the air gap between the retaining plate (2) and magnet housing (8) and the test dimension $X = 0.9 \text{ mm}$.
With latching manual release:
Unscrew both lock nuts (21) by 3 turns, thereby creating the test dimension $X = 2 \text{ mm}$.
- 1.10 After fitting the fan cowl, screw the manual-release rod (14) into manual-release bracket (13) and tighten.

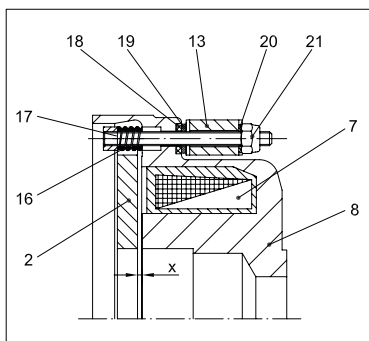


Fig 12: Assembly of the manual release

Function

The manual release bracket (13) is pressed by the compression springs (16) into the neutral position. The brake can be released by axial manipulation. For the model with a latching manual release, the manual release bracket is fixed by screwing the manual release rod (14) into the appropriate bore in the brake housing with the brake released.

To release the latch, turn the manual release rod back again.

Brakes

Manual release for spring loaded brake with DC solenoid

Model ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A

The hand release allows the manual release of the brake. By operating the manual release, the pressure plate (Pos. 6) is pulled against the magnet housing (Pos. 1). An air gap between the brake disc (Pos. 2) and the pressure plate (Pos. 6) results, the brake is open and the rotor shaft can rotate freely.



ATTENTION!

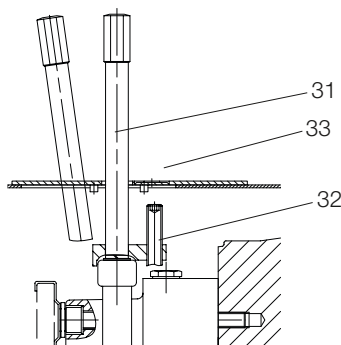
The settings of the manual release may not be changed.

The manual release lever and locking arrangement are supplied separately as a by-pack kit.

Assembly of the manual release lever

Screw the manual release lever Pos. 31 and the locking arrangement Pos. 32 (if applicable) into the manual release. For brake types ES(X) and ZS(X) insert the protective guard Pos. 33 over the lever into the fan cowl slot. If the locking arrangement is present, break out the perforation in the protective guard.

ES(X)...A / EH(X)...A/ZS(X)...A	Lever thread	Driving torque reference value [Nm]
010	M5	5
027/040	M6	8
070/125/200/300	M8	18
250/400/500/800	M12	25



Caution:

Manual release lever and grip protection in the accessory kit. If the manual release lever (Pos. 31) and grip protection (Pos. 33) are not installed together, the operator must allow for suitable contact protection at the opening in the fan cowl provided for movement of the manual release lever.

Assembly and disassembly of the manual release

The assembly of the manual release is only possible when the brake was ordered specifically for this option.

When changing the brake torque, the manual release must be re-assembled as follows:

Caution!

Before assembly of the manual release, the brake must be disassembled from the motor and electrically disconnected!

- Before the pressure plate is assembled, the O-Rings (Pos. 20) and the discs (Pos. 21) must be placed in the in the countersinks in the magnet housing.
- Assemble the pressure plate (Pos. 6) with the hollow bolts (Pos. 9).

Caution!

The hollow bolts must lie under the sealing face of the magnet housing.

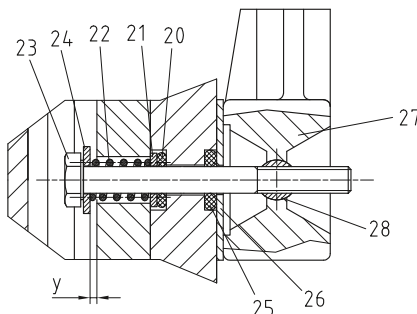
Brakes

Manual release for spring loaded brake with DC solenoid

Model ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A

Setting of the manual release

- Push the washer (Pos. 24) and then the spring (Pos. 22) onto the bolt (Pos. 23). Then push all parts, from the inside looking at the sealed coil, into the bores in the pressure plate and magnet housing (Pos. 1).
- Push the O-Rings (Pos. 25) over the threads of the mounting bolts (Pos. 23) and press them into the countersink of the magnet housing (Pos. 1).



- Push the plate (Pos. 26) over the screw thread.
- Push the bolts (Pos. 28) into the manual release lever.
- Screw the screw (Pos. 23) into the bolt (Pos. 28).
- Tighten both hexagon bolts (Pos. 23) until the pressure plate (Pos. 6) lies uniformly on the magnet housing (Pos. 1).

- Release both hexagonal bolts (Pos. 23) by X turns (see Table Setting the Manual Release). The adjusting dimension „y“ for the manual release is set this way.



ATTENTION!:

The adjusting dimension „y“ must be set uniformly and may not be readjusted later.

- Mark the setting using locking varnish on the hand release bolts (Pos. 28).
- After assembly of the fan cowl, screw the manual release lever in the manual release and tighten.

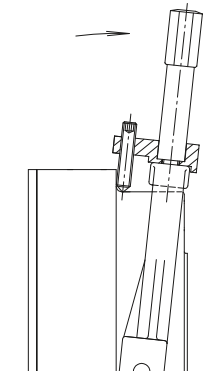
Type	Setting the Manual Release		
	Adjusting Dimension Y [mm]	"Unscrew bolts by X turns"	Wrench size Manual release lever [mm]
ES(X)010	1	1.5	8
ES(X)027/EH(X)027	1	1.5	10
ES(X)040/EH(X)070	1	1.3	10
ES(X)070	1	1	12
ES(X)125/EH(X)125	1.2	1.2	12
ES(X)200/EH(X)200	1.2	1.2	12
ES(X)250	1.5	1.2	19
ZS(X)300	1.2	1.2	12
ZS(X)500	1.5	1.2	19
EH(X)400	1.5	1.2	19
ZS(X)800	1.5	1.2	19

Brakes

Manual release for spring loaded brake with DC solenoid
Model ES(X)010A ... ES(X)250A, ZS(X)300A, ZS(X)500A

Locking manual release

After the manual release has been operated, this can be locked by seizing the threaded bolt against the magnet housing.



Assembly of the locking:

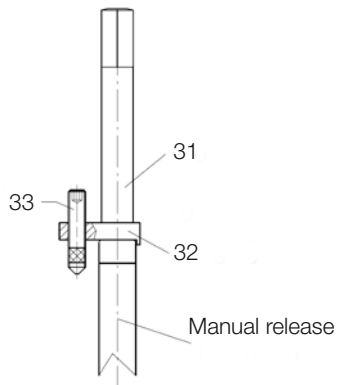
The existing manual release of the brakes ES(X)010 to ES(X)250 and the ZS(X)300 bis ZS(X)500 can be converted to locking manual release through assembly of the plate (Pos. 32) and the threaded bolt (Pos. 33).

- Disassemble lever (Pos.31).

Caution!

The existing spring ring on the lever thread must be removed

- Locate plate (Pos. 32) with threaded bolt (Pos. 33) between the hand release and the lever.
- Tighten lever (Pos. 31)



Brake	Wrench size of the threaded bolt
ES(X)010 to ES(X)027/EH(X)027	2.5
ES(X)040/EH(X)040	2.5
ES(X)070 to ES(X)200/EH(X)200	4
ES(X)250	5
ZS(X)300	4
ZS(X)500	5
EH(X)400	5
ZS(X)800	5

Geared Motors

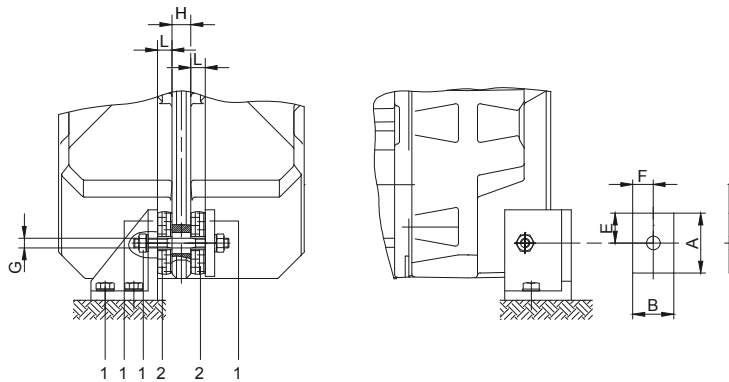
Gear units designed with torque arms and rubber buffers for series BF

1. Installation of the rubber buffer.

The rubber buffers provided are to be fixed in accordance with the drawings N-BF-DST, N-BK-DST or N-BS-DST and tightened to the pretension required.

2. As part of the specified maintenance intervals, the correct tensioning and conditions of the rubber buffers is to be checked and the same changed should faults be apparent. For dynamic applications, this procedure must be performed independent from the standard intervals, every 3.000 hours machine usage.

Note:
Play in the rubber buffers can result in damage to the gearwheels and bearings.



- 1 not included in delivery
- 2 rubber buffer pretensioned

G maximum screw diameter

Material: Natural rubber
Hardness: 50 +/-5 Shore A
Dimensions of the cross hole: See dimensional drawing of the respective gear

Gearbox	Position	A	B	C	D	E	F	G	H	L
BF06	Position 0	30	30	12	12	15	15	M10	10	10
BF10	Position 1	48	32	15	14	24	16	M10	16	13.5
BF20	Position 1	48	32	15	14	24	16	M10	18	13
BF30	Position 2	63	43	20	14	31.5	21.5	M10	18	17
BF40	Position 2	63	43	20	14	31.5	21.5	M10	20	16.5
BF50	Position 3	88	60	25	22	44	30	M18	24	21.5
BF60	Position 3	88	60	25	22	44	30	M18	28	21
BF70	Position 4	123	88	30	26	61.5	44	M20	30	25.5
BF80	Position 5	133	103	35	26	66.5	51.5	M20	40	30
BF90	Position 5	133	103	35	26	66.5	51.5	M20	50	29.5

Dimensions in millimetres (mm)

Geared Motors

Gear units designed with torque arms and rubber buffers for series BK

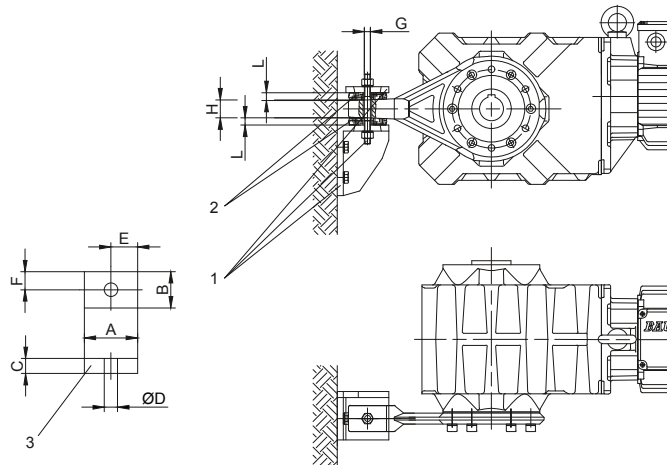
1. Installation of the rubber buffer.

The rubber buffers provided are to be fixed in accordance with the drawings N-BF-DST, N-BK-DST or N-BS-DST and tightened to the pretension required.

2. As part of the specified maintenance intervals, the correct tensioning and conditions of the rubber buffers is to be checked and the same changed should faults be apparent. For dynamic applications, this procedure must be performed independent from the standard intervals, every 3.000 hours machine usage.

Note:

Play in the rubber buffers can result in damage to the gearwheels and bearings.



- 1 not included in delivery
2 rubber buffer pretensioned

- 3 rubber buffer
G maximum screw diameter

Material:

Natural rubber

Hardness:

50 +/-5 Shore A

Dimensions of the cross hole:

See dimensional drawing of the respective gear

Gearbox	Position	A	B	C	D	E	F	G	H	L
BK06	Position 0	30	30	12	12	15	15	M10	10	10
BK08	Position 1	48	32	15	14	24	16	M10	19	13.5
BK10	Position 1	48	32	15	14	24	16	M10	19	13.5
BK17	Position 1	48	32	15	14	24	16	M10	19	13
BK20	Position 1	48	32	15	14	24	16	M10	19	13
BK30	Position 2	63	43	20	14	31.5	21.5	M10	30	17
BK40	Position 2	63	43	20	14	31.5	21.5	M10	30	17
BK50	Position 3	88	60	25	22	44	30	M18	36	21.5
BK60	Position 3	88	60	25	22	44	30	M18	38	21
BK70	Position 4	123	88	30	26	61.5	44	M20	40	25.5
BK80	Position 5	133	103	35	26	66.5	51.5	M20	45	30
BK90	Position 5	133	103	35	26	66.5	51.5	M20	45	29.5

Dimensions in millimetres (mm)

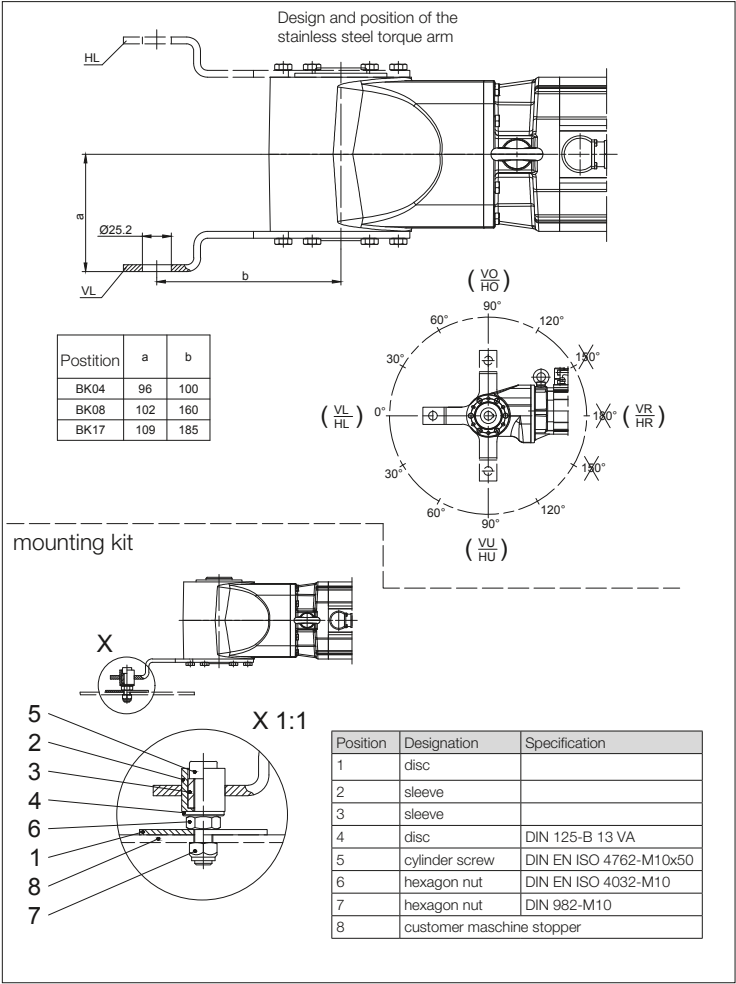
Geared Motors

Gear units designed with torque arms made of stainless steel for series BK

The torque arm is supplied as a set, consisting of a mounting kit and the torque arm in an accessory bag with the product.

Installing the Mounting Kit
The supplied mounting kit must be mounted as shown below.

Note:
This torque arm is not suitable for switching operation. For switching operation, please contact Bauer Gear Motor.



The mounting kit contains 6 stainless steel screws for mounting the torque arm to the gear housing. The tightening torques of the screws must be observed.

Tightening torques of the fixing screws

Size	Tightening torques [Nm]
M6	8
M8	22
M10	43

Geared Motors

Gear units designed with torque arms and rubber buffers for series BS

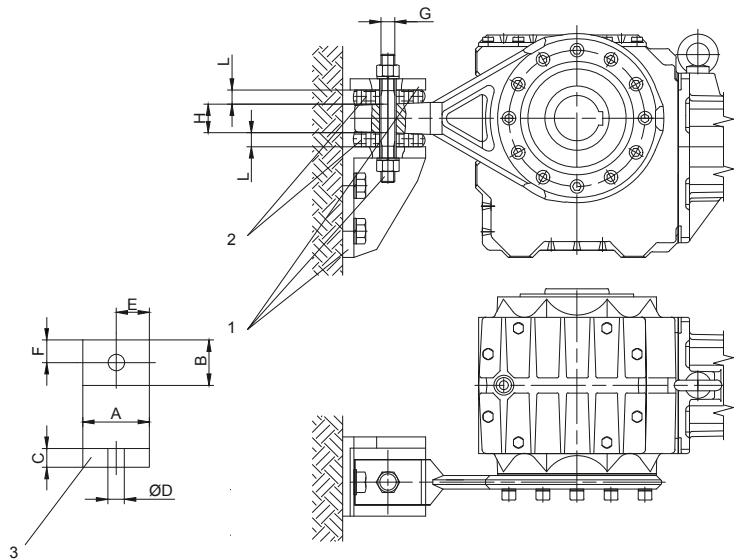
1. Installation of the rubber buffer.

The rubber buffers provided are to be fixed in accordance with the drawings N-BF-DST, N-BK-DST or N-BS-DST and tightened to the pretension required.

2. As part of the specified maintenance intervals, the correct tensioning and conditions of the rubber buffers is to be checked and the same changed should faults be apparent. For dynamic applications, this procedure must be performed independent from the standard intervals, every 3.000 hours machine usage.

Note:

Play in the rubber buffers can result in damage to the gearwheels and bearings.



- 1 not included in delivery
2 rubber buffer pretensioned

- 3 rubber buffer
G maximum screw diameter

Material:

Natural rubber

Hardness:

50 +/-5 Shore A

Dimensions of the cross hole:

See dimensional drawing of the respective gear

Gearbox	Position	A	B	C	D	E	F	G	H	L
BS02	-	-	-	-	-	-	-	M8	6	-
BS03	Position 0	30	30	12	12	15	15	M8	10	10.5
BS04	Position 0	30	30	12	12	15	15	M8	10	10.5
BS06	Position 0	30	30	12	12	15	15	M10	10	10
BS10	Position 1	48	32	15	14	24	16	M10	19	13
BS20	Position 2	63	43	20	14	31.5	21.5	M10	30	17.5
BS30	Position 2	63	43	20	14	31.5	21.5	M10	30	17
BS40	Position 3	88	60	25	22	44	30	M18	38	22

Dimensions in millimetres (mm)

The backstop locks the geared motor in a certain direction of rotation (indication of direction with view of the mounted side of the gear unit).

Mounting

The backstop is assembled on the end shield of the motor.

The inner ring with mounted on clamping part insert is located on the extended rotor shaft. This clamping part insert consists of caging in which the individually spring-loaded clamping parts are guided. The clamping parts lay flush

on the outer ring. The end guard protects against contact and the penetration of foreign objects.

Operation

When the geared motor is started, the clamping parts disengage and do not make contact until the speed of the motor drops below approximately 640/min (D..08), 740/min (D..09,D..11), 665/min (D..13 bis D..18) after disconnection or a power failure. The clamping parts then slowly rise and lock a reversing movement at the moment of rest.

The power transmission in locked state goes from the rotor shaft via the inner ring to the clamping parts and from there via the outer ring to the end shield and the housing of the geared motor.

Supply connection

The standard three phase current motors are normally connected for anti-clockwise rotation when looking at the front of the fan shaft end and with the phase sequence L1 - L2 - L3. The actual phase sequence of the mains is to be selected in such a way that the motor starts in the freewheeling direction. For the first test start, it is advisable to connect particularly larger motors in star connection to protect the backstop as far as possible.

If a brief test connection finds that the motor is not connected in direction of rotation, but in the blocked direction, two mains leads are to be exchanged as with any normal change of direction of rotation. After a wrong connection, check fuses and motor protection switches and check for correct terminal board connection as indicated on the rating plate.



Safety information:

Mounting, connection, adjusting and maintenance work may only be carried out taking into account the safety information given on the accompanying information sheet No. 122 and of the operating instructions for the backstop.

Assembly of the freewheeling mechanisms may only be carried out by trained specialist personnel taking into account the installation information!

The machinery must be secured against rollback before work is performed on the backstop!

Geared Motors

Assembly of standard motor with C-Adapter (IEC and NEMA)

Safety Information

Connection and maintenance work may only be carried out taking into account the safety information on pages 4/5.

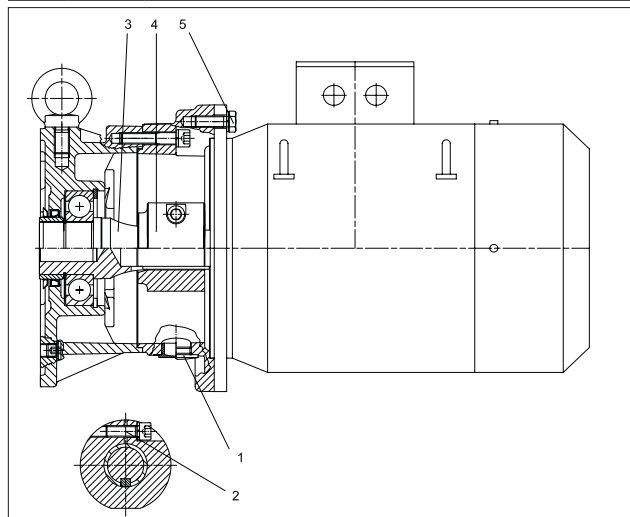
Motor Attachment

The assembly of standard motors in the size range IEC 56 to IEC 280 and NEMA 56 C to NEMA 405 TC using the C-Adapter is to be performed according to the following plan:

- I. Remove assembly plug 1.
- II. Arrange the clamping ring so that the clamping screw 2 is under the assembly hole. Loosen the clamping screw 2 so much that the clamping ring 4 does not tighten on the intermediate shaft 3 anymore.
- III. Arrange the motor and rotor shaft to the gear side mounting face.
- IV. To make assembly easier, bring the motor and gearbox together in the vertical position (motor pointing upwards).
- V. Insert motor shaft into the intermediate shaft without forcing.
- VI. Tighten motor fitting screws 5.
- VII. Tighten clamping screw 2.
- VIII. Insert assembly plug 1.

Tightening torques of screws

IEC	NEMA	Threads DIN 13	M [Nm]
56	-	M6	12.3
63	-	M6	12.3
71	56	M6	12.3
80	-	M8	29.8
90	145	M8	29.8
112	184	M8	29.8
132	215	M12	102
160	256	M12	102
180	286	M12	102
200	-	M16	252
225	326	M16	252
250	365	M16	252
280	405	M16	252



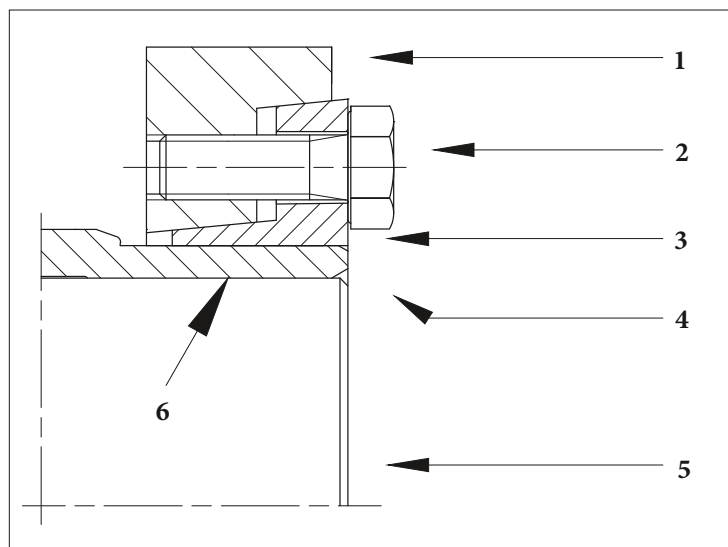
The shrink-disc is supplied ready to be installed and should therefore not be dismantled. The shrink-disc may not be clamped without fitting the shaft. The shaft must be fitted or the hub pushed onto the shaft in the area of the shrink-disc seat.

The tightening bolts must then be tightened evenly around the circumference until the front side face of the outer ring and inner rings is flush. The state of clamping can thus be monitored visually.

To disassemble, loosen all screws evenly in sequence. If the outer ring does not automatically separate from the inner ring, a number of tightening bolts may be removed and screwed into the neighbouring forcing-off threads.

Any rust on the shaft in front of the hub must be removed before the shaft is removed or the hub is taken from the shaft.

Removed shrink-discs only need to be dismantled, cleaned, and regreased before refitting if they are very dirty. In this event, a solid lubricant with a coefficient of friction of $\mu = 0.04$ or better is to be used.



- 1 Outer Ring
- 2 Tightening Bolts
- 3 Inner Ring

- 4 Hub
- 5 Shaft
- 6 grease free

The oil expansion tank is used in critical drive constellations to maintain the build-up of pressure in the gear box resulting from the volumetric expansion of the oil within the permissible range during operation.

The oil expansion tank and the components required for installation are supplied as a three-piece accessory set delivered with the drive.

The oil expansion tank is installed on the gear box in the machine on site when the drive has been brought to its intended mounting position, as stated on the nameplate.

Accessory set, part 1: (tank unit with pre-installed holding plate and connecting hose with ring stud as shown in figure 1).

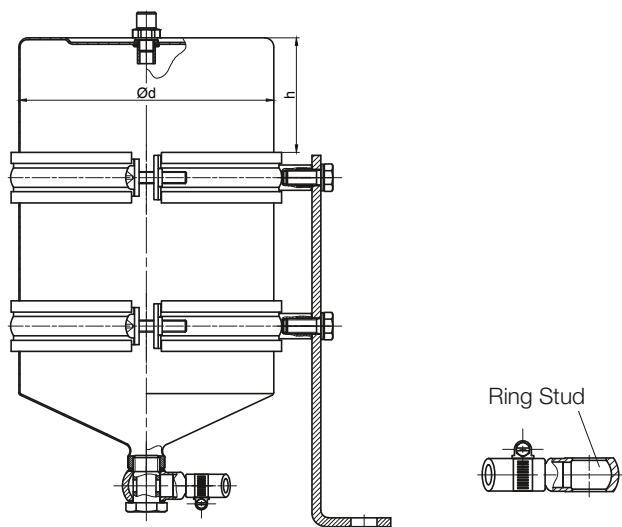


Figure 1: Oil expansion tank

Tank diameter d [mm]	Height positioning h [mm]
ca. 100	5
ca. 112	77

Table 1: Adjustment of height of oil expansion tank

Accessory set, part 2: (Mounting set with nut and lock washer)

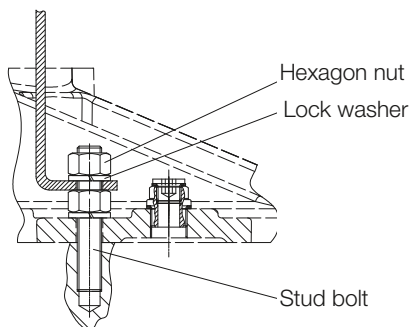


Figure 2: Mounting on the gear housing

The gear is delivered with pre-installed stud bolt, lock washer and hexagon nut.

Place tank unit with retaining plate (= accessory set, part 1) on stud bolt and mount with accessory set, part 2 (consisting of lock washer and hexagon nut) as shown in figure 2. Tighten hexagon nut on stud bolt to 100 Nm for M12 or 250 Nm for M16.

To avoid noise generation, care must be taken during positioning that the container unit is mounted contact-free to the motor housing. If it is necessary to align the connection hose for this or another reason, only the complete tank should be turned in the pipe clamps. To do so, sufficiently loosen the pipe clamps by opening the tightening bolts; see information label on the tank (figure 3).

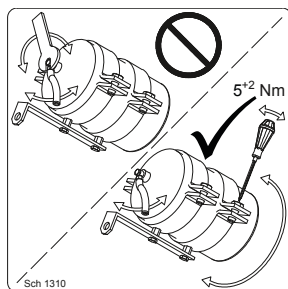


Figure 3: Sch 1310

After aligning the tank while observing the correct height positioning h , see figure 1 and table 1, re-tighten the tightening bolts of the clamps to $5 + 2$ Nm.

Accessory set, part 3: (hollow screw and two sealing washers)

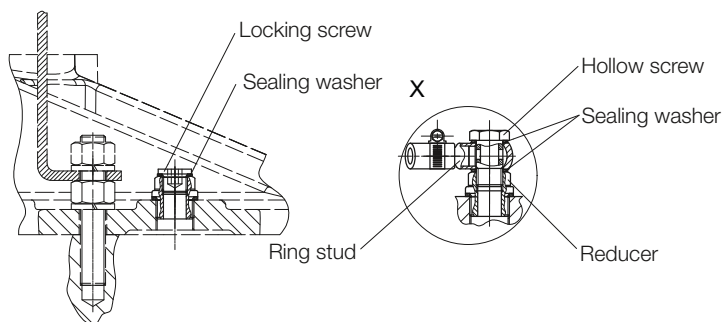


Figure 4: Mounting the connecting hose to gear housing

To ensure the expansion tank functions correctly, the free end of the connection hose must be connected to the gear via the ring stud. To do so, remove the screw plug M16 x 1.5 marked with a red dot and the sealing ring below that; they are no longer required for operation. In the now open threaded hole of the reducer, the ring stud of the connection hose must be screwed in according to figure 4, detail X with new sealing rings and the hollow screw from mounting set 3, taking into account the arrangement of the components. Tightening torque for hollow screw 27 Nm.

If geared motors are to be stored for an extended time before start-up, increased protection against damage by corrosion or humidity can be achieved by observing the following information. Since the actual loading depends very strongly on local conditions, the time data can be regarded only as a guide value. It should also be noted that this data does not include any extension of the warranty term. If disassembly is necessary before start-up according to this information, it is recommended that the nearest BAUER franchised workshop or representatives be called in.

Geared motor condition and storage space

The plugs supplied by the works in all entry holes on the terminal box are to be checked for damage caused during transportation and for correct positioning and replaced if necessary.

Any vent valves which are present are to be removed and replaced with a suitable cover screw for the length of storage.

If any damage is caused in transit to the paintwork or if there are bare areas where the anti-rust paint has come away, such as on shafts or flanges, this should be repaired.

The storage space should be dry, well-ventilated and vibration-free. If the temperature in the space exceeds the normal range of approximately -20°C to $+40^{\circ}\text{C}$ for an extended period of time or varies strongly frequently, it could even become necessary to employ the measures before start-up specified in section "Measures before start-up" after shorter storage times.

Measures during the storage period

The drives should be turned 180° , at intervals of approximately one year, so that lubricant covers the bearings in the gearing, previously located in an upper position, and the cog wheels. The levels of lubricant on the bearings must also be inspected at the same time. The lifespan of lubricant in the bearings will be reduced in the event of lengthy storage. If you see that the lubricant is contaminated, it must be changed.

Turning the drive unit does not have to be carried out if the gear unit enclosure is completely filled with lubricant as the result of a special agreement. In this case, the lubricant level before start-up is to be reduced to the desired value as defined in the operating instructions and the lubrication information plate.

Measures before start-up

Motor component

- Insulation measurement
Measure the insulation resistance of the winding with commercially available measuring apparatus (e.g. with a magneto) between all winding parts and between the winding and the enclosure.

- Measured value above 50 megohm: no drying necessary, new condition
Measured value under 5 megohm: drying advised
Measured value approximately 1 megohm: lowest permissible threshold
- Drying the winding by standstill stator heating without disassembly
Connection to stepless or tapped variable alternating current voltage up to approximately 20 % maximum of the rated voltage. Heating current max. 65 % of the rated current according to the rating plate. Observe heating up for first 2 to 5 hours; reduce heating voltage if necessary.
Heating duration approximately 12 to 24 hours until insulation resistance rises to desired value.
- Dry the winding in the oven after disassembly
Dismantle the motor in the appropriate manner
Dry the stator winding in a well ventilated drying oven at between 80 °C and 100 °C for approximately 12 to 24 hours until the insulation resistance rises to the desired value.
- Lubricating the rotor position
If the storage period exceeds approximately 2 to 3 years, or the temperatures were very unfavourable throughout a shorter storage period as described in "Geared motors with three phase cage rotors" paragraph 3, the lubricant in the rotor positions must be checked and refilled if necessary. For checking, a partial assembly on the fan side is sufficient, where the rolling contact bearing becomes visible after removal of the fan cowl, fan and bearing flange (end shield).

Gear unit component

- Lubricant
If the storage period exceeds approximately 2 to 3 years, or the temperatures were very unfavourable throughout a shorter storage period as described in "Geared motors with three phase cage rotors" paragraph 3, the lubricant in the gear unit must be changed. For detailed instructions and lubricant recommendations please see chapter lubrication quantity.
- Shaft seals
When changing the lubricant, the function of the shaft seals between the motor and gear unit as well as on the output shaft must also be checked. If a change in shape, colour, hardness or sealing effect is determined, the shaft seals must be replaced appropriately under observance of the after-sales manual.
- Gaskets
If lubricant is draining out at the connecting points on the gear unit enclosure, the sealing compound must be replaced as described in the after-sales manual.
- Vent valve
If a vent valve was replaced with a cover screw when storing, the vent valve must be refitted to the point of removal.

Bauer does not make any representations, warranties or guarantees, expressed or implied, as to the accuracy or completeness of this manual or any statements, technical information and recommendations contained herein, or in any other documentation supplied by Bauer in connection with the use of the geared motor or gear unit (the "Product"). Before using the Product you should determine its suitability for your intended use. You bear all risk in connection with the use of the Product. You are reminded that all warranties as to merchantability and fitness for purpose are excluded from the contract under which the Product and this manual have been supplied to you. Bauer's only obligation in this respect is, at its option, to repair or replace any product proven to be defective. Neither Bauer nor any of its affiliates or any of their respective directors, officers, employees or agents shall be liable in contract, tort or in any other manner whatsoever to any person for any direct or indirect loss, damage, injury, liability, cost or expense of any nature (whether loss of profit or otherwise), including without limitation incidental, special, direct or consequential damages arising out of or in connection with the use of this manual.

Photos/Graphics: Bauer-Archiv, Chr. Mayr, Precima

Bauer Gear Motor GmbH

Eberhard-Bauer-Straße 37

73734 Esslingen

Germany

Tel.: +49 711 3518-0

Fax: +49 711 3518-381

www.bauergears.com

Motor-data sheet

Motor item no.: 51718583

Manufacturer: Bauer

Description (1)	Value (1)	Description (2)	Value (2)
Motor type:	DPE09XA4-D/C2-SP	Gearbox type:	BF40-74W
Motor power [KW]:	1,1	Gear ratio:	111,1
Motor speed (n1) [1/min]:	1440	Output speed (n1) [1/min]:	13,0
Voltage [V]:	400	Torque - output: [Nm]:	800
Frequency [Hz]:	50	Weight [kg]:	66
Nominal current [A]:	2,4	Lubricant type:	CLP220
Starting current [A]:		Lubricant amount [l]:	4,9
Insulation class:	F		
Protection class [IP]:	65		
Ex-protected:	Nein		
Power factor cos phi:	0,78		
Efficiency:	84,4		
Efficiency class:	IE3		
Motor technology:	ASM		

Notes:

BAUER

Bauer Antriebstechnik GmbH
Postfach 10 02 08
D-73726 Esslingen
Eberhard-Bauer-Straße 36 – 60
D-73734 Esslingen
Telefon (0711) 35180
Telefax (0711) 3518381
www.bauer-gear-motors.de

BAUER

Einzelteilverzeichnis
und Schnittzeichnung

Parts list
and sectional drawing

Liste des pièces
et dessin en coupe

Despiece y sección

Спецификация деталей
и схема в разрезе

ET-BF 40

Getriebe-Typ

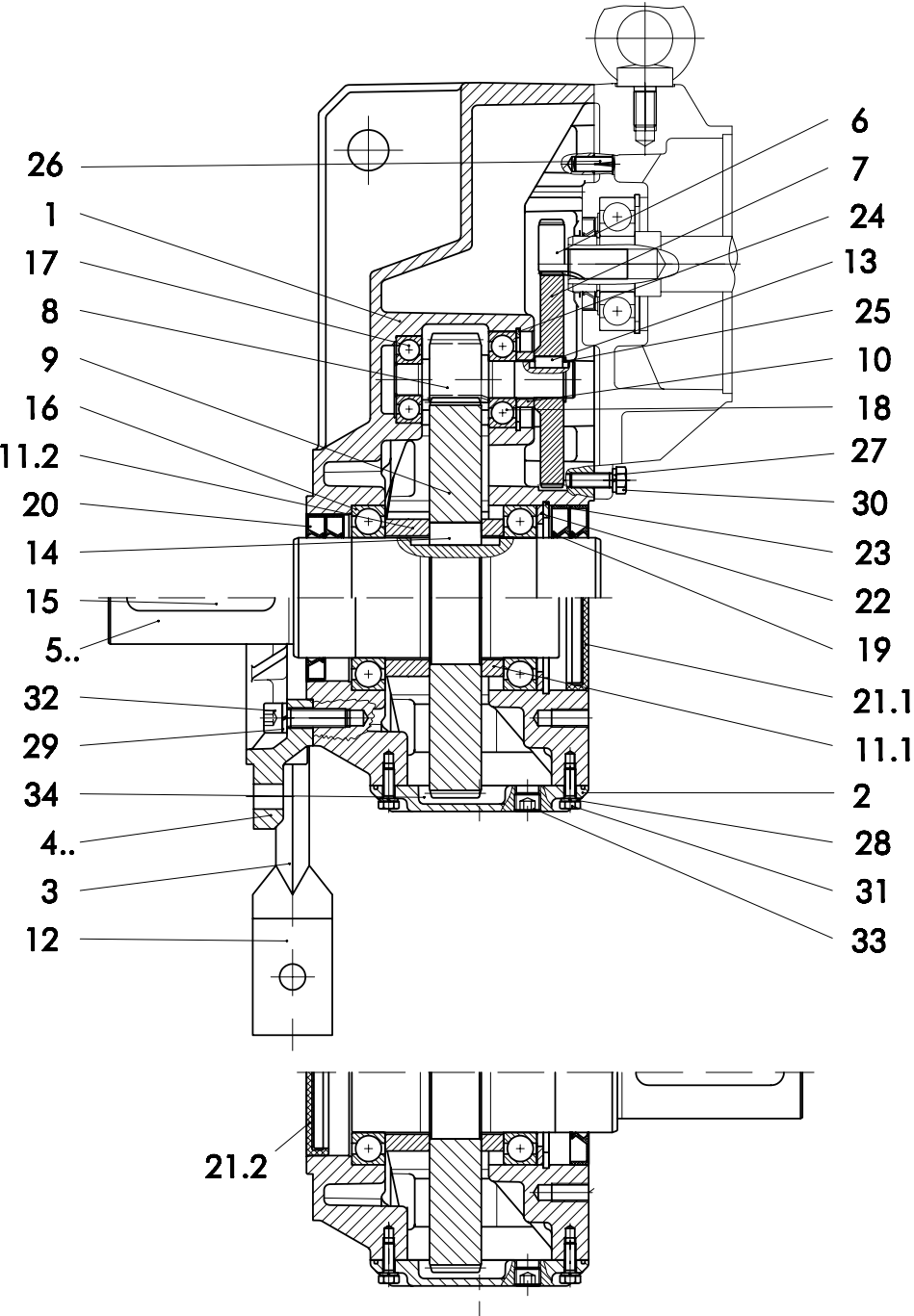
Gear unit type

Réducteur type

Reductor tipo

Редуктор тип





Teil-Nr.	Part-No. / Pièce No. / Pieza № / Деталь №					
	Bestell-Nr. / Order No. / Comm. No. / Código № / Заказ №					
	Stück / Quantity / Pièces / Cantidad / Шт.					
			Benennung	Désignation	Désignation	Наименование
34	—		Schmierstoff	Lubricant	Lubrifiant	Смазочное средство
33	2506530	2	Verschlusschraube	Cover screw	Vis de fermeture	Резьбовая пробка
32	1016121	0/6	Zylinderschraube	Fillister head screw	Vis à tête cylindrique	Винт с цилиндрической головкой
31	1365703	6	Sechskantschraube	Hexagon bolt	Vis à tête hexagonale	Винт с шестигранной головкой
30	1316656	8	Sechskantschraube	Hexagon bolt	Vis à tête hexagonale	Винт с шестигранной головкой
29	1011626	0/6	Federring	Spring washer	Rondelle ressort	Пружинное кольцо
28	1010484	6	Federring	Spring washer	Rondelle ressort	Пружинное кольцо
27	1010492	8	Federring	Spring washer	Rondelle ressort	Пружинное кольцо
26	1015915	2	Paßkerbstift	Dowel pin	Goupille cannelée	Установочный штифт
25	1000179	1	Sicherungsring	Retainer ring	Circlip	Стопорное кольцо
24	1015966	1	Sicherungsring	Retainer ring	Circlip	Стопорное кольцо
23	1000306	1	Sicherungsring	Retainer ring	Circlip	Стопорное кольцо
22	2502569	1	Stützscheibe	Supporting plate	Rondelle d'appui	Упорная шайба
21.2	1448366	0/1	Verschlusskappe	Sealing cover	Couvercle de protection	Запорная крышка
21.1	1448374	0/1	Verschlusskappe	Sealing cover	Couvercle de protection	Запорная крышка
20	2504162	0/1/2	Wellendichtring	Shaft seal	Bague d'étanchéité d'arbre	Манжета
19	2502861	0/1/2	Wellendichtring	Shaft seal	Bague d'étanchéité d'arbre	Манжета
18	1015672	1	Rillenkugellager	Ball bearing	Roulement	Подшипник
17	1010166	1	Rillenkugellager	Ball bearing	Roulement	Подшипник
16	2502828	2	Rillenkugellager	Ball bearing	Roulement	Подшипник
15	1164104	0/1	Paßfeder	Key	Clavette	Призматическая шпонка
14	2502992	1	Paßfeder	Key	Clavette	Призматическая шпонка
13	2503000	1	Paßfeder	Key	Clavette	Призматическая шпонка
12	1387481	0/2	Gummipuffer	Rubber buffer	Cales élastiques	Резиновый компенсатор
11.2	2503948	1	Distanzhülse	Spacer	Douille d'espacement	Распорная втулка
11.1	2503930	1	Distanzhülse	Spacer	Douille d'espacement	Распорная втулка
10	2503794	1	Distanzhülse	Spacer	Douille d'espacement	Распорная втулка
9	*	1	Zahnrad	Gear	Roue dentée	Зубчатое колесо
8	*	1	Zahnrad	Gear	Roue dentée	Зубчатое колесо
7	*	1	Zahnrad	Gear	Roue dentée	Зубчатое колесо
6	*	1	Ritzel	Pinion	Pignon	Шестерня
5.3	2503131	0/1	Arbeitswelle	Output shaft	Arbre secondaire	Рабочий вал
5.2	2503123	0/1	Arbeitswelle	Output shaft	Arbre secondaire	Рабочий вал
5.1	2503115	0/1	Arbeitswelle	Output shaft	Arbre secondaire	Рабочий вал
4.2	2503859	0/1	Flansch	Flange	Bride	Фланец
4.1	2503841	0/1	Flansch	Flange	Bride	Фланец
3	2503883	0/1	Drehmomentsstütze	Torque restraining arm	Appui de couple	Упор против проворачивания
2	2503905	1	Gehäusedeckel	System cover	Couvercle de carter	Крышка корпуса
1	*	1	Anbauehäuse	Attachment housing	Boîtier de montage	Приставной корпус

BAUER

Bauer Antriebstechnik GmbH
Postfach 10 02 08
D-73726 Esslingen
Eberhard-Bauer-Straße 36 – 60
D-73734 Esslingen
Telefon (0711) 35180
Telefax (0711) 3518381
www.bauer-gear-motors.de

BAUER

Einzelteilverzeichnis
und Schnittzeichnung

Parts list
and sectional drawing

Liste des pièces
et dessin en coupe

Despiece y sección

Спецификация деталей
и схема в разрезе

ET-D 09

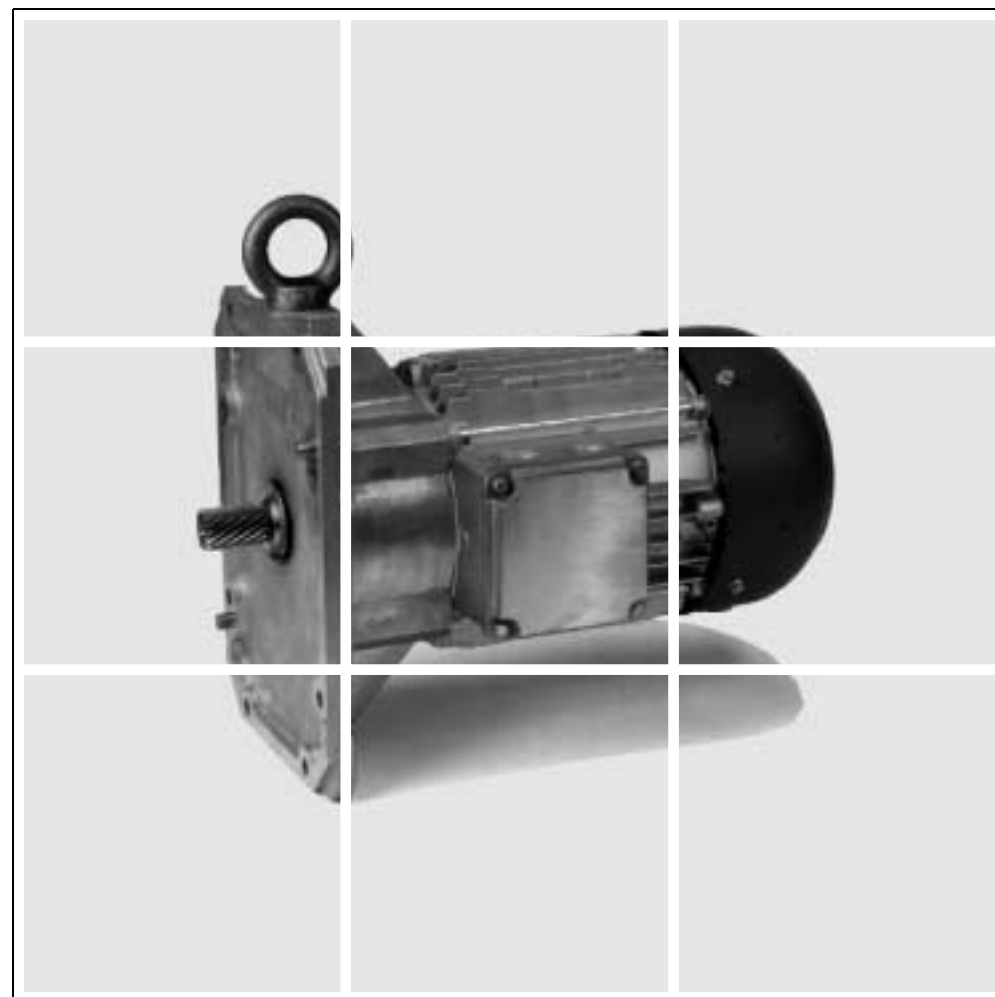
Motor-Typ

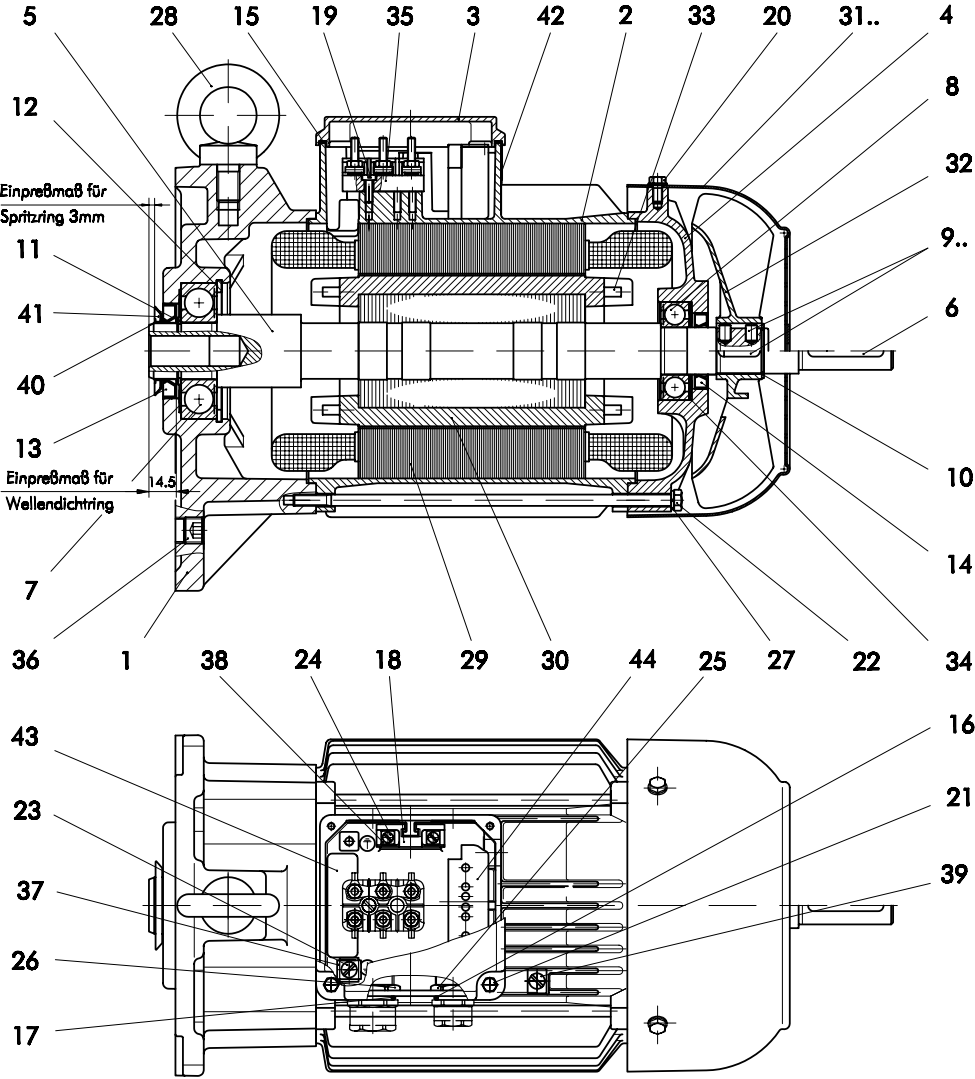
Motor type

Moteur type

Motor tipo

Двигатель тип





Teil-Nr.	Part-No. / Pièce No. / Pieza № / Деталь №				
	Bestell-Nr.	Order No.	Comm. No.	Código №	Заказ №
Stück / Quantity / Pièces / Cantidad / Шт.					
	Benennung	Designation	Désignation	Designación	Наименование
44	*	0/1	Rectifier, komplett	Redresseur	Выпрямитель пер. напр. в комплекте
43	2515067	1	Füllstück	Packing piece	Проставка
42	*	1	Leistungsschild	Rating plate	Шильдик
41	2502267	1	Spritzring	Centrifugal washer	Маслоотражающее кольцо
40	2513480	1	Speedi-Sleeve-Hülse	Speedi-Sleeve spacer	Втулка „Speedi-Sleeve“
39	1242351*	0/1	Schutzleiterklemme	Stator earth terminal	Клемма заземления
38	1271407*	0/2	Anschlußscheibe	Connection disc	Присоединительная клемма
37	2504197	2	Klemmbügel	Clamping unit	Клеммная скоба
36	*	2	Verschlußschraube	Cover screw	Резьбовая пробка
35	*	1	Klemmbrett, vollständig	Terminal board, complete	Клеммная колодка в сборе
34	1024205	1	Kula-Ausgleichscheibe	Kula-Shim	Кула-Компенсирующая шайба
33	-		Wuchtgewicht	Balancing weight	Противовес
32	1013921	1	Flügelrad	Fan	Вентилятор
31.2	1319965*	0/1	Lüfterhaube	Fan cowl	Колпак вентилятора
31.1	1013971	0/1	Lüfterhaube	Fan cowl	Колпак вентилятора
30	*	1	Preßgußläufer	Pressure diecast rotor	Запрессованный ротор
29	*	1	Statorpaket	Statorpacket	Статорный пакет
28	*	1	Ringschraube	Eye bolt	Рым-болт
27	1010484	4	Federring	Spring washer	Пружинное кольцо
26	1491792	1	Sechskantmutter	Hexagon nut	Шестигранная гайка
25	1307550	1	Sechskantmutter	Hexagon nut	Шестигранная гайка
24	1232657*	0/2	Zylinderschraube	Allen screw	Винт с цилиндрической головкой
23	1011715	1	Zylinderschraube	Allen screw	Винт с цилиндрической головкой
22	2506611	4	Sechskantschraube	Hexagon bolt	Шестигранный болт
21	2519372	4	Gewindeform-Schraube	Thread forming screw	Фасонный болт
20	1397591	4	Gewindeform-Schraube	Thread forming screw	Фасонный болт
19	1397508	1	Gewindeform-Schraube	Thread forming screw	Фасонный болт
18	2506327*	1	Thermistor-Klemme	Thermistor terminal	Клемма термисторов
17	2513676	1	O-Ring	O-Ring	Уплотнительное кольцо круглого сечения
16	2513684	1	O-Ring	O-Ring	Уплотнительное кольцо круглого сечения
15	2506319	1	Dichtung	Gasket	Прокладка
14	1329740	1	Wellendichtring	Shaft seal	Манжета
13	2505088	1	Wellendichtring	Shaft seal	Манжета
12	1000381	1	Sicherungsring	Retainer ring	Стопорное кольцо
11	1011278	1	Sicherungsring	Retainer ring	Стопорное кольцо
10	1011251	1	Sicherungsring	Retainer ring	Стопорное кольцо
9.2	1021541*	0/1	Paßfeder	Key	Призматическая шпонка
9.1	2504979	0/2	Zylinderstift	Parallel pin	Цилиндрический штифт
8	1028961	1	Rillenkugellager	Ball bearing	Подшипник
7	2505096	1	Rillenkugellager	Ball bearing	Подшипник
6	1015788*	0/1	Paßfeder	Key	Призматическая шпонка
5	*	1	Läuferwelle	Rotor shaft	Вал ротора
4	2504448	1	B-Lagerschild	End shield	Щит подшипника
3	2506262	1	Klemmenkastendeckel	Terminal box cover	Крышка клеммной коробки
2	2504758	1	Ständergehäuse	Stator housing	Корпус статора
1	*	1	Systemdeckel	System cover	Передний щит